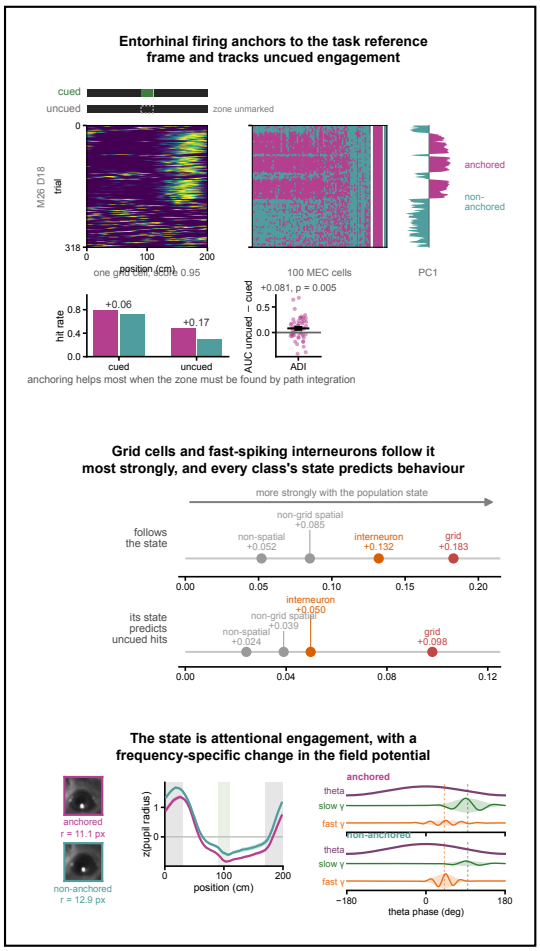
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# Entorhinal grid codes are gated by attention

## Graphical abstract




## Highlights

- Anchoring of entorhinal firing to the task frame is a population state
- Grid cells and fast-spiking interneurons follow it most strongly
- Pupil constriction tracks the state, identifying it as focused engagement
- The grid attractor survives the loss of anchoring; it is left uncorrected


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## In brief

Clark et al. show that the anchoring of entorhinal spatial firing to a task reference frame is a property of the population rather than of single cells. Every cell class follows this state, grid cells and fast-spiking interneurons most strongly, yet synaptically connected pairs share it no more than matched unconnected pairs, and the same state is present in every structure recorded. It tracks attentional engagement, indexed by pupil constriction, and predicts success when the reward zone must be found by path integration. The grid attractor keeps running when the state is lost: what the state sets is whether it is corrected to the task.

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# Entorhinal grid codes are gated by attention

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## SUMMARY

Spatial firing of neurons in the medial entorhinal cortex (MEC) is not fixed to the external world. During a location memory task on a virtual linear track, the firing of a single neuron can be anchored to the task reference frame on some trials and not on others, and this anchoring predicts whether the animal localises a reward that can only be found by path integration. What governs the switching is unknown. Here we show that anchoring is a property of the population rather than of single cells, and that the population state is not built by the entorhinal circuit that expresses it. Using four-shank Neuropixels recordings from 8 mice, with open-field recordings on the same days so that cell identity is assigned independently of the task, we find that trial-by-trial anchoring is correlated across simultaneously recorded neurons and organised into blocks of tens of trials. Measured against each cell's own null, every cell class follows this population state, but grid cells and putative interneurons do so roughly twice as strongly as spatial cells without grid structure. Grid cells and interneurons also agree with each other, and their pairing is confirmed in millisecond spike timing, where interneuron-to-grid is the strongest monosynaptic pathway we detect. Yet synaptically connected pairs share the anchoring state no more than unconnected pairs matched for identity and distance, indicating that the coordination is not propagated through the local circuit. Consistent with the state being imposed from outside, pupil diameter tracks it, and performance improves monotonically as the pupil narrows, identifying the anchored state as one of focused task engagement. Every structure we recorded - including visual cortex and cerebellum - carries a coherent anchoring state of its own, and corrected for how reliably each is estimated these states are indistinguishable. What differs is the cell type rather than the structure. The local field potential changes with the state in a frequency-specific way, with theta and mid gamma falling while slow gamma rises, and theta's coupling to running speed roughly halves - yet the fast-spiking speed cells that carry the septal speed signal, identified in the open field, show no change in their speed coding between states. Reading the grid population as a bump on a torus, we find that the attractor survives the loss of anchoring: the bump remains well formed in both states, so the non-anchored state is an attractor that is running uncorrected rather than one that has come apart. A continuous attractor in which engagement gates a landmark correction onto the ring reproduces this, together with the direction and size of the theta and slow-gamma changes, and it predicts that fast gamma should track the moment-to-moment positional error - which we confirm in the recordings, in the fast band only. We suggest that the entorhinal circuit does not generate the anchoring state but contains cell types tuned to read it, and that what the state sets is not whether a grid code exists but whether it is corrected to the task.

## INTRODUCTION

Grid cells in the medial entorhinal cortex (MEC) are thought to provide a metric for space that supports navigation and memory. A standard view is that this metric is continuously available and stably anchored to the external world. Several observations challenge that view. On circular tracks, grid firing remains periodic but is no longer anchored to the environment, so that it is informative about distance travelled rather than position. Populations of MEC neurons can spontaneously remap between representations within a session. During a virtual-track location memory task, grid firing switches between a task-anchored mode, in which fields recur at the same track positions, and a task-independent mode, in which they do not, and this switching is associated with behaviour: anchoring predicts successful localisation of a reward zone when the animal must path-integrate to find it, but not when a visual cue marks it.

That dissociation makes the switching itself worth explaining. If anchoring determines whether an internal position estimate is usable, then whatever controls anchoring controls when the grid code contributes to behaviour. Two broad accounts are available. Under the first, anchoring is set within the entorhinal circuit - by the balance of visual and self-motion input to the grid network, or by interactions among its own neurons, in which case the cells that share an anchoring state should be those that are connected to one another. Under the second, anchoring reflects a variable that is not entorhinal at all, with the circuit responding to a signal generated elsewhere, in which case shared anchoring should track what a cell is rather than what it is wired to, and the same signal should be legible outside MEC.

Distinguishing these accounts requires three things that have not previously been available together: simultaneous recordings of enough neurons to define a population state trial by trial, independent assignment of cell identity so that the categories cannot be shaped by the dynamics under study, and recordings that extend beyond MEC. We obtained these using four-shank Neuropixels 2.0 probes targeted to span the mediolateral extent of MEC, with the same neurons recorded in an open arena on the same day, and with the probe shanks also sampling the pre-, para- and postsubicular complex, visual cortex, and in some animals cerebellum.

We show here that anchoring to the task reference frame is a population-level state that every entorhinal cell class follows, but that grid cells and putative interneurons follow about twice as strongly as spatial cells without grid structure. These two classes are synaptically coupled, but their synaptic coupling does not account for their shared anchoring, which indicates that the state is imposed on the population by a common input. Consistent with this, the state tracks pupil-indexed task engagement, and it is present in every structure we recorded, with only the strength of the readout differing by cell type. We conclude that the entorhinal circuit

does not construct the anchoring state but contains neurons tuned to it.

## RESULTS

We recorded from four-shank Neuropixels 2.0 probes targeted to sample the mediolateral extent of the MEC, in mice exploring an open arena and then performing a location memory task on a virtual linear track (Figure 1A-C). Mice were trained for 26.6 +/- 6.3 days (range 19-41) before recording (Figure 1D). On cued trials the reward zone was marked by a visual cue; on uncued trials the cue was absent and the zone could only be localised efficiently by path integration. Cell identity was assigned from the open-field session: neurons with grid firing fields in the arena ([n]/[n]) typically showed repeating localised firing on the track, whereas neurons with non-grid spatial firing in the arena ([n]/[n]) more often showed ramping profiles (Figure 1E). Across the dataset we analysed 6,455 MEC neurons from 61 sessions in 8 mice.

### Anchoring to the task reference frame is a property of the population

A cell whose firing is anchored on some trials and not others could be switching independently of its neighbours, in which case anchoring is a property of single neurons, or it could be switching together with them, in which case anchoring is a property of the population. These possibilities make different predictions for simultaneously recorded cells: under the first, trial-by-trial anchoring labels should be uncorrelated across cells; under the second, they should be correlated, and the correlation should exceed what is expected from each cell's overall tendency to be anchored.

To distinguish them we classified each trial of each cell as anchored or non-anchored by comparing that trial's rate map against the cell's own circularly shifted null (see Methods). We then took the first principal component of the resulting cell x trial label matrix and compared it against a circular-shift null that preserves each cell's anchored fraction, so that a cell which is anchored on most trials cannot generate apparent population structure on its own. The first principal component exceeded this null by approximately twofold (Figure 1F). Anchoring was organised in time as well as across cells: blocks of tens of trials shared a state, and the population crossed between states over approximately three trials (Figure 1G). We refer to the per-trial fraction of anchored cells as the population anchoring state.

To ask whether this state is behaviourally relevant we compared stopping outcomes between anchored and non-anchored trials. Mice were more likely to stop in the reward zone on anchored trials, and the advantage was larger on uncued trials, where the zone can only be localised by path integration, than on cued trials, where a visual landmark is available. Quantified per session as the difference between the uncued and cued anchoring benefit, which we refer to as the Anchoring Dependence Index, the effect was positive (+0.094, p = 0.012). This reproduces, at the level of a population state, the selective association between anchoring and path integration-dependent performance previously reported for individual grid cells.

Together, these data indicate that anchoring to the task reference frame is a coordinated state of the recorded population rather than an independent property of single neurons, and that it matters most for behaviour when the animal must rely on an internal estimate of position.

### Grid cells and putative interneurons follow the population state most strongly

The population state is an average over cells and so does not by itself indicate which cells enter it. We considered two possibilities. If anchoring is a property of spatial coding as such, then all spatially tuned cells should follow the state. If instead it is a property of a particular subnetwork, then following should be restricted to specific cell types.

Testing this requires a definition of cell identity that cannot have been shaped by the phenomenon under study. Firing on the linear track changes with engagement and task demand, which is what we are trying to explain, so classifying cells on the same session would allow those dynamics to define the categories and then rediscover them. We therefore assigned identity entirely from the open field - grid, non-grid spatial and non-spatial classes from open-field rate maps - on a different environment and a different behaviour.

It also requires an appropriate null. Chance for this comparison is not zero: a cell that is anchored on most trials will correlate with a population that is also anchored on most trials, and we measured the resulting bias directly by circularly shifting each cell's label sequence. The null sits near +0.15 rather than at zero, and each cell was therefore compared against its own shifted distribution.

Measured this way, every identity exceeded chance, and they differed in degree (Figure 2D). Grid cells led (mean |r| = 0.342 against a chance level of 0.157, excess +0.185, $p = 1 \times 10^{-11}$, above chance in 7 of 7 mice), followed by putative interneurons (0.280 against 0.149, excess +0.131, $p = 3 \times 10^{-16}$, 7 of 7), then non-grid spatial cells (0.228 against 0.147, excess +0.081, $p = 6 \times 10^{-22}$, 8 of 8) and non-spatial cells (0.197 against 0.141, excess +0.056, $p = 4 \times 10^{-16}$, 7 of 7). Grid cells and interneurons did not differ from one another (p = 0.54) and both exceeded the other two classes ($p <= 8 \times 10^{-6}$, Holm-corrected), while non-grid spatial cells in turn exceeded non-spatial ones ($p = 2 \times 10^{-6}$). The same ordering appears cell by cell: 43% of grid cells and 37% of interneurons individually beat their own null, against 24% of non-grid spatial and 19% of non-spatial cells, all well above the 5% false-positive rate (Figure 2E). Because the two leading classes are also the two most theta-rhythmic, we asked whether theta rhythmicity predicts following independently of cell type, using a theta index scored in the open field so that it cannot have been shaped by the session it is used to explain (Figure 2K). It does. The index contrasts theta-band power in a cell's instantaneous firing rate against the power in the adjacent bands, and is reliable across the two open-field sessions of a day (Spearman rho = 0.865, 11,810 cells). It separates the identities - median 0.229 in putative interneurons, 0.088 in grid cells, 0.046 in non-grid spatial and 0.020 in non-spatial cells - and it predicts agreement with the population axis beyond them: +0.173 per unit index alone ($p = 1 \times 10^{-11}$) and +0.136 with identity in the model ($p = 2 \times 10^{-6}$; mixed model with mouse and session random effects). The relationship also holds within classes taken separately, in three of the four: grid +0.353 (p = 0.037), non-grid spatial +0.147 ($p = 9 \times 10^{-5}$) and putative interneurons +0.116 (p = 0.028), with only non-spatial cells failing to reach significance. Theta rhythmicity is therefore a predictor of following in its own right rather than a restatement of which classes are rhythmic. Anchoring is therefore not confined to the attractor-associated identities, but it is strongest in them, and the gap between grid cells and non-grid spatial cells - cells that are spatially tuned but carry no grid structure - is more than twofold.

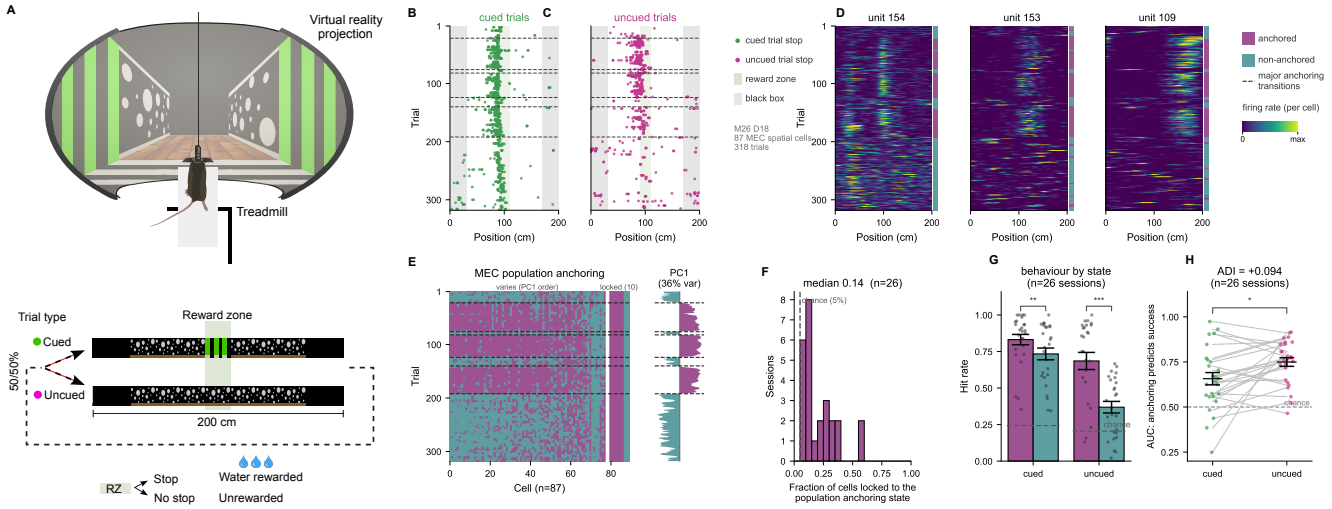



**Figure 1. A population anchoring state in medial entorhinal cortex improved path integration-based navigation**

(A) The task: head-fixed mice run a 200 cm virtual linear track and stop in a reward zone, visually cued on half the trials and uncued on the rest, where it can be localised only by path integration. (B, C) Stops against track position for one session, cued and uncued trials separately. (D) Three example units as trial x position rate maps: firing anchors to the track on some trials and not others. (E) The MEC anchoring raster for the same session, cells against trials, with the first principal component of that matrix beside it. PC1 exceeds a circular-shift null that preserves each cell's anchored fraction by roughly twofold, and blocks of tens of trials share a state. (F) The anchored fraction per session, with the median. (G) Hit rate by state, cued and uncued trials separately. The dashed line is chance, derived as follows. A stop is the onset of a period below the task's own 3 cm/s reward criterion, and a trial is a hit if any stop falls in the 20 cm reward zone. For each trial the whole set of its stops is then shifted rigidly by a distance drawn uniformly from 0 to 200 cm and wrapped around the track, and the trial is scored a hit if any shifted stop lands in the zone; the chance level is that score averaged over 500 shifts per trial and over trials. The shift keeps how many times the animal stopped and how those stops were spaced, and destroys only where they sat relative to the reward zone, so it asks how often this trial would have hit had the zone been elsewhere. It comes to 0.24 on cued and 0.21 on uncued trials over 62 sessions, and it is not 20/200 because most trials carry several stops, which come in clusters that a rigid shift moves together. Trials with no stop are kept, as misses under both the data and the null. Chance varies widely between sessions (0.02 to 0.66) with how much the animal stops, so the line is a mean; within a session it does not differ between the two states, and subtracting each state's own chance widens the gap in G rather than narrowing it. (H) The Anchoring Dependence Index, the uncued minus cued anchoring benefit: +0.094, p = 0.012.

That both classes exceed chance does not establish that they are coordinated with each other, since two populations could track a common signal independently. We therefore correlated every pair of simultaneously recorded cells and averaged by identity pair, again as an excess over an independent circular-shift null so that a pair which is simply anchored most of the time scores nothing for it. Grid-interneuron agreement reached +0.109 (29 sessions, $p = 1.4 \times 10^{-6}$) against +0.058 for the remaining off-diagonal pairs, and was indistinguishable from either within-group value (grid-grid +0.115, interneuron-interneuron +0.115). Aligned to a transition in the population state, all four identities switched together over approximately three trials, and identity did not predict switch timing (p = 0.15).

We carried out three further analyses to establish what a non-anchored trial is. First, a non-anchored trial could reflect a firing field that has moved rather than one that has gone. Correlating each trial's map against the cell's anchored template at every circular offset, the correlation at zero offset collapsed (+0.33 to +0.07) while the best correlation over offsets fell only slightly (+0.56 to +0.53), which superficially suggests displacement. However, two unrelated cells reach +0.500 under the same best-of-offsets search, and non-anchored trials sat only 0.029 above that floor with a mean best offset of 47 cm against the 50 cm expected from a uniform choice on a 200 cm track. Non-anchored trials therefore show loss of spatial correspondence with no evidence of a coherent field elsewhere.

Second, anchoring concerns where cells fire rather than how much. Firing rate differences between states were small and their sign depended on the measure: −0.090 in log when rate was averaged over position bins, −0.028 when computed as spikes per second of running, and +0.036 when trials were additionally speed-matched (all $p < 0.005$, n = 6,517). An effect of this size that reverses under a change of denominator and a behavioural control is not a rate effect.

Third, under the gate a cell whose two clusters both fall on the same side of its null is labelled anchored on every trial or on none, and its correlation with the population axis is then undefined. These cells are 24.8% of the population in sessions where the state actually switches, and the split is identity-dependent and runs in opposite directions: putative interneurons locked on (109 always-anchored against 2 never, consistent across mice), non-spatial cells locked off (39 against 293), while grid and non-grid spatial cells were balanced. Non-grid spatial cells dominate both locked groups only because they are approximately 70% of the population, and they lock at their base rate.

Speed coding accounts for the locking, and does so in the direction the identity split predicts. A cell tuned to running speed has a profile that repeats on every trial, because the speed profile along the track is itself stereotyped; the classifier compares each trial's rate map against the cell's own template, so such a cell is called anchored on every trial and never varies. Entorhinal speed cells are predominantly fast-spiking interneurons, which is the class that locks on. Scoring speed modulation in the open field, so that it cannot have been shaped by the task dynamics being explained, the interneurons that lock on are markedly more speed-modulated than those that vary (median open-field speed score +0.141 against +0.065, p = 0.005; Figure 2I), and the same gradient holds in principal cells (+0.042 against +0.029, $p = 8 \times 10^{-5}$). The locked-on population is therefore not a separate phenomenon but the speed-coding population, seen through a classifier that asks whether firing repeats across trials.

Together, these data indicate that the population anchoring state is followed by grid cells together with local inhibition, rather than by spatial tuning in general, and that these two

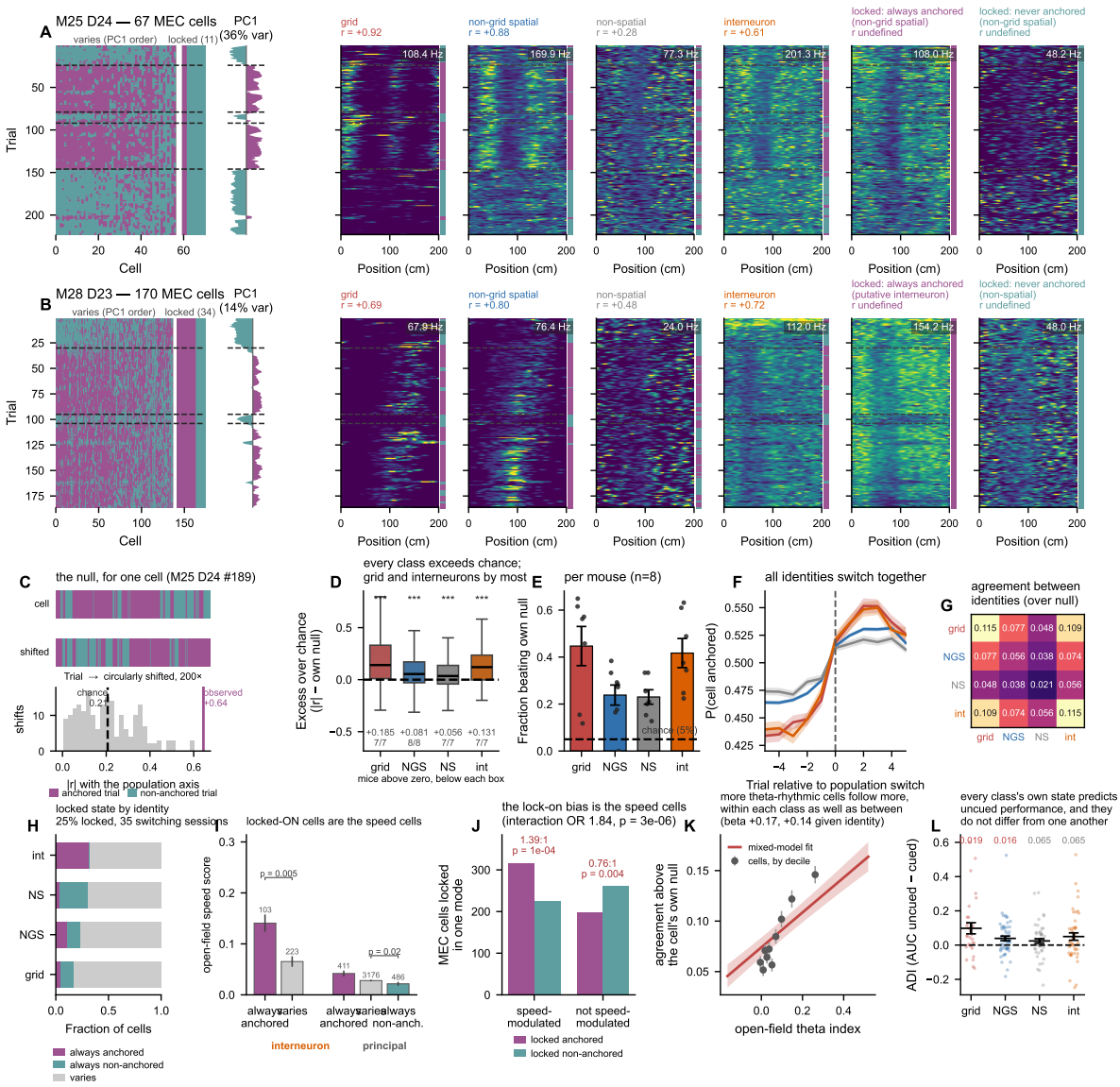



**Figure 2. Grid cells and putative interneurons follow the population state most strongly**

(A, B) Two example sessions: the MEC anchoring raster with PC1, and example single cells of each identity, including cells locked into one mode. (C) The null for one cell, its labels circularly shifted 200 times against the population axis, with the observed value marked. Chance lies near +0.15 rather than at zero, because a cell anchored on most trials correlates with a population that is also anchored on most trials. (D) Agreement with the population axis by identity, as excess of |r| over each cell's own null. Every class exceeds chance and they differ in degree: grid +0.185 (p = 1e-11) and putative interneuron +0.131 (p = 3e-16) against non-grid spatial +0.081 (p = 6e-22) and non-spatial +0.056 (p = 4e-16). Grid and interneuron do not differ (p = 0.54) and both exceed the other two (Holm p <= 8e-6). (E) The fraction of cells individually beating their own null, per mouse: 43% of grid cells and 37% of interneurons against 24% and 19%, all above the 5% false-positive line. (F) Agreement in the trials around a population transition. All four identities switch together over about three trials, and identity does not predict switch timing (p = 0.15). (G) Agreement between identities as a 4 x 4 matrix over its own null; grid-interneuron reaches +0.109, indistinguishable from either within-group value. (H) Which identities never leave one mode: interneurons lock anchored (108:2) and non-spatial cells lock non-anchored (39:293). (I) Open-field speed score for interneurons that lock anchored against those that vary: +0.141 against +0.065, p = 0.005. (J) The same as a count across entorhinal cells: speed-modulated cells lock anchored 1.39:1 (p = 1e-4) while cells with no open-field speed tuning lock the other way at 0.76:1 (p = 0.004), interaction odds ratio 1.84 (p = 3e-6), so the lock-on bias is carried by the speed cells. (K) Open-field theta index against agreement with the population axis, pooled over all cells. Points are deciles of the index with their standard error and the line is the mixed-model fixed effect with a band of one standard error. The slope is +0.173 per unit index alone (p = 1e-11) and +0.136 with identity in the model (p = 2e-6), and holds within grid (+0.353, p = 0.037), non-grid spatial (+0.147, p = 9e-5) and interneuron (+0.116, p = 0.028) classes, so theta rhythmicity predicts following in its own right rather than restating cell type. (L) Whether each class's own state predicts behaviour, as the Anchoring Dependence Index: AUC(uncued) minus AUC(cued) for a state built from five randomly drawn cells of that class, so a class with 92 cells in a session is not compared against one with five. Every class is positive -- grid +0.098, putative interneuron +0.050, non-grid spatial +0.039, non-spatial +0.024, Holm p <= 0.065 -- and no pair of classes differs (smallest p = 0.12 across the 27 sessions carrying all four). The ordering follows panel D, but this figure cannot show that one class carries the behavioural index more than another.

classes form a coordinated subnetwork rather than two populations independently reflecting a shared signal.

## Grid cells are enriched medially and superficially in MEC

If grid cells and interneurons form a subnetwork, it should have an anatomical address. We localised neurons by reconstructing the position of each of the four shanks of every probe and registering recorded cells to a standardised atlas. The best-fit recording plane was fitted to device contacts rather than to cell positions, because per-cell mediolateral position is a per-shank nominal value plus jitter, and fitting to it would fit the approximation rather than the probe geometry (Figure 3B and C).

Grid cells were enriched medially within session (median rho = −0.078 against grid membership, negative in 18 of 27

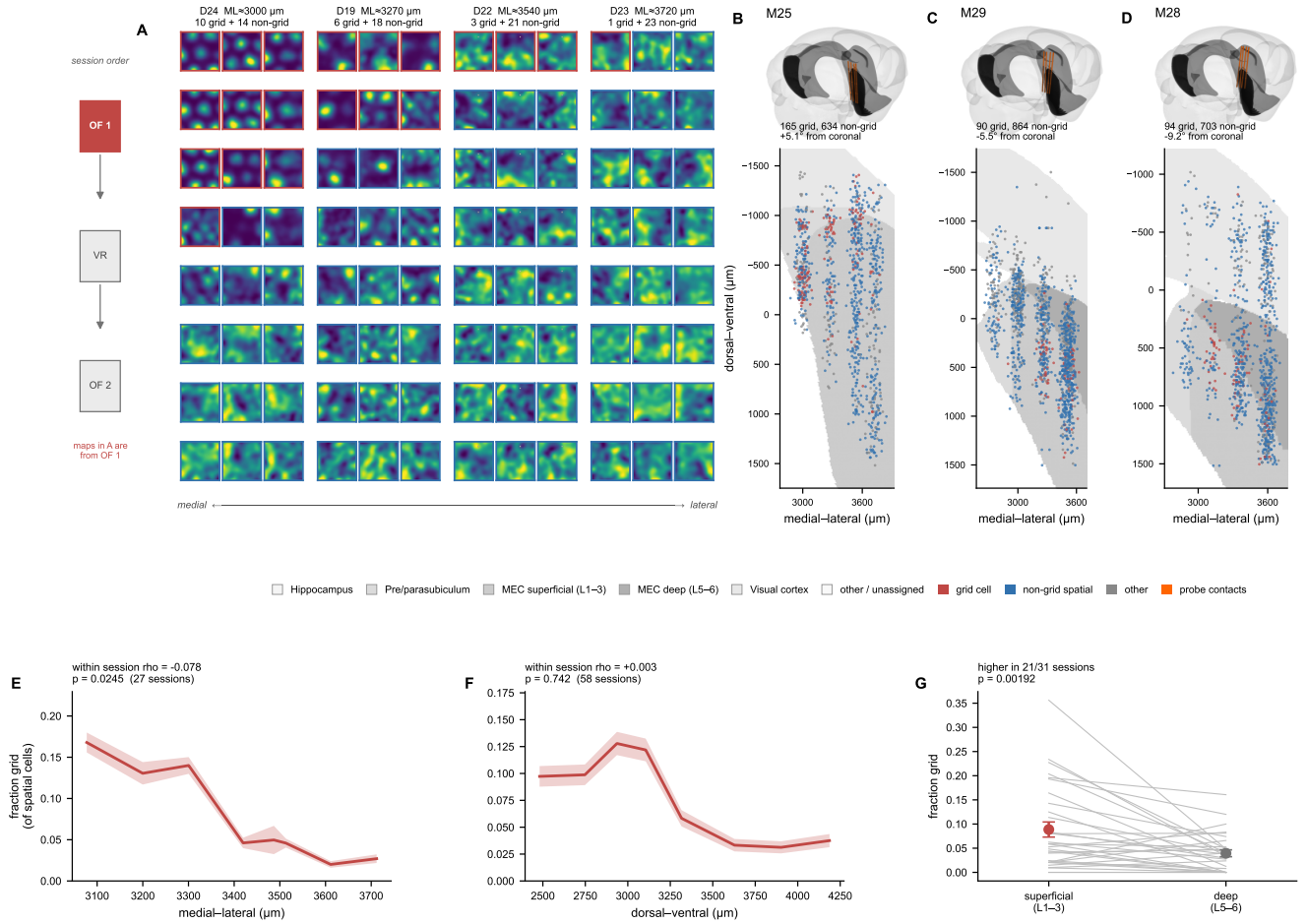



**Figure 3. Grid cells are enriched medially and superficially in MEC**

(A) Open-field rate maps from four M25 sessions whose probes sat at different mediolateral positions, sampled so the grid to non-grid ratio drawn is the ratio recorded. Every map comes from OF1, the arena session preceding the task. (B-D) Best-fit histological slices through three mice, each under a view of the probe position, with cells coloured by open- field identity over the Allen region annotation. (E-G) Identity against mediolateral position, dorsoventral position and layer. Grid cells are enriched medially (median within- session rho = -0.078, p = 0.025) and superficially (8.9% against 3.9%, p = 0.002). Regions are shaded in greys throughout, as in Figure S7, so that only the cells carry colour: deep MEC (L5-6) darkest, then superficial MEC (L1-3), pre/parasubiculum, visual cortex and hippocampus.

multi-shank sessions, p = 0.025) and superficially (8.9% of superficial spatial cells against 3.9% of deep, higher in 21 of 31 sessions, p = 0.002), whereas dorsoventral and anterior-posterior position predicted identity no better than chance (p = 0.90 and 0.10). The gradient is visible without a statistic: open-field rate maps from four sessions in one animal whose probes sat at approximately 3000, 3270, 3540 and 3720 µm, with grid and non-grid cells drawn in the ratio actually recorded, yield 10 grid cells of 24 at the medial end and 1 of 24 at the lateral end (Figure 3A).

## Cells that follow the population state are enriched medially, within the superficial grid population

Following was organised on the same axis and in the same direction, but testing this required care about which sessions can answer the question. Pooling cells across sessions produces a large gradient (medial 0.205 against lateral 0.120, rho = −0.15, p = 2 x $10^{-25}$), but pooling is not safe here, because probe placement differs markedly between animals - 16.5% medial in one, 63.1% in another - so a cell's mediolateral coordinate carries information about which session it came from. The test must be made within session, where every cell shares one probe placement, and on sessions that have mediolateral variation to test: position is assigned per shank, so a single-shank session contributes only noise, and 33 of 61 sessions span under 50 µm. Restricted to the 27 sessions spanning at least 200 µm, two shank pitches or more, the gradient held (median rho = −0.109, negative in 19 of 27 sessions, p = 0.034; Figure 4G), and it held for agreement above chance as well (median rho = −0.061, p = 0.023; Figure 4E). Aligning cells to the border between MEC and parasubiculum rather than to probe placement showed that following above chance was highest within about 200 µm of the border and roughly halved beyond 300 µm (pooled rho = −0.20, p = 2 x $10^{-31}$), although within session this relation did not reach significance (median rho = −0.043, p = 0.20, 25 sessions), and parasubicular cells next to the border followed less than the MEC cells beside them (0.152 against 0.190, p = 0.0002; Figure 4F). Dorsoventral position (p = 0.99, 59 sessions) and probe depth (p = 0.92) showed nothing (Figure 4G), and for those axes the session-selection criterion made no difference, since every session varies along the shank - which indicates that the mediolateral result is specific to the axis rather than produced by the restriction. Layer behaved differently from the mediolateral axis: although it predicts identity, it did not predict following within session (median rho = +0.028 against superficial membership, 25 of 39 sessions positive, p = 0.154; superficial +0.158 against deep +0.139 compared within session, 24 of 36 sessions, p = 0.077). Pooled across sessions the superficial bias in following does reach significance (+0.162 against +0.134, p = 0.004), but pooling is no safer here than it was mediolaterally, and the

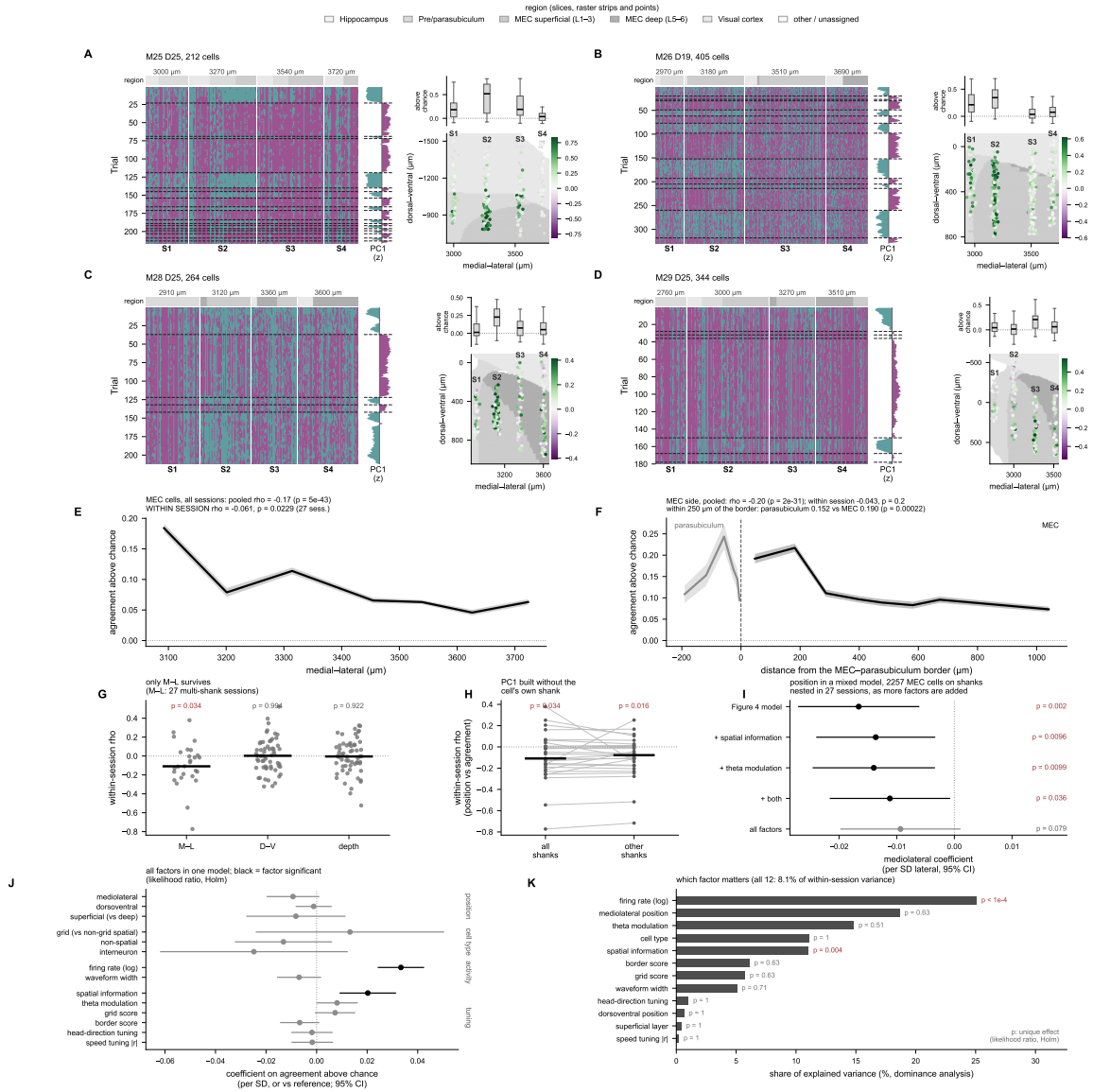



**Figure 4. The responsive subnetwork is medial, within the superficial grid population**

Following the population state, rather than identity, organised along the mediolateral axis of MEC. (A-D) Four multi-shank sessions, each shown two ways. A-C are the three strongest of the 27 multi-shank sessions by within-session rho on MEC cells, against a median of -0.11, so they show the gradient clearly rather than typically; D, from a fourth animal, is a session in which it is absent. Left, the trial-by-cell anchoring raster split by shank from medial to lateral, cells within a shank ordered dorsal to ventral, under a strip giving each cell's region, beside z-scored PC1, with dashed lines where PC1 changes sign and holds the new sign for at least three trials. Shanks are numbered S1 to S4 from medial to lateral, on each slice above the shank's most dorsal cell and beneath each raster block. Right, the best-fit slice with each cell coloured by its agreement with the population axis above chance: |r| minus the mean |r| of the cell's own circular-shift null, as in Figure 2D, so a cell locked to the opposite state also counts as following. The scale diverges from white at chance, green above and purple below. Cells are jittered slightly along the mediolateral axis on the slice (SD 15 um) so that cells sharing a shank do not overlap. Above the slice, agreement above chance for each shank as a boxplot at the shank's own position on the slice's axis. Cells from every region on these shanks are shown, and the per-shank means and rho quoted below include the parasubicular and visual cells. Regions are shaded in greys throughout, as in Figure S7, so that only the cells carry colour: deep MEC (L5-6) darkest, then superficial MEC (L1-3), pre/parasubiculum, visual cortex and hippocampus. (A) M25 D25, 212 cells over four shanks spanning 720 um and anchored on most trials. Agreement is 0.31, 0.52, 0.37 and 0.04 from medial to lateral (rho = -0.255, p = 0.0016). (B) M26 D19, 405 cells over four shanks spanning 720 um, in which PC1 changes sign 13 times. Following is concentrated on the two medial shanks (mean agreement 0.31 and 0.44) and nearly absent from the two lateral ones (0.09 and 0.09; rho = -0.467, p = 6e-20). (C) M28 D25, 264 cells over four shanks spanning 690 um and the most evenly balanced multi-shank session (49% of trials anchored, 4 sustained PC1 transitions). Counted over every region its most medial shank looks unresponsive (mean agreement 0.13; rho = -0.107, p = 0.14), but the region strip shows that shank lies entirely in the parasubiculum. Across its three MEC shanks agreement falls from 0.36 to 0.12 and 0.18 (MEC only, rho = -0.291, p = 0.0007). (D) M29 D25, 344 cells over four shanks spanning 750 um, from a fourth animal, with no gradient: agreement across its three MEC shanks is 0.16, 0.27 and 0.17 from medial to lateral (MEC only, rho = -0.015, p = 0.86, rank 18 of 27), and counted over every region the trend runs slightly the other way (rho = +0.126, p = 0.076). This animal's multi-shank sessions show the gradient weakly at best (strongest, rho = -0.19). (E) Mediolateral position against agreement above chance, MEC cells pooled over sessions (rho = -0.17, p = 5e-43); within session the gradient holds over the 27 sessions spanning at least two shank pitches (median rho = -0.061, p = 0.023). (F) The same against distance from the MEC-parasubiculum border, which aligns sessions to an anatomical landmark rather than to probe placement. The border is taken from each animal's best-fit slice through the atlas, as the parasubiculum pixels adjacent to MEC; a cell's distance is its shortest distance to it, signed by the cell's own atlas label (MEC positive). Five animals have enough border in their slice to measure (3,702 cells), and positions are per-shank, so distances are no finer than the shank pitch. Agreement above chance is highest in MEC within about 200 um of the border and roughly halves beyond 300 um (MEC side, pooled rho = -0.20, p = 2e-31), but within session the relation does not reach significance (median rho = -0.043, p = 0.20, 25 sessions), so the pooled curve partly reflects differences between sessions. Within 250 um of the border, parasubicular cells agree less than MEC cells (0.152 against 0.190, p = 0.0002). (G) Within-session rho for raw agreement on each axis: mediolateral position (median -0.109, p = 0.034, 27 sessions), dorsoventral position (p = 0.99, 59 sessions) and probe depth (p = 0.92); only mediolateral position survives. (H, I) Controls, on the MEC cells of the 27 multi-shank sessions. (H) The

gradient survives an axis to which a cell's own shank does not contribute: each cell scored against a PC1 built only from the other shanks (median within-session rho = -0.078, negative in 19 of 27 sessions, p = 0.016, against -0.109, p = 0.034 on the usual axis). (I-K) What predicts following, in a mixed model with shanks nested in sessions (so that position, which varies mostly between shanks, is tested at the level of shanks), on the 2,257 MEC cells on 167 shanks with every factor measured. (I) The mediolateral coefficient as factors are added. In the model of identity, layer and firing rate it is -0.017 per SD lateral (p = 0.002); lateral cells are also less theta-modulated (r = -0.36) and carry less spatial information (r = -0.21), and with both added it falls to -0.011 (p = 0.036), and with every factor to -0.009 (p = 0.079). (J) Coefficients of all twelve factors in one model, continuous factors per SD; black, factors whose removal worsens the fit (likelihood ratio, Holm across factors). (K) Which factor matters most. Several overlap by construction (putative interneurons are classed on waveform and firing rate, grid cells on grid score), so a factor's unique share understates any that duplicate one another; factors are instead ranked by dominance analysis, each one's added variance averaged over every combination of the others (4,096 models, least squares with session fixed effects), which divides shared variance between them. Together the twelve explain 8.1% of within-session variance: firing rate 25% of that, mediolateral position 19%, theta modulation 15%, cell type and spatial information 11% each, and dorsoventral position, layer, head-direction and speed tuning about none. p, the unique effect as in J. Firing rate predicts following within non-grid spatial and non-spatial cells separately (p = 0.0009, 0.004; rate x type interaction p >= 0.29), so it is not a proxy for interneurons, and grid score does so within non-grid spatial cells (p = 0.02). Between sessions, in the mixed model, session accounts for 17.5% of the variance and shank for 7.9%, against 4.8% for all twelve factors.

within-session tests are the ones we rely on. Three multi-shank sessions show the gradient on their own. In one, agreement with the population axis was 0.31, 0.52, 0.37 and 0.04 across the shanks from medial to lateral (212 cells, 720 μm span, rho = −0.255, p = 0.0016; Figure 4A). In a second animal, in a session whose population state changed 13 times, following was concentrated on the two medial shanks (mean agreement 0.31 and 0.44) and nearly absent from the two lateral ones (0.09 and 0.09; 405 cells, 720 μm span, rho = −0.467, p = 6 x $10^{-20}$; Figure 4B). A third session, the most evenly balanced between the two states, shows why these profiles have to be read by region: counted over every region its most medial shank appears unresponsive (0.13), but that shank lies entirely in the parasubiculum, and across its three MEC shanks agreement falls from 0.36 to 0.12 and 0.18 (264 cells, rho = −0.291, p = 0.0007 within MEC; Figure 4C). These three are the strongest of the 27 multi-shank sessions, against a median within-session rho of −0.11, and are shown because they display the gradient clearly rather than because they are typical. Not every session shows it: in a fourth animal, a session with 134 MEC cells over three MEC shanks showed no gradient (0.16, 0.27 and 0.17 from medial to lateral; rho = −0.015, p = 0.86; Figure 4D), and that animal's sessions showed the gradient weakly at best.

Two caveats apply. First, because mediolateral position predicts identity and identity predicts following, the following gradient could be an expression of the identity gradient. Restricting to non-grid cells left the effect size almost unchanged (median rho = −0.100) but it did not reach significance at 26 sessions (p = 0.111), so we report both and claim neither over the other. Second, given the curved shape of MEC and the flat plane of a four-shank probe, mediolateral organisation could in principle be a laminar sampling artifact. It partly is. Superficial layers supply 90.3% of entorhinal recording sites in the most medial quartile and 59.8% in the most lateral, a gradient present in every animal (within-mouse median rho = −0.447, negative in 6 of 6, p = 0.031), and the same bias carries through to the cells analysed (Figure S5). Stratifying the mediolateral tests by layer separates the two results. The identity gradient does not survive it: within superficial cells the effect roughly halves and loses significance (median rho = −0.051, p = 0.24 at 25 sessions) against −0.078 (p = 0.025) across all layers. The following gradient does survive, and is if anything stronger within superficial cells (median rho = −0.144, p = 0.043, 22 sessions) than across all layers. Deep layers are sampled by too few sessions to test alone. We therefore treat the mediolateral gradient in following as the more robust of the two and the gradient in identity as partly confounded with layer - which reverses the order of confidence the first caveat above would suggest.

Two further checks support the gradient. First, each cell's agreement is measured against an axis to which its own shank contributes, so a shank that dominated the axis could appear more responsive by construction; scoring every cell against a PC1 built only from the other shanks preserved the gradient (median within-session rho = −0.078, negative in 19 of 27 sessions, p = 0.016; Figure 4H). Second, a mixed model with shanks nested in sessions, which tests position at the level of shanks rather than cells, found that position predicted agreement beyond identity, layer and firing rate (−0.017 per SD lateral, p = 0.002; 2,257 MEC cells on 167 shanks with every property measured; Figure 4I).

We then asked which property of a cell matters most, entering twelve together: mediolateral and dorsoventral position, layer, identity, firing rate, waveform width, and open-field spatial information, grid score, head-direction tuning, border score, speed tuning and theta modulation (Figure 4J, K). Several of these overlap by construction - putative interneurons are classed on waveform and firing rate, grid cells on grid score - so we ranked them by dominance analysis, which averages each factor's added variance over every combination of the others and so divides shared variance between them rather than discarding it. Firing rate ranked first (25% of the explained within-session variance; unique effect p < $10^{-4}$, Holm-corrected), consistent with spikes making a cell's trial labels more reliable, and it was not a proxy for interneurons: it predicted following within non-grid spatial and non-spatial cells taken separately (p = 0.0009 and 0.004), with no interaction with identity. Mediolateral position ranked second (19%) and theta modulation third (15%), followed by identity and open-field spatial information (11% each). Position, theta modulation and identity overlap heavily: lateral cells are less theta-modulated (r = −0.36) and carry less spatial information, and with every factor in the model the position coefficient fell to −0.009 per SD (p = 0.079). Theta modulation itself survived firing rate and spatial information (p = 0.009) and weakened only alongside position, so the gradient in following and the theta-modulation effect of Figure 2K are partly one observation. Dorsoventral position, layer and head-direction, border and speed tuning explained essentially nothing. Most strikingly, in the mixed model all twelve properties together explained 4.8% of the variance in agreement above chance, against 17.5% for the session a cell was recorded in and 7.9% for its shank: how strongly a cell follows the population axis depends far more on which population it sits in than on what kind of cell it is.

Together, these data indicate that the cells that follow the population state occupy a specific region of MEC. The mediolateral gradient belongs to following itself. The superficial bias belongs to identity: layer separates grid cells from other spatial cells but does not separate followers from non-followers within session, so the responsive subnetwork is medial, and lies within the superficially biased population that grid cells occupy, rather than being independently superficial. At this sample size we cannot separate an anatomical gradient in following from the anatomical

gradient in identity.

### Monosynaptic connectivity does not account for shared anchoring

That grid cells and interneurons agree is a trial-level statistic and remains compatible with two populations independently tracking a common signal. Putative monosynaptic connections allow the same question to be asked in millisecond spike timing. We reasoned that if the anchoring state is propagated through the local circuit, then pairs of neurons that are synaptically connected should share the state more strongly than pairs that are not.

Connections were identified from cross-correlograms (Figure 5). Interneuron-to-grid was the strongest pathway in the dataset (3.93% of testable ordered pairs, 5.1-fold over within-session identity shuffling, p = 0.0005) and grid-to-grid the strongest excitatory one (0.217%, 2.6-fold, p = 0.004). Detecting the inhibitory half required extending the standard criteria, which test for a causal peak and therefore detect excitation only, onto the lower tail of the same Poisson test. Two features of the resulting connections indicate they are not an artifact of that extension: they are slower than excitatory connections (median 3.0 against 1.0 ms), and they originate from waveform-classified interneurons at 3.1 times their population share, neither of which the detector can produce by construction. The pairing seen in the trial-level statistics is therefore present in the wiring.

It nonetheless does not carry the state. Compared against unconnected pairs from the same session, of the same identity pair and at a similar probe distance (+/-40 μm), connected pairs shared the anchoring state no more strongly: grid-interneuron +0.176 against +0.187 (n = 193, p = 0.39) and interneuron-interneuron +0.090 against +0.090 (n = 116, p = 0.90). Both comparisons resolve effects of 0.045 and 0.043 at 80% power, approximately a third of the size of the identity effect they are testing against, so these are informative nulls rather than absence of evidence. The grid-grid comparison (n = 27, minimum detectable effect 0.157) is underpowered and we report it as such rather than as a null.

Together, these data indicate that anchoring travels with what a cell is rather than with what it is wired to. This is the result on which the rest of the paper turns: a state that is shared by cells of a given identity, but not more strongly by the subset of those cells that are synaptically connected, is unlikely to be propagated cell to cell within the circuit that expresses it, and is more consistent with a common input to which grid cells and local inhibition are more sensitive than other cells are.

### Transitions in the population state coincide with changes in attentional engagement

If the state is imposed from outside the local circuit, the question becomes what the population is listening to, and a global behavioural variable is the natural candidate. Pupil diameter is the standard non-invasive index of the arousal and engagement axis and was available on 80 of 81 virtual-reality sessions.

Pupil radius differed between the two states by approximately 0.24 SD and changed at transitions: entering the anchored state it fell by 0.414 z, and entering the non-anchored state it rose by 0.493 z ($p = 1.7 \times 10^{-7}$ and $4.4 \times 10^{-8}$; Figure 6). Pupil radius did not change instantaneously, with the single trial spanning a transition accounting for only 16-31% of the total change. Because the classifier's median filter smooths each cell's label sequence over +/-2 trials, these data cannot distinguish a genuinely graded change from an abrupt one whose onset is imprecisely timed, and we make no claim about which it is.

A constricted pupil is read as low arousal when it accompanies disengagement, and we therefore asked directly how pupil size relates to performance. Binning trials by within-session pupil z into deciles, the hit rate fell at every step from 0.826 in the smallest decile to 0.391 in the largest (14,011 trials, 58 sessions), and the anchored fraction fell with it. A mixed model gave −0.123 per z ($p = 8 \times 10^{-271}$). Adding a quadratic term did not produce an inverted U: the quadratic coefficient was positive (+0.018, $p = 3 \times 10^{-15}$), placing the minimum of the fitted curve at z = +3.4, beyond the observed data, so that within the range the animal occupies the relationship is monotonically decreasing. The relationship was not a consequence of running: mean speed was flat across pupil deciles (37-40 cm/s) and controlling for within-session speed left the coefficient unchanged (−0.128, quadratic +0.011).

The shift is tonic rather than localised, and this is what identifies it as a change in engagement rather than attention paid at particular points on the track. Comparing pupil radius between states in each of 50 position bins, the difference has the same sign in every bin, reaches significance in 48 of them, and varies little along the track (mean -0.266 z, range -0.432 to -0.136, SD across bins 0.075 z). It does not trend with position (Spearman rho = -0.264, p = 0.064), and the reward zone is if anything slightly less affected than the rest of the track (-0.244 against -0.269). A state in which the animal attended selectively to task-relevant locations would concentrate the difference at those locations; instead the whole traversal is shifted.

Together, these data indicate that the anchoring state tracks a global behavioural variable indexed by pupil diameter, and that the anchored, constricted-pupil state is the one in which the animal performs. Because performance improves monotonically as the pupil narrows, we interpret this state as one of focused task engagement - a selective, exploitative mode - rather than as low arousal in the sense of drowsiness, which the behaviour excludes.

### One state across structures, and a visual readout of the task frame that entorhinal cortex leaves

If the anchoring state is a global behavioural variable rather than an entorhinal computation, it should be legible outside MEC. Asking this properly means not making entorhinal cortex the frame of reference: scoring other regions against the MEC axis can only measure how much they resemble MEC and cannot ask whether they have a state of their own. Each region therefore built its own first principal component from its own cell x trial label matrix, and the axes were compared with one another.

We show the states themselves before the averages (Figure S6), for the two sessions that bracket the range among those recording all three structures. The block structure is shared and visible in the rasters and in single-cell rate maps without any statistic, while entorhinal-visual agreement runs from +0.82 down to +0.15, which is the context in which the population averages should be read.

Every region had a coherent state of its own. Split-half reliability - the axis built twice from disjoint halves of a region's own cells, size-matched to 8 cells per half so that MEC's hundreds of cells carry no advantage over visual cortex's tens - was +0.447 in MEC (27 sessions, p = 1.5 x

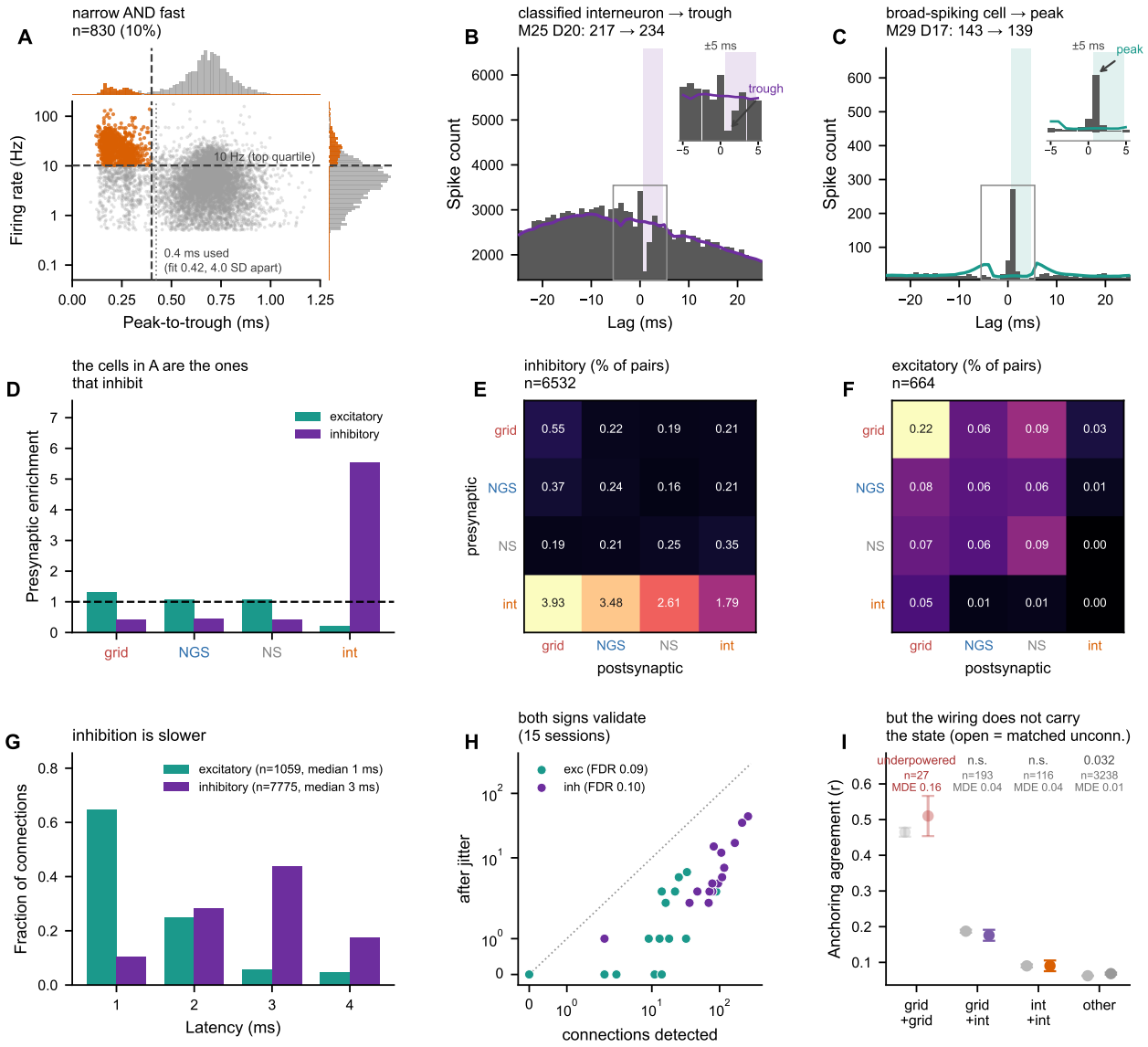



**Figure 5. Monosynaptic connectivity does not account for shared anchoring**

(A) The interneuron call: waveform peak-to-trough duration against firing rate, with the marginal distributions and the fixed 0.4 ms boundary. (B) What a classified interneuron does to its target, a causal trough in the cross-correlogram. (C) What a broad-spiking cell does, a causal peak on the same scale. (D) Presynaptic enrichment by identity: inhibitory connections originate from the cells panel A called interneurons at 3.1 times their population share, which the detector cannot produce by construction since it never sees the waveform. (E, F) The inhibitory and excitatory connection matrices, each as a percentage of the ordered pairs an identity contributes. Interneuron-to-grid is the strongest pathway in the dataset (3.93%, 5.1x over identity shuffling, p = 0.0005) and grid-to-grid the strongest excitatory one (0.217%, 2.6x, p = 0.004). (G) Latency by connection type: inhibition is the slower, median 3.0 against 1.0 ms. (H) Detection against a jitter null, for both signs. (I) Whether connected pairs share the anchoring state more than unconnected pairs matched for session, identity pair and probe distance. They do not (grid-interneuron p = 0.39, interneuron-interneuron p = 0.90). Both comparisons resolve effects a third the size of the identity effect; grid-grid is underpowered and is greyed.

$10^{-8}$), +0.474 in the subicular complex (10 sessions, p = 0.0039) and +0.330 in visual cortex (21 sessions, p = 2.4 x $10^{-5}$).

Before comparing these axes with one another we asked whether a single axis is an adequate description of a region's label matrix at all, since a first principal component exists whatever the data contain and its existence does not establish that the rest is noise rather than a second state. The natural statistic, the fraction of variance the first component explains, cannot answer this across regions: variance explained by the leading component of an n x T matrix rises as n falls, so a comparison between regions that contribute hundreds and tens of cells would largely report how many cells each contributed, in the direction that makes the least well sampled region appear the most one-dimensional. We therefore scored components out of sample, building the axis from one half of a region's cells and measuring the variance it explains in the other half, which a direction fitted to noise cannot do at any cell count (Figure S8).

Scored this way across 60 sessions, an axis built from 8 cells explained 0.052 of held-out entorhinal variance against a circular-shift null of 0.023, with an excess over null of +0.061 in the subicular complex and +0.020 in visual cortex. Comparing components at the same cell count, the excess was +0.0314 for the first and +0.0006 for the second, a ratio of approximately fifty, with the third to fifth at null. Because principal components are orthogonal by construction, a population carrying two independent states would have both generalise to held-out cells. We note that the second component does clear its null (p = 4.3 x $10^{-4}$), so the appropriate claim is that the label matrix is effectively rather than strictly one-dimensional. We note also that the null itself

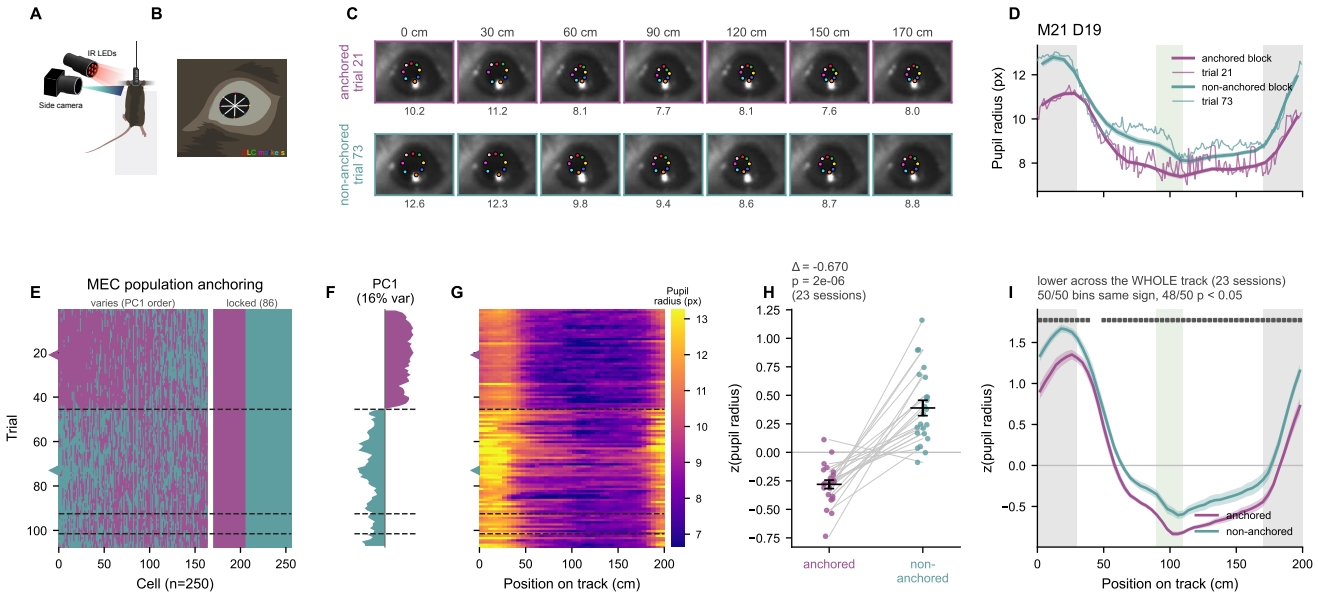



**Figure 6. The state tracks attentional engagement**

(A) The recording arrangement, with infrared illumination and the side camera. (B) The eye with the DeepLabCut perimeter landmarks from which pupil radius is computed. (C) One anchored and one non-anchored trial as seven moments of a single traversal, with pupil radius printed beneath each frame. (D) Mean pupil radius against track position for the two state blocks of this session, with the two example trials overlaid. (E) The MEC anchoring raster for the same session, cells that vary ordered by PC1 loading and cells locked into one mode past a gap. (F) PC1 of the same label matrix. (G) Pupil radius for every trial, on the trial axis of E and F, so a state block can be read across. (H) Pupil radius by state, paired within session over 23 sessions: -0.670, p = 2e-06. (I) Pupil radius against track position for the two states. The difference has the same sign in all 50 position bins and reaches significance in 48, with no trend along the track (rho = -0.264, p = 0.064) and the reward zone if anything less affected than the rest, so this is a tonic shift rather than attention paid at particular places.

rises with cell count, because circularly shifted cells retain their own block structure and their leading component captures shared low-frequency structure that is estimated better as more cells are included, which is why components are compared against a shuffle null at matched cell count rather than against the value expected for a random direction.

Together, these observations indicate that a single axis is an adequate description of each region's anchoring labels, and that the population anchoring state is therefore a well-posed object rather than a convenient summary of something higher-dimensional.

Whether these are the same state cannot be read from a raw correlation between two axes, because such a correlation confounds a different state with no state: an axis estimated from few noisy cells is itself noisy, and will correlate poorly with anything. Dividing by the geometric mean of the two reliabilities separates them. Corrected this way, MEC and the subicular complex agreed at +0.920 and MEC and visual cortex at +0.863; neither differed from 1 (p = 0.098 and 0.515) and the two did not differ from each other (p = 0.66). The raw gradient, 0.46 against 0.31, therefore reflects how noisily each region estimates the state rather than which state it estimates. This correction amplifies noise by approximately 2.6-fold for the visual pair, giving a bootstrap confidence interval of [0.49, 1.23] with 8 of 19 sessions exceeding 1, so "not different from 1" excludes differences larger than about 0.3 and should not be read as a demonstration of identity.

What differs is the cell type rather than the region. At matched cell count MEC's own state was only marginally more coherent than visual cortex's (+0.417 against +0.334, paired within session, p = 0.065). Per cell, with every cell scored out of sample against an axis built from a random half of the entorhinal population, grid cells exceeded visual cortex by +0.149 ($p = 9.9 \times 10^{-6}$, 26 of 32 sessions) and interneurons by +0.144 ($p = 1.3 \times 10^{-6}$, 29 of 33), while other entorhinal cells exceeded it by only +0.050. The cell-type contrast is three times the regional one, and the regional one does not reach significance. Scoring out of sample is necessary here rather than leave-one-out, because leave-one-out removes a cell's own contribution but still measures an entorhinal cell against an axis built from its correlated network, whereas subicular and visual cells were never in that axis at all.

This also resolves an otherwise anomalous behavioural result. A population state built from visual cortex carries an Anchoring Dependence Index of +0.148 (p = 0.0014, 26 sessions), larger than the entorhinal state's +0.085 (p = 0.0015, 60 sessions), and the index is unchanged when the state is computed from MEC cells alone (p = 0.81). If the state is one global variable legible from any sufficiently sampled structure, this is expected rather than puzzling, and we therefore claim no behavioural specificity for the entorhinal readout. These session counts matter, because the same index computed on the speed-matched subset (Figure S3) reaches significance in MEC and not in visual cortex, which invites the opposite reading. That subset applies three requirements beyond the ones used here: at least 20 trials of each cue type, at least 10 trials in each anchoring state within each cue type, and at least 20 trials surviving the speed match with both outcomes still present. It therefore selects sessions that genuinely switch, and leaves 21 entorhinal and 10 visual sessions rather than 60 and 26. The mean index stays higher in visual cortex in both analyses (+0.148 against +0.085 here, +0.133 against +0.120 after matching); what changes is which reaches p < 0.05, and that follows from the loss of sessions rather than from a difference between regions.

Together, these data indicate that the anchoring state is global rather than entorhinal, and that the specificity lies in the cell type that expresses it: one state, present in every structure we recorded, read out most strongly by grid cells

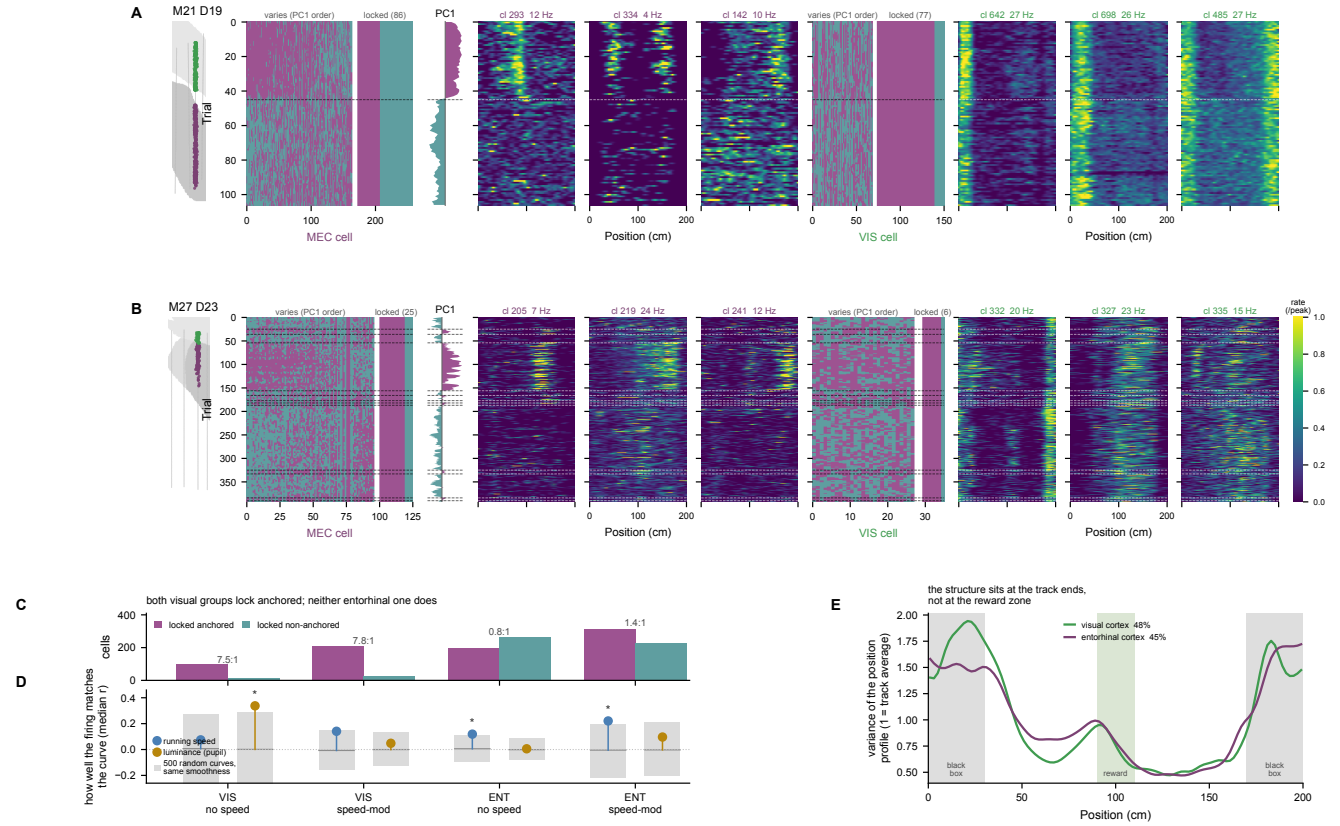



**Figure 7. Visual cortex holds the task reference frame through entorhinal switches**

(A, B) One session per row: the session's best-fit slice through the atlas with recorded cells coloured by structure, then the entorhinal anchoring raster, the session PC1, and the visual raster, all on one trial axis. Both rasters are ordered the same way, cells that vary by PC1 loading and cells locked into one mode past a gap, and both are cut by the same dashed lines, the trials on which PC1 changes sign for at least three trials. The entorhinal population reorganises at those lines and the visual locked block, 77 of 146 cells in A, does not. Each raster is followed by three of its own cells on the same trial axis: the entorhinal three are the largest PC1 loadings among cells that vary and their fields appear in one state only, the visual three are cells that never switch and show the same field in both. Cells on the slices are spread by 15 um of Gaussian jitter along the mediolateral axis for display only, as in Figure 4, so that cells on one shank do not overlap. (C) The census: which cells lock anchored, by structure and by open-field speed tuning. Both visual groups lock anchored, 7.5:1 without speed tuning and 7.8:1 with it, against 0.8:1 and 1.4:1 in entorhinal cortex, so the visual enrichment does not depend on the speed split. (D) What those same cells follow. Each group's median correlation with running speed and with a luminance proxy is compared against 500 phase-randomised surrogates of the same regressor, which keep its smoothness while destroying its relation to position; the block is the middle 95% of the surrogates and the dot is the real regressor. Entorhinal cells follow speed whether or not they are speed-tuned in the open field (+0.12, p = 0.016 and +0.22, p = 0.028) and the light in neither case; visual cells without open-field speed tuning follow the light (+0.34, p = 0.010) and not speed, while speed-tuned visual cells follow neither. Random 98-cell draws from the 308 reach +0.34 in 0.6% of cases, so the open-field classification selects the luminance-following cells specifically. (E) Where the position structure of these cells sits: variance of the profile against position, normalised to its track average, with the 30 cm black box at each end shaded. Nearly half of it falls inside the black boxes, which cover 30% of the track, and the reward zone is where it is scarcest.

together with local inhibition. We note that the cerebellum, included as a distant structure with no parahippocampal connectivity that could explain a shared state, carries a matching state in both sessions in which enough cells were recorded to build an axis. It was intended as a negative control and is not one.

A state shared across structures does not require that every structure leaves the task reference frame at the same moments, and the cells that never leave it turn out not to be distributed as the shared state would suggest. Figure 2J showed that the entorhinal excess of cells locked permanently into the anchored mode is carried entirely by cells that are speed-modulated in the open field; among entorhinal cells with no open-field speed tuning the lock is if anything the other way, 199 permanently anchored against 262 permanently non-anchored. Repeating that count in every recorded structure turns up one exception. In visual cortex, cells with no open-field speed tuning lock into the anchored mode 98 against 13, a sevenfold excess where entorhinal cortex shows none (odds ratio 9.9 against the entorhinal cells, $p = 4 \times 10^{-19}$), and the excess is present in every one of the 15 sessions contributing such cells. It does not depend on the speed restriction either: among visual cells that ARE speed-modulated in the open field the ratio is 7.8:1, against 0.8:1 and 1.4:1 for the two entorhinal halves (Figure 7C). These are cells that hold their field across the trials on which the entorhinal population reorganises, and they are not explained by the speed signal that accounts for the entorhinal lock.

Seen session by session, the dissociation is visible without a statistic (Figure 7A, B). Within one session the entorhinal raster reorganises at each sign change of PC1 while the visual raster does not, and the locked visual block runs to 77 of 146 cells in the clearest example. The entorhinal cells with the largest loadings on the population axis have a field in one state and little or nothing in the other; the visual cells have the same field in both.

What those visual cells are doing is not a second anchoring signal. Where these cells fire is not peculiar to visual cortex: clustering each population on the shape of its position profile, peak positions are far from uniform in both structures (KS $p = 2 \times 10^{-13}$ and $5 \times 10^{-13}$) and concentrate in the same places, the first 50 cm and the last 25 cm of the track. What separates them is what the firing follows. Two variables are

confounded with position by the task itself: running speed, because the animals slow into the reward zone and run fastest after it, and screen luminance, which was not recorded but is indexed inversely by pupil radius against position - pooled over 23 sessions the pupil is most constricted at 106 cm, inside the reward zone, and most dilated near 22 cm, a swing of 2.2 z.

Testing either against zero would not be informative, because both are smooth functions of position and any smooth function of position resembles a smooth firing profile. We therefore compared each population's median correlation with each regressor against 500 phase-randomised surrogates of that regressor, which preserve its power spectrum and so its smoothness while destroying its relationship to position, and we split each structure by open-field speed tuning at this point rather than when selecting the cells (Figure 7D). Entorhinal cells follow the track's speed profile and not its light, whether or not they are speed-modulated in the open field (+0.12, p = 0.016 without speed tuning; +0.22, p = 0.028 with it; luminance p = 0.90 and 0.50). Visual cells are the opposite, and only in one half of the population: those without open-field speed tuning follow the light and not the speed (luminance median r = +0.34, p = 0.010; speed p = 0.79), while the speed-modulated visual cells follow neither. The same half carries the beacon result - reward-zone firing differs between cued and uncued trials at p = 0.025 across those 98 cells, against p = 0.063 across all 308 - so the scene-following visual cells are a subset of the persistently anchored ones rather than all of them. This is not a subset chosen for giving the answer: random draws of 98 from the 308 reach a median correlation of +0.11 with a 97.5th percentile of +0.27, and 0.6% of them reach the +0.34 the untuned cells give, so the open-field speed classification - made in a different environment on a different day, and never exposed to the track - selects the luminance-following visual cells specifically. Pooling the halves instead leaves the entorhinal speed result intact (+0.17, p = 0.018) and visual cortex with no detectable relationship to either variable.

We are explicit about what this test can and cannot say. It establishes that each relationship is present, but not how large it is: cross-validated variance explained does not clear the same null for any regressor in any population (best p = 0.075), so we do not quantify the share of the profile either confound accounts for. What can be said positionally is where the structure of these cells lies (Figure 7E). Nearly half of the variance of their position profiles falls inside the 30 cm black box at each end of the track - 48% for the visual cells and 45% for the entorhinal ones, against the 30% of the track those boxes occupy - while the reward zone is the one place it is scarce. The persistently anchored cells of both structures are organised around the ends of the track, where each traversal begins and ends, rather than around the goal.

We therefore read these visual cells as following features of the scene - its luminance structure and the beacon - which repeat on every lap, rather than as carrying an anchoring signal of their own. The consequence is the point of interest. The task reference frame remains legible in visual cortex throughout, including on the trials during which the entorhinal population has left it, so the information MEC would need in order to stay anchored is available and is not being used. Whatever governs the entorhinal state is therefore not a failure of sensory access.

### The population state is associated with frequency-specific changes in the local field potential

If anchoring is a network state rather than a property of individual cells, it should be visible in the local field potential. We considered two possibilities. A uniform change in the amplitude of the field - arising from a change in the spiking contribution to the LFP, or from electrode referencing - predicts that power moves in the same direction at all frequencies. A change in the balance of inputs or in local processing predicts that it does not.

To distinguish these we computed Welch spectra per trial from the entorhinal LFP across the full range from 1.9 to 150 Hz rather than integrating over bands fixed in advance, because a band split whose "gamma" runs from 30 to 80 Hz cannot detect a change that reverses sign inside it. Spectra were whitened by multiplying power by frequency so that peaks are visible against the 1/f background, and z-scored within session so that no recording's absolute amplitude dominates. Across the 26 sessions with at least 8 trials in each state, the difference between states changed sign across the gamma range (Figure 8E and F): theta (6-10 Hz) fell by 0.094 (p = 0.020, lower in 20 of 26 sessions) and mid gamma (60-100 Hz) by 0.108 (p = 0.0056, 19 of 26), while slow gamma (30-48 Hz) rose by 0.105 (p = 0.0013, 21 of 26). Delta rose (+0.160, p = 0.036) and beta did not change (+0.063, p = 0.28). The reversal is orderly rather than a scatter of band results: the difference profile crosses zero at a median of 67 Hz, and crosses within the gamma range in 25 of the 26 sessions, so the two gamma effects are two sides of one spectral rotation rather than two independent findings. The pooled difference profile was positive to approximately 60 Hz and negative above it, with 25 of 26 sessions showing a crossing in that range at a median of 67 Hz (Figure 8J). Because the difference reverses sign, no uniform change in amplitude, referencing or spiking contribution can account for it.

Because the proposed account of these band changes places fast-spiking interneurons at the centre of the gamma rhythm, we measured spike coupling to gamma directly, with pairwise phase consistency and with the field taken at least 120 µm from each cell so that a cell's own waveform could not stand in for the field (Figure S11). Coupling is present and local, falling from the cell's own channel to 120 µm away and to near zero at 1.2 mm, but it does not distinguish interneurons: they are no more coupled than principal cells in either band (p = 0.60 and p = 0.92 over sessions) and grid cells are the most coupled class in both. What does hold is the phase relationship - in 60-100 Hz gamma interneurons follow the principal cells by a median 45° ($p = 1 \times 10^{-4}$, later in 25 of 31 sessions), the direction pyramidal-interneuron gamma predicts, with no such lag at 30-48 Hz. None of this coupling changes with the anchoring state (four comparisons, Holm-corrected p >= 0.17).

We checked that the slow-gamma increase is not mains contamination, since the 50 Hz line falls within that band. Excluding the 48-52 Hz bins strengthened the effect (+0.109, $p = 3.6 \times 10^{-4}$) and the 48-52 Hz band tested alone did not reach significance (p = 0.063).

Theta's nesting of gamma changed for one band only. Quantified as Tort's modulation index over 18 phase bins with a phase-shuffle null subtracted, theta-fast-gamma coupling fell in the anchored state by 15.9% ($p = 5 \times 10^{-6}$, lower in 23 of 26 sessions), while theta-slow-gamma coupling did not change (−7.9%, p = 0.71, 11 of 26) (Figure 8G). The

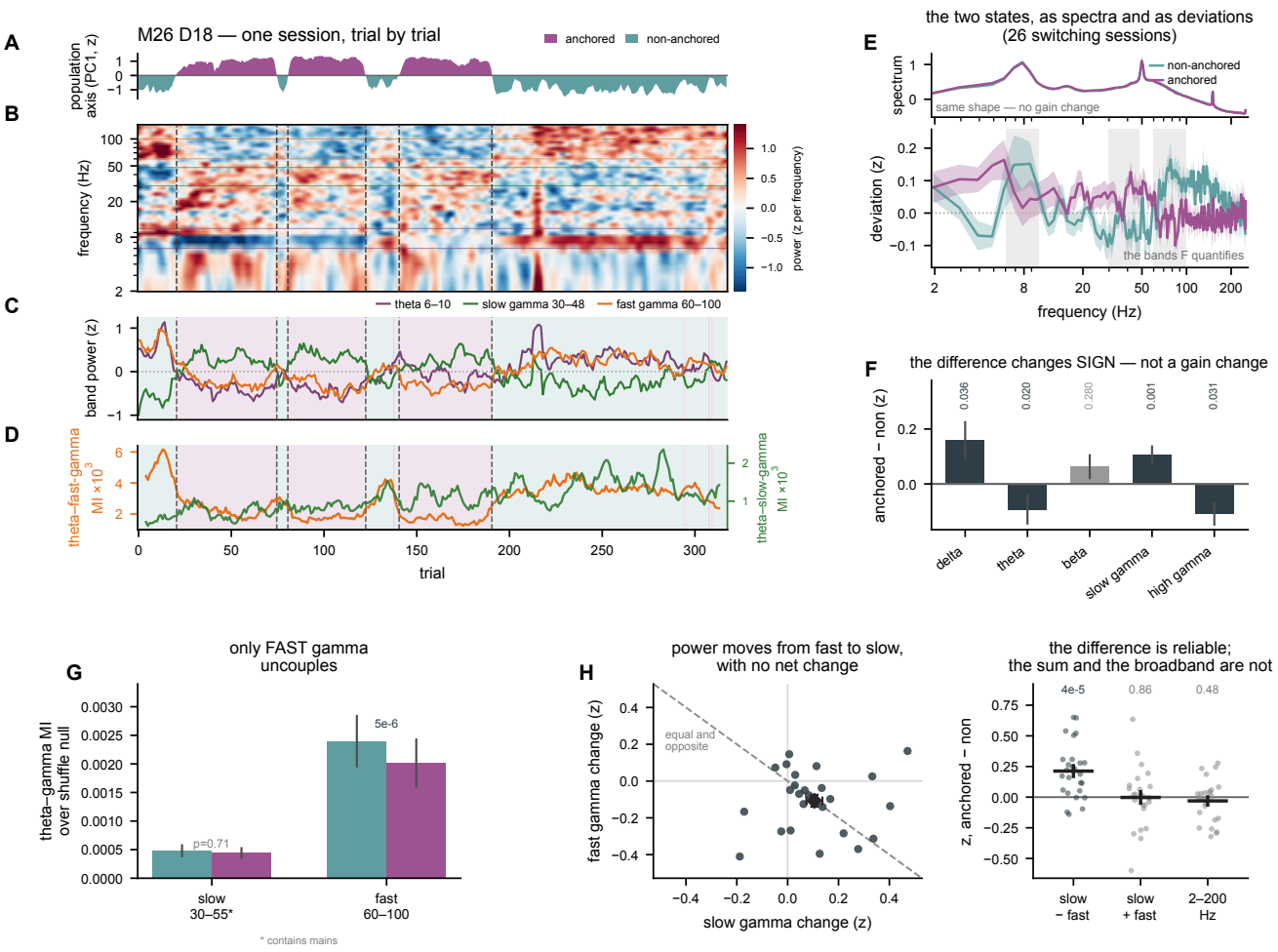



**Figure 8. A frequency-specific change in the entorhinal field potential**

(A-D) One session trial by trial, both states shaded behind every panel and dashed lines where PC1 changes sign for at least three trials. (A) The population axis. (B) The spectrogram, 1.9-150 Hz in 56 log-spaced bands, smoothed along the trial axis only. (C) The three band traces. (D) Theta-gamma modulation index for both bands on twinned axes, fast on the left and slow on the right, which they need because slow-gamma MI is about five times smaller. (E) The two states as spectra across 26 switching sessions, the spectra above and the deviation between them below. The deviation crosses zero at a median of 67 Hz and does so in 25 of the 26 sessions. (F) The same difference band by band: theta falls (-0.094) while slow gamma rises (+0.105), so the difference reverses sign across the spectrum. (G) Theta-gamma coupling by band: fast gamma uncouples (-15.9%, p = 5e-6) and slow gamma does not. (H) The redistribution, tested pairwise within session. Left, each session's slow-gamma change against its fast-gamma change, with the equal-and-opposite line and the mean with its SEM. Right, the three contrasts drawn directly, one point per session. The difference between bands is large and reliable (slow minus fast +0.213, p = 4e-5) while the sum is not distinguishable from zero (-0.002, p = 0.86), nor is the change across 2-200 Hz (-0.031, p = 0.48): power moves from fast to slow with no net change. The sum is zero on average rather than session by session, and the balance is per Hz -- weighting each band by its width moves the sum to -0.36 (p = 0.12) -- so what is supported is that power moves from fast to slow, not that the total is conserved. Where in the theta cycle each frequency peaks is shown in Figure S10.

band whose power falls is therefore also the band that uncouples from theta, while the band whose power rises remains nested as before. We note that the slow-gamma null is the weaker of the two, since its absolute modulation index is approximately five times smaller and its band contains the mains notch, and we therefore report this as an unsupported difference rather than as a demonstrated invariance.

Tested pairwise within session, the two changes are equal and opposite. The difference index (slow minus fast) was positive in 21 of 26 sessions (+0.213, p = 4.4 x $10^{-5}$), while their sum was not distinguishable from zero (−0.002, p = 0.86), and neither was the change across the whole 2-200 Hz range (−0.031, p = 0.48) (Figure 8H). The same contrast in log power rather than in per-frequency z gives a sum of +0.0000 (p = 0.75), so the balance is not an artifact of z-scoring each frequency. Power therefore moves from fast to slow with no detectable net change. Two qualifications. The sum is zero on average and not session by session: the scatter about the equal-and-opposite line is wide, and a session that gains slow gamma is not reliably one that loses fast gamma by the same amount. And the balance is per hertz - the fast band is 40 Hz wide against slow's 18, and weighting each band by its width moves the sum to −0.36 (p = 0.12). What the data support is that power moves from fast to slow, not that the total is conserved.

Theta amplitude also fell in the anchored state, although marginally (deltaz = −0.224, p = 0.033 across the 31 sessions with speed-matched trials in both states; an amplitude fall of 1.66%, p = 0.032). This is not a shift of the rhythm out of the analysis window: instantaneous frequency was unchanged (−0.016 Hz, p = 0.54) and the spectral peak moved by −0.110 Hz (p = 0.016), far too small to move power out of a 4 Hz band.

The relationship between theta and behaviour changed more robustly than its amplitude. Theta amplitude and frequency both track running speed, and that coupling was approximately halved during anchored epochs: on the 19 sessions with at least 15 trials in each state, the within-session regression slope of theta amplitude on speed fell from 0.374 to 0.218 (p = 0.045) and of theta frequency on speed from

0.0126 to 0.0070 Hz per cm/s ($p = 0.012$), with the corresponding correlations falling from 0.424 to 0.244 ($p = 0.0015$, 15 of 19 sessions) and 0.398 to 0.185 ($p = 3.4 \times 10^{-4}$, 16 of 19). Mice do run more uniformly when anchored (speed SD 6.07 against 8.58 cm/s, $p = 2.2 \times 10^{-5}$), which attenuates a correlation on its own, but several observations indicate that this is not the explanation. The regression slope is insensitive to range restriction and fell by the same factor; the per-session change in correlation did not track the per-session change in speed variability (rho = −0.01, $p = 0.92$); restricting both states to their common speed range left the slope difference intact (0.12 against 0.34, $p = 0.008$); and a circular-shift null on the state vector, which preserves run structure and within-session drift, gave $p = 0.005$. That the gain changes rather than only the level distinguishes this from a state in which the animal simply runs differently.

Because theta's coupling to speed changes while the rhythm itself does not, we asked directly whether the cells that carry the speed signal into entorhinal cortex change their speed coding with state. Entorhinal speed cells are reported to be largely fast-spiking inhibitory neurons that inherit a speed signal from the medial septum, so the prediction is specific: if the anchored state reflects a reduction in that septal speed pathway, the affected cells should be fast-spiking speed cells. We identified them entirely outside the virtual-reality session, from waveform class and the open-field speed score, and divided all entorhinal cells into four groups on those two criteria alone.

The classification reproduced the reported anatomy. Speed modulation was enriched approximately twofold among putative interneurons (58.7% against 41.5% of principal cells, odds ratio 2.01, $p = 1.1 \times 10^{-13}$), the median open-field speed correlation was three times higher in interneurons than in principal cells ($p = 5 \times 10^{-23}$), and the relationship was continuous rather than categorical, with narrower waveforms and higher firing rates both predicting stronger speed modulation (rho = −0.137 and +0.118, $p < 10^{-18}$). The open-field labels also transferred to the task: on the virtual track, cells classified as speed interneurons in the arena had a median speed slope of 0.138 Hz per cm/s against 0.012 for non-speed interneurons, 0.010 for speed principal cells and 0.0008 for non-speed principal cells, a twelvefold separation obtained from a label assigned in a different environment.

Within those cells, speed coding did not change with the anchoring state. On the two states' common speed support, the speed slope of open-field-defined speed interneurons was 0.1523 when anchored against 0.1361 when not (+11.9%, $p = 0.121$, 29 sessions) - if anything steeper when anchored, which is the opposite of the predicted direction - and the rate-normalised gain was unchanged (+6.8%, $p = 0.565$). Neither of the other informative groups changed either (speed principal cells −22.2%, $p = 0.121$; non-speed interneurons −6.9%, $p = 0.67$). This null is informative rather than merely negative: the comparison resolves a 17% change in slope and an 18% change in gain at 80% power, against observed values of 12% and 7%, so a halving of speed drive of the size seen in the field potential is excluded in the cells that carry it. One group, non-speed principal cells, reached nominal significance (−39.2%, $p = 0.026$) on a near-zero baseline slope of 0.0023; across the sixteen comparisons made here, two results below 0.05 are approximately what chance provides, and we do not interpret it.

Speed modulation also did not predict how strongly a cell followed the population state. Pooled across entorhinal cells the two were weakly correlated (rho = +0.066, $p = 3 \times 10^{-6}$), but entering both predictors in one model leaves cell class carrying the effect and speed modulation carrying none (putative interneuron +0.069, $p = 1.2 \times 10^{-6}$; speed modulation +0.005, $p = 0.21$). The pooled correlation is a composition effect: interneurons are both the most speed-modulated cells and the strongest followers, so pooling manufactures an association between two properties that are independent within class.

Finally, we asked whether discrete gamma events rather than continuous power carry the state or the task demand, reasoning that if gamma cycles correspond to discrete processing episodes then uncued trials, which require path integration, should contain more of them. Counting supra-threshold envelope excursions lasting at least one gamma cycle, expressed per theta cycle across 10,802 trials from 44 sessions, the raw comparison appeared to support this: fast-gamma events were more frequent on uncued trials (+0.0057 per theta cycle, $p = 0.016$). The effect did not survive control. Adding trial speed and theta frequency to the model reversed its sign (−0.0054, $p = 4 \times 10^{-5}$), and adding band power removed it entirely (+0.0003, $p = 0.68$). Event counts here track power and running rather than task demand, and we report this prediction as tested and not supported. The same analysis did reveal an organisation by track position: slow-gamma event rate peaked at 92 cm, within the reward zone, in 40 of 44 sessions between 70 and 130 cm, surviving control for speed and band power (+0.356 Hz for 80-110 cm against the rest of the track, $p = 5 \times 10^{-8}$), while fast-gamma events showed no such peak. This did not interact with cue condition ($p = 0.93$) or with anchoring state ($p = 0.77$), and so stands beside the state result rather than explaining it.

Together, these data indicate that the population anchoring state is accompanied by a spectrally structured change in the entorhinal field potential, in which theta and mid gamma fall while slow gamma rises, and by a reduction in the coupling of theta amplitude and frequency to running speed.

## DISCUSSION

Our results indicate that anchoring of entorhinal spatial firing to a task reference frame is governed by a global behavioural state that the entorhinal circuit reads rather than builds. We find that anchoring is a coordinated property of the recorded population rather than of single neurons (Figure 1), that it is expressed most strongly by grid cells together with putative interneurons, at roughly twice the level of spatial cells without grid structure (Figure 2), and that these cells occupy a medial position within MEC (Figure 4), inside the superficially biased population that grid cells occupy (Figure 3). Although grid cells and interneurons are synaptically coupled, with interneuron-to-grid the strongest pathway we detect, synaptically connected pairs do not share the anchoring state more strongly than matched unconnected pairs (Figure 5), which argues that the coordination is not propagated through the local circuit. Consistent with the state being imposed from outside it, transitions coincide with changes in pupil-indexed engagement (Figure 6), the same state is present in every structure we recorded with only the strength of its readout differing by cell type, while visual cortex holds a readout of the task reference frame through the very trials on which the entorhinal population abandons it (Figure 7), and it is accompanied by frequency-specific changes in the local field potential (Figure 8).

### Shared anchoring tracks cell identity rather than synaptic partnership

Coordinated switching among grid cells is usually interpreted through continuous attractor dynamics, in which recurrent connectivity enforces a shared network state. Our monosynaptic analysis does not support that interpretation for anchoring specifically. Two grid cells agreed at approximately the same level whether or not a synapse was detectable between them, and the grid-interneuron comparison would have resolved an effect a third the size of the identity effect it was testing against. We emphasise what this does and does not show. It does not show that the entorhinal circuit lacks attractor dynamics, nor that recurrent connectivity is irrelevant to grid firing; our sampling recovers only a small fraction of the connections present, and detectable monosynaptic connections are a biased and sparse sample of the circuit. What it shows is that, within the pairs we can test, synaptic partnership adds nothing to identity in predicting shared anchoring. The most economical account of that result is a common input to which grid cells and local inhibition are more sensitive than other entorhinal cells are.

The identity dependence is itself informative about the candidate input. Putative interneurons followed the state as strongly as grid cells and locked almost unanimously into the anchored mode, while non-spatial cells locked out of it. Neuromodulatory and septal projections to MEC target inhibitory populations preferentially, and a common input arriving through local inhibition would be expected to produce exactly this pattern: strong expression in interneurons and in the principal cells they most densely innervate, weak expression elsewhere.

The same dissociation appears a second time, with a different candidate property. Because entorhinal speed cells are predominantly fast-spiking interneurons, and interneurons are the strongest followers of the state, speed modulation is an obvious candidate for what the followers have in common. It is not: pooled across cells the two are weakly associated, but within class the association disappears entirely, and in a model containing both, cell class carries the effect while speed modulation carries none. What predicts a shared anchoring state is what a cell is, not whether it is synaptically connected to another follower and not how strongly it is tuned to running speed. Two functional properties have now been tested in place of identity and neither substitutes for it.

### A global state with a cell-type-specific readout

Asking whether the anchoring state is entorhinal required abandoning the entorhinal axis as the frame of reference. When each region was allowed to describe the state using only its own cells, every structure we recorded carried a coherent state, and once corrected for the reliability with which each region estimates it, those states were indistinguishable from one another. The regional difference that appeared in raw correlations was a difference in estimation noise rather than in what was being estimated.

We are cautious about how far this licenses a claim of identity. Disattenuation divides by the geometric mean of two reliabilities, which for the entorhinal-visual pair amplifies noise approximately 2.6-fold, so the comparison excludes differences larger than about 0.3 and no more. Sessions also differ substantially, with entorhinal-visual agreement running from +0.82 to +0.15 among sessions recording all three structures, and one session in which the two axes are anti-correlated. The claim the data support is that the state is not specific to entorhinal cortex, not that every structure expresses one identical variable.

The cerebellar result deserves comment because it was intended as a control and behaved as a finding. Cerebellar cells have no parahippocampal connectivity that would explain a shared anchoring state, and we included them expecting them to sit at zero. In both sessions with enough cells to build an axis, the cerebellar state matched the entorhinal one. Two sessions cannot establish this, and we report it as the weakest point at which the globality claim can be inspected rather than as independent support for it.

What remains entorhinal is the readout. The contrast between grid cells and visual cortical cells was three times the contrast between MEC and visual cortex as regions, and the regional contrast did not reach significance. On the behavioural side there is no entorhinal specificity at all: a state read from visual cortex predicted path integration-dependent performance at least as well as one read from MEC. We therefore make no claim that the entorhinal readout is what drives behaviour, and note that this is the expected result if a single global variable is legible from any sufficiently sampled structure.

### What the field potential adds, and what it does not settle

The spectral result constrains mechanism more than the amplitude result does. A change confined to overall amplitude would be compatible with many uninteresting explanations, including a change in how much spiking contributes to the field or in electrode referencing. That the difference reverses sign within the gamma range excludes those, because each acts in one direction at all frequencies. The accompanying loss of theta-fast-gamma coupling, with theta-slow-gamma coupling preserved, points in the same direction: the band that falls in power is the band that also uncouples.

We resist a stronger reading. Slow and fast gamma in the entorhinal-hippocampal system have been associated with different input streams and, in one influential framework, with internally guided retrieval and externally driven encoding respectively. An animal localising a reward zone by path integration is in the internally guided case, and that is also the condition in which anchoring matters most for behaviour, so the direction of our effects is the one that framework would predict for a shift toward internally guided processing in the engaged state. But those band assignments derive largely from hippocampal work, our recordings are entorhinal, and we have not shown that the two bands here carry the inputs the framework assigns them. We also note that although the two gamma changes are equal and opposite on average, the per-hertz balance does not survive weighting by bandwidth, so we describe power as moving from fast to slow rather than as conserved between the bands.

The decoupling of theta from running speed is robust to every control we applied but remains mechanistically unexplained. The most natural reading, that the velocity signal driving the entorhinal network is down-weighted when position is pinned by landmarks, predicts changes in the cellular speed code that we do not observe - and the sharpest version of that test, restricted to the fast-spiking speed cells that carry the septal speed signal and defined entirely from the open field, excludes a change a third the size of the field-potential effect in the very population the

account requires it in. We tested the gamma-as-discrete-processing-episode hypothesis directly and it was not supported: apparent differences in event counts between cued and uncued trials tracked running speed and band power rather than task demand.

### Limitations

Three limitations bear on the central claims. First, the monosynaptic null is a null, and although both informative comparisons resolve effects a third the size of the identity effect, absence of a wiring contribution within detectable pairs does not exclude one within the much larger set of connections we cannot detect. Second, the non-entorhinal recordings sit on the same probe as the entorhinal ones, so for the field-potential analyses volume conduction cannot be excluded, and a shared effect was always going to be ambiguous where a divergent one would have been decisive. Third, the classifier's median filter smooths each cell's label sequence over +/-2 trials, so our data cannot distinguish a graded transition from an abrupt one that is imprecisely timed, and the temporal structure of state transitions should be regarded as unresolved.

### Ideas and speculation

If the anchoring state is a global variable with a cell-type-specific readout, then the relevant question for grid cell function shifts from what the entorhinal circuit computes to what it is listening to, and when. One implication is that the availability of a usable grid code may be set by processes that have nothing to do with space. A mouse that is disengaged has a grid network that is not anchored to the task, and therefore cannot localise a reward by path integration, even though nothing about its spatial circuitry has changed. If this generalises, variation in path integration ability between individuals may reflect variation in engagement rather than in path integration machinery as such, which would be relevant to the interpretation of spatial deficits in ageing and in neurodegenerative disease.

A second implication concerns experimental design. Cue-rich virtual environments, which are widely used to study grid cells, engage behaviours that our results suggest can be solved without anchored grid firing. In such environments the anchoring state may vary without behavioural consequence, and grid activity recorded there may be only loosely coupled to what the animal is doing.

Finally, the identity of the common input remains open, and two septal candidates that seemed natural are now both constrained by the data. Medial septal GABAergic neurons are the obvious one, since their inhibition disrupts grid periodicity and phase precession together, but the direction of our field-potential result does not fit an account in which the non-anchored state is reduced septal drive: theta amplitude is lower in the anchored state, which is the state in which the code is best locked to the track. Spike-theta phase locking is also unchanged between states, which is what an intact pacemaker looks like.

That suggested a cell-type distinction, since the septal pacemaker and the speed signal reaching entorhinal cortex are carried by different projections, and a reduced speed-theta gain with preserved spike-theta timing resembles a change in the speed pathway with the pacemaker intact. The speed-pathway account is now also difficult to sustain. Entorhinal speed cells are predominantly fast-spiking inhibitory neurons, and we confirm that here; but those cells' speed coding does not change between anchoring states, and the comparison has the power to exclude a change a third the size of the one the field potential shows. If anything their speed tuning is marginally steeper in the anchored state, which is the wrong direction. The speed signal arriving on the cells that carry it is therefore preserved while its expression in the field potential is not, and the locus of the decoupling lies somewhere other than the afferent speed drive or the local cells receiving it.

A third septal projection is constrained by neither result. The medial septum and diagonal band send cholinergic as well as GABAergic and glutamatergic axons to entorhinal cortex, more densely to its medial than its lateral division, terminating in the superficial layers where the grid network and its feedback inhibition lie. Neither observation that narrows the other two candidates bears on this one: a change in cholinergic tone requires no change in the septal pacemaker, whose signature - preserved spike-theta phase locking - we observe, and no change in the speed signal arriving on the cells that carry it, which we also observe. It predicts the direction of the field-potential result rather than being embarrassed by it. Cholinergic tone drives theta, suppresses delta and favours fast over slow gamma, so a lower tone in the anchored state predicts lower theta, higher delta, lower fast gamma and higher slow gamma, which is the pattern we find; and because acetylcholine modulates gamma power rather than gamma frequency, it further predicts that the gamma peak should not move between states, which it does not.

Our pupil measurement is better suited to this reading than it would ordinarily be. Rapid pupil dilations track noradrenergic activity, whereas sustained dilations during locomotion are accompanied by sustained cholinergic activity (Reimer et al., 2016); our animals run throughout, and the anchoring state is a sustained variable held across blocks of tens of trials. The timescale fits for the same reason. Neuromodulatory tone varies over tens of seconds to minutes, which is the scale of the blocks we observe, whereas a state propagated synaptically within the entorhinal circuit has no particular reason to vary on that scale - a point independent of the connectivity null, which concerns which pairs share the state rather than how slowly it changes.

We offer this as the septal candidate our data do not exclude rather than one they establish. Pupil diameter indexes a correlate of cholinergic activity and not acetylcholine itself, and nothing here separates cholinergic tone from any other variable covarying with sustained engagement. The direction also runs against the familiar reading in which cholinergic activation accompanies attention: here the code is best anchored, and the animal performs best, when the pupil is narrow. Testing it requires measuring or manipulating cholinergic tone directly, which these recordings cannot do.

A different locus is suggested by what the two gamma bands are likely to be. Gamma rhythms are generated by fast-spiking inhibitory neurons, in loops whose frequency is set by the kinetics of the inhibitory limb, so two bands that move in opposite directions are more naturally two loops than one rhythm of varying amplitude. Our monosynaptic data describe such a loop directly: interneuron-to-grid is the strongest pathway we detect, grid-to-grid the strongest excitatory one, and the inhibitory limb is the slower of the two. The same fast-spiking population supplies the dominant inhibition onto grid cells, carries the speed signal, and generates the rhythms. That offers a reason for these cells to follow the state which does not depend on their coupling to grid cells: on this view they do not report the state so much

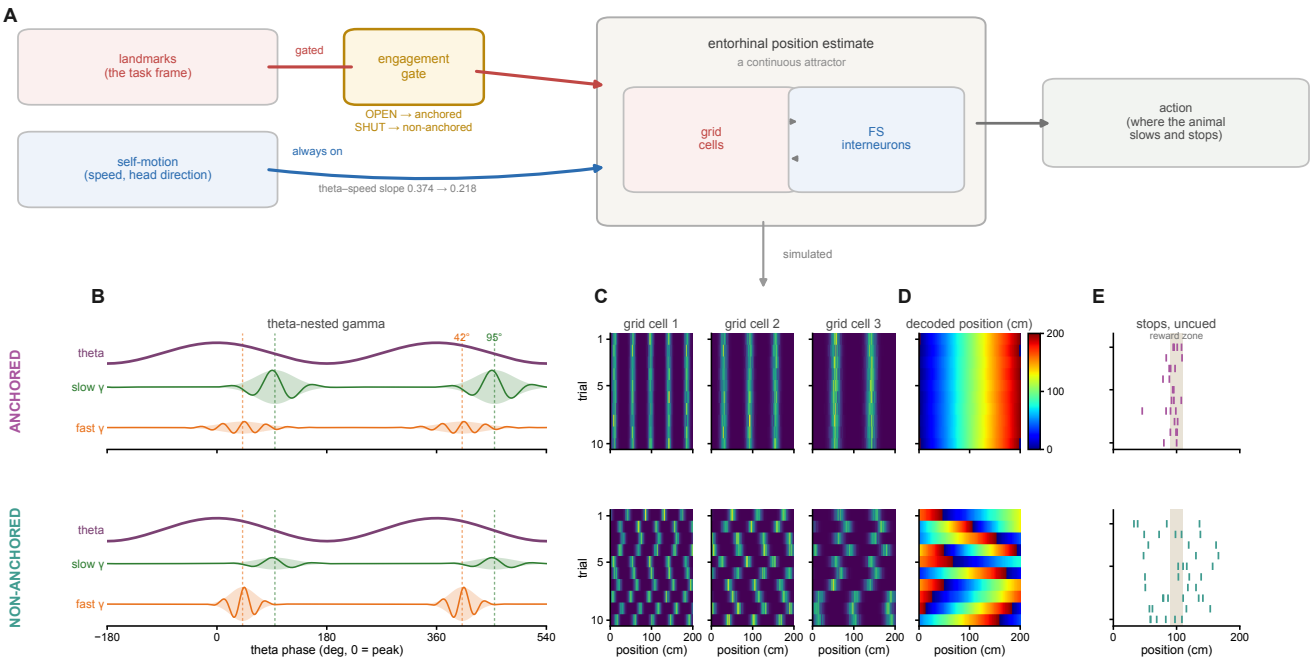



**Figure 9. Proposed engagement gate for position estimation in the entorhinal cortex**

A proposed mechanism, placed with Ideas and speculation rather than with the Results. Panel A is a schematic and the values on it are measured; B is drawn from measured values; C, D and E are simulated. (A) The circuit. A self-motion drive is always present and a landmark drive carrying the task frame passes through an engagement gate; the gate is what the anchoring state is, open meaning the estimate is re-referenced to the task frame on every traversal and shut meaning it is not. In this task engagement goes with a constricted pupil, not a dilated one (Figure 6). The estimate is carried by grid cells and fast-spiking interneurons, the two populations that follow the state most strongly (Figure 2), and drives where the animal slows and stops. The slope on the self-motion arrow is the hinge: theta's entrainment to running speed roughly halves when the gate opens (0.374 to 0.218, p = 0.045) while the open-field speed cells code speed just as well in both states (+11.9%, p = 0.12), so the velocity signal is intact and the rhythm is heeding it less. (B) Theta-nested gamma in the two states, each band at its measured preferred theta phase -- 95 degrees for slow gamma and 42 degrees for fast, with 0 degrees the theta peak (Figure S10). The phases are identical in the two rows because the state does not move them (paired within session, 6.6 and 2.8 degrees, p = 0.33 and 0.16). What changes is amplitude, slow gamma up and fast gamma and theta down (Figure 8), and how tightly the fast burst is confined to its phase, drawn broader when anchored because theta-fast coupling weakens 15.9% (p = 5e-6) while theta-slow does not (p = 0.71). (C) Three simulated periodic grid cells of different spacing over ten traversals. With the gate open the landmark drive re-references the attractor each lap and fields recur at the same places; with it shut the phase is drawn afresh lap by lap. The phase is drawn afresh rather than progressively displaced because non-anchored trials show no coherent field at any offset and neighbouring traversals do not align, so the correction is missing rather than moved. (D) Position decoded from a continuous attractor fed by those cells, drawn as the decoded position itself, wrapped onto the track. Open, it runs 0 to 200 cm identically every lap; shut, it starts from an arbitrary offset and integrates away within the lap. (E) Stops on uncued traversals, with the reward zone at 90-110 cm shaded. Uncued because the zone carries no visual cue there, so stopping in it requires the estimate to be on the track, and because that is where anchoring buys most of its measured behavioural advantage (Figure 1). The circuit is drawn once rather than twice because the measurements say it is the same in both states: spike coupling to gamma does not move with the state (Holm p >= 0.17), nor does the laminar theta phase gradient, and connected pairs share the state no more than matched unconnected ones (Figure 5).

as constitute it.

The state would then be the balance between a recurrent entorhinal loop and a feedforward sensory one, set by neuromodulatory tone rather than by either circuit, and the polarity is the one we observe. Cholinergic tone favours feedforward input over recurrent transmission, and the anchored state is the constricted-pupil state, which should therefore be the recurrent-dominant one: slow gamma higher, fast gamma lower and less tightly nested in theta. That vision should be down-weighted in the state where performance is best is not paradoxical in this task. When the gain between locomotion and the visual scene is altered, mice follow self-motion rather than vision (Tennant et al., 2018), and on uncued trials the reward zone carries no visual marker at all. The account does not require the two bands to trade against one another exactly, which matters because the per-hertz balance we observe does not survive weighting by bandwidth: modulation of feedforward transmission and of recurrent gain would act on different targets, and need not conserve a fixed amount of power between them.

We state this as a hypothesis rather than a conclusion, because its weakest step is an assignment borrowed from elsewhere. The association of slow and fast gamma with internally and externally guided processing derives from hippocampal recordings, where fast gamma indexes entorhinal input to CA1; that framework cannot be carried into entorhinal cortex unexamined, since there the same band cannot index its own input. The assignment could be earned directly in these recordings rather than assumed: in the sessions where entorhinal cortex, the subicular complex and visual cortex were recorded together, the two bands should differ in which structure they cohere with. Until that is done, the account identifies a locus consistent with our data rather than one our data establish.

We had expected phase precession to decide between the pacemaker and speed-pathway accounts, since they predict opposite outcomes for it. That test is not yet decisive in our hands. Precession is detectable in the expected direction - slopes are negative in 59% of cells ($p = 3 \times 10^{-9}$) - but at a magnitude far below what the phenomenon should produce, with a circular-linear correlation barely separable from the chance level for the available spike counts, and a minimum detectable state difference several times the size of the entire signal. Both the naive and the field-matched comparisons are null, and neither null is informative. We attribute this to pooling spikes across field traversals, which is close to worst-case in a dataset where field position varies between trials by construction; a per-traversal analysis is the obvious

next step, and until it is done we draw no conclusion from precession in either direction.

## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work the authors used Claude (Anthropic) to write and debug analysis code, to compute the statistics reported here from the authors' own recordings, to generate the figures, and to draft and revise manuscript text. The scientific questions, experimental design, data collection and analysis decisions were the authors'. Every reported result was produced by code that the authors reviewed, and the authors reviewed and edited all generated text. The authors take full responsibility for the content of this publication.

## STAR METHODS

### Animal details

All experiments were carried out under a UK Home Office project licence, were approved by the Animal Welfare and Ethical Review Board of the University of Edinburgh College of Medicine and Veterinary Medicine, and conformed with the UK Animals (Scientific Procedures) Act 1986 and the European Directive 86/609/EEC. 7 male and 1 female C57BL/6J mice were used in this study (age range [TODO]). Mice were housed with littermates in a holding room with a 12 hour light-dark cycle, temperature range of 20-24 °C and humidity of 45 to 65%. Cages included bedding and nesting material, a half-dome cardboard shelter, a colourful toy and a running wheel. Mice had ad libitum access to food and water. Before headpost/Neuropixel implants, mice were single housed to avoid damage to the recording device.

### Head-fixed behaviour in virtual reality

Mice were head-fixed similar to previous experiments ([TODO: refs]). Mice with surgically attached low-profile RIVETS headposts (Form3 resin printer, in Grey Pro Resin) were connected to a custom-milled stainless steel RIVETS clamp (Ronal Tool Company, Inc) situated above a cylindrical polystyrene treadmill. We projected a virtual track generated using Blender3D (blender.com) onto a torus-shaped screen (Talbot Designs Limited) that surrounded the animal. A rotary encoder (Pewatron, E6-2500-472-IE) relayed treadmill rotation to Blender via an Arduino Due to update the virtual projection in correspondence to the animal's movements. A lickport (Harvard Apparatus) was placed in front of the animal which could dispense sucrose water rewards (10%; 5-10 ul per reward). A custom Arduino Due received input from Blender3D to signal the opening of a solenoid valve (NResearch, 225PNC1-21) to allow the reward to flow.

### Handling

After headpost attachment, mice were handled twice a day for 7 days before training began. Littermates were placed via tube-transfer in a playground arena containing tubes, running wheels, climbing ropes, and colourful toys to promote exploration. The experimenter placed their hands in the playground on the first day of handling without actively touching the mice. From day 2 onwards, once mice began to climb onto the experimenter's hand within the arena, mice were cupped and removed from the arena for 10-20 seconds before being placed back into the arena. This continued with increased duration of handling over time until mice showed no signs of distress when being handled. Once achieved, mice were weighed daily to record a baseline weight prior to water restriction. When mice had at least 3 successive days of recorded weights, water was removed from their cage for 24 hours. From this point, mice were given 0.75-1.5 mL water per day to maintain a bodyweight of 85% the baseline weight, which was normalised to the expected daily growth for the animal's age.

### Habituation

Mice were then slowly introduced to the virtual reality setup. On habituation days 1 and 2, the mouse was set in a perched position within a holding cup for 5 minutes, with the lick port positioned between the VR display and the mouse. The experimenter manually moved the treadmill to simulate movement through the VR environment. On habituation days 3 and 4, the mouse was placed non-head-fixed on the running wheel for 10 minutes with the lickport positioned in front of the running wheel between the VR display and the mouse. Again, the running wheel was manually controlled by the experimenter to facilitate movement through the VR environment. On habituation days 5 and 6, the mouse was head-fixed for 10 and 15 minutes respectively from its surgically attached headpost to the RIVETS clamp. To encourage the mouse to move forward through the VR environment, the experimenter coaxed the mouse by slowly controlling the running wheel until the mouse took over with its own movements. After each habituation session, mice were returned to the playground for 15 minutes of free play.

### Location memory task

On training days, mice were collected from the holding room 30-60 minutes before recording, handled briefly and placed in the playground. Mice were head-fixed in the virtual reality setup for 30 minute sessions. We projected a virtual track that the mouse could navigate through. The track had a length corresponding to 200 cm of running wheel travel distance, with a 60 cm track zone, a 20 cm reward zone, a second 60 cm track zone and a 60 cm black box to separate successive trials (30 cm at the beginning and end of each trial). Once the mouse reached the end of a trial, it was teleported back to the start. The distance visible ahead of the mouse was 50 cm. The reward zone was either marked by distinct vertical green and black bars on cued trials, or was not marked by a visual cue at all on uncued trials. Trials were delivered in a pseudo random fashion such that all mice on a given session experienced the same trial sequence. The probability of each trial type was equal on any given trial, e.g. cued ($p = 0.5$) or uncued ($p = 0.5$). Compared to habituation, mice were required to stop within the reward zone (register a speed $< 3$ cm/s, calculated on a rolling average of 6 frames at 60 frames per second) to receive a reward. As mice initially moved slowly on the first couple of training days, rewards were typically dispensed every trial. As mice sped up across training days, the mice needed to develop a selective stopping strategy to maximise their rewards within the 30 minute session for both trial types. Once mice demonstrated a clear selective stopping strategy for the reward zone, mice were chronically implanted with multi-shank Neuropixels 2.0 silicon probes to record single unit activity while the expert mouse performed the task.

### Dual-context location memory task

Once all recording sites in the brain were recorded, mice advanced to a similar but more demanding dual-context location memory task. Mice were again required to stop at a reward zone either marked or not by a visible cue, but now the location of the reward zone was conditioned on the visual context of the trial. The track length was extended to 230 cm to accommodate an additional reward zone. Context 1 remained similar to the original task with a reward zone 60 cm away from the black box and the same visual pattern on the track walls of white circles on a black background. Context 2 however

required the mouse to stop at a reward zone 90 cm from the black box. Within this context the mouse experienced a different wall pattern of black circles on white walls. Both contexts conveyed the same amount of optic flow information about distance travelled. Contexts were delivered in blocks of 20 trials, alternating between context 1 and context 2. Within a block, the first trial was always a cued trial while all successive trials had an equal probability of the next trial being cued ($p = 0.5$) or uncued (non-beaconed; $p = 0.5$). Mice were trained on the dual-context task for 5-6 days while single unit activity was recorded simultaneously.

### Surgeries

Headpost attachment. Inhalation anaesthesia was induced using 4-5% isoflurane / 95-96% oxygen, and sustained at 1-2% isoflurane / 98-99% oxygen with a flow rate of 1 L/minute. Carprofen and buprenorphine were given subcutaneously at the recommended dosage (20 mg/kg and 0.1 mg/kg respectively) for analgesia, in addition to 1 mL saline to maintain hydration. An oval incision was made to expose the top of the skull. The exposed skull was scored with a scalpel blade to increase the bonding surface area for the future Neuropixel 2.0 implant. The edges of the oval incision were glued in place on the skull (Vetbond tissue adhesive, 3M) and a low-profile RIVETS headpost (3D printed in Grey Pro Resin on a Form3 3D printer) was attached along the incision perimeter and held in place with UV curing resin cement (RelyX™ Unicem, 3M). We aimed to achieve a window through the headpost with both bregma and lambda landmarks clearly visible, with at least 5 mm clearance anterior and posterior for the craniotomy and stereotaxic head straightening in the future surgery. Silicon was also applied to the headpost window to cover the exposed skull until the future surgery. After the surgery, mice recovered for ~20 minutes on a heat mat and had unlimited access to Vetergesic jelly (0.5 mg/kg of body weight buprenorphine in raspberry jelly) for 48 hours.

Chronic Neuropixels 2.0 implantation. Prior to surgery, we assembled the NP2 reusable Apollo drive (https://doi.org/10.7554/eLife.98522.2; https://github.com/Coen-Lab/chronic-neuropixels) using the recommended 3D printer materials. The reference and ground were shorted together with a fine silver wire attached to a grounding screw (M1 x 4 mm screw; AccuGroup SFE-M1-4-A2). Probes reused across animals were cleaned and stored in preparation for surgery (Neuropixel 1.0 user manual; https://www.neuropixels.org/support). Anaesthesia induction, analgesia and hydration were all applied as in the headpost surgery. The silicon over the headpost window was removed to expose the skull. An indent was bored on the left hemisphere in the centre of the intraparietal skull plate to accommodate a single grounding screw. A craniotomy was made from 3 mm lateral to the midline along the lambdoid suture, ending at 4 mm lateral to the midline. The dura was removed and DuraGel (Cambridge Neurotech) inserted to seal the craniotomy. The probe was loaded onto a stereotaxic handle oriented rostrally 15° from vertical and lowered over the craniotomy site. The most medial shank of the multi-shank probe was positioned 3 mm lateral of the midline, with 250 µm spacing between adjacent shanks up to roughly 3.8 mm. The ground screw was secured into the bored indent over the left hemisphere. The probe was brushed with a pipette tip 10 times with DiI (V22889, Thermo Fisher). The probes were manually advanced at < 5 µm $s^{-1}$ to an angled depth of 3 mm or until resistance was met. If resistance was met, probes were retracted by 10 µm. The probes were then secured by cementing the sacrificial Apollo drive docking module to the skull and headpost. Exposed electrical connections were tucked into the payload lids and taped over with electrical tape until recording. Mice were given the same recovery regime as previously described.

### Chronic electrophysiological data collection

Electrophysiological signals were received onto a headstage for filtering, amplifying, multiplexing and digitizing. A 3 metre interface cable connected the probe to a National Instruments acquisition system, which relayed signals to a data acquisition card on a desktop computer running OpenEphys. The Neuropixel acquisition plugin recorded voltage traces at 30 kHz from 384 simultaneous recording sites, in addition to an auxiliary input channel receiving aperiodic transistor-transistor logic (TTL) pulse information from an Arduino Due for data synchronisation with behavioural data recorded in Blender. The same TTL pulse information was relayed to Blender and recorded at 60 Hz, as well as an infrared LED visible to a camera capturing the side profile of the head-fixed mouse.

### Offline spike sorting

The electrophysiological data was processed using SpikeInterface (Buccino et al. 2020) and Kilosort (Pachitariu et al. 2024). The data and probe geometry were loaded using SpikeInterface. Bad and noisy electrode channels were then removed and a phase shift correction was applied, accounting for mismatched sampling times across Neuropixel probes. Both steps were originally developed by the IBL (Banga et al. 2022). The preprocessed recording was passed to Kilosort v4.0.16 with default parameters and motion correction turned off. Kilosort applies a bandpass filter and local whitening, then detects and sorts individual spikes into clusters using a template matching algorithm. After sorting we use SpikeInterface to compute cluster location by monopolar triangulation (Boussard et al. 2021); locations are combined with the registered anatomy to label the brain region of each cluster.

Clusters were curated with bombcell (Fabre et al. 2023), which classifies each unit from its waveform and spike train into good, mua, noise or non_soma. We keep units labelled good or mua and discard noise and non_soma; mua is retained because the population measures used here are built from trial-level label sequences rather than from single-unit isolation, and excluding multi-unit clusters would discard a substantial fraction of the recorded population for no gain in the quantity being measured. Where single-unit isolation does matter - the putative monosynaptic connections of Figure 4, whose cross-correlogram test is sensitive to the spike-sorting errors that split or merge units - the restriction is stated with that analysis. Bombcell is the only curation applied; no additional quality-metric thresholds are imposed.

### Explantation, histology and probe localisation

Following experiments, inhalation anaesthesia was induced using 4-5% isoflurane / 95-96% oxygen, and sustained at 1-2% isoflurane / 98-99% oxygen with a flow rate of 1 L/minute. The Apollo drive payload was disconnected from the docking module and the probes slowly removed from the brain. Mice were then given a lethal dose of sodium pentobarbital (Euthatal, Meridal Animal Health, UK), and transcardially perfused with PBS followed by 4% paraformaldehyde (PFA). Mouse heads were removed and left in 4% PFA solution for at least 2 days. Brains were then extracted and placed in 4% PFA solution for another 2 days, then transferred to 30% sucrose solution for at least 24 hours. Brains were frozen and sliced into 60 µm sagittal sections using a freezing microtome. Sections were stained with DAPI, mounted and imaged using widefield microscopy (Zeiss Axioscanner), using a blue channel for DAPI and an orange-yellow channel for DiI fluorescence. Sections were aligned and registered to the Allen Mouse Brain Atlas using the SHARP-Track package in MATLAB (https://github.com/cortex-lab/allenCCF/tree/master/SHARP-Track). This involved marking the DiI fluorescence to reconstruct the

trajectory that each shank took in the brain. In cases where the probes passed through the cortex, we used additional information recorded across the shank to calculate the shank trajectory. We evaluated noise profiles, theta power (6-12 Hz) and curated cluster density to calculate likely exit points of the probe through the cortical surface.

### Reconstruction of probe locations

By measuring 250 µm between adjacent tips in 8 animals, we targeted the medial entorhinal cortex across the medial-lateral axis, starting medially (around 3 mm medial of the midline) at the MEC-parasubiculum border, which has been shown to display a sharp anatomical division of grid cells (Obenhaus et al. 2022), out to lateral regions of the MEC (around 3.75 mm lateral of the midline), close to the borderlands of the LEC. We targeted dorsally to afford adequate sampling of grid fields in a 1 m x 1 m open field and accurate classification of grid cells.

### Anatomical coordinates and the mediolateral axis

Each cluster carries two sets of coordinates, both shipped with the dataset, and both are used:

- coord_CCFs_x/y/z - Allen CCF coordinates, with coord_CCFs_x the mediolateral axis. Used for the anatomical distributions in Figure 2. - coord_SCs_x/y/z - the same locations expressed relative to the midline. Used for the medial/lateral split, and via coord_SCs_x/y as the distance metric for ordering covariate cells in the encoding models.

The two are related by an exact constant offset, verified across all 49,747 registered clusters and every clark2025 session:

so coord_SCs_x is signed distance from the midline (negative in the left hemisphere, where all probes were implanted) and the CCF midline lies at x = 5700 µm. Analyses that threshold on mediolateral position negate coord_SCs_x first, so the working variable throughout is absolute distance from the midline. Across entorhinal clusters this spans 3000-3759 µm (median 3510), consistent with the surgical targeting of 3.0-3.8 mm lateral.

Medial/lateral boundary. Cells are classified medial at < 3400 µm from the midline and lateral at >= 3400 µm, dividing the entorhinal sample 43.2% / 56.8% (11,300 / 14,876 clusters). The boundary separates the layer-defined subdivisions in the expected order - ENTm1 90.9% medial, ENTm2 56.5%, ENTm3 30.8%, ENTm5 18.3%, ENTm6 16.1%, ENTl6 0% - which is a consistency check on the registration rather than an independent justification of the value.

### Open field recordings

The open arena consisted of a metal box with a square floor area, removable metal walls and a metal frame (frame parts from Kanya UK: C01-1, C20-10, A33-12, B49-75, B48-75, A39-31, ALU3), with an A4-sized cue card in the middle of one of the metal walls. For open-field exploration sessions, mice were placed in the arena while tethered via a Neuropixel cable and were prompted to explore with small crumbled pieces of chocolate Weetos, for a maximum of 20 minutes per session.

### Analysis of behaviour during the location memory task

To quantify trial performance, we used a number of metrics based on previous work (Tennant et al. 2022; Pettit et al. 2024). We classified trials into discrete behaviours that describe the speed profile and engagement for any given trial. Trials were first classified into rewarded and unrewarded based on whether the mouse stopped (speed < 3 cm/s) within the reward zone, initiating a reward to be dispensed. To discriminate between rewarded trials in which mice ran and slowed to a stop (demonstrating stop selectivity for the reward zone) and rewarded trials in which mice maintained a slow speed throughout the track but still initiated rewards, trials in which the mouse's average speed outside of the reward zone was > 10 cm/s were classified as hits, while the remainder were removed from behavioural analysis. Unrewarded trials were split into try and run trials by comparing their average speeds in the reward zone to hit trials: the 95th percentile of speeds in the reward zone for hit trials was used to discriminate between try trials (< 95th percentile speed) and run trials (> 95th percentile speed).

Session-level stopping strategy. Trial-level performance describes single trials; sessions were additionally summarised by the stopping strategy the animal used, which distinguishes solving the task by sight from solving it by path integration.

For each session and each trial type, stops were binned by track position into 20 bins and the resulting profile normalised within trial - each trial's stops sum to one before averaging across trials - so that a trial with twenty stops contributes no more than a trial with one. Pooling raw stops instead allows the heaviest-stopping trials to dominate the profile and can manufacture an apparent reward-zone preference out of dwell time rather than choice. Stops inside the 30 cm start box were excluded, as they precede any decision about the zone.

Stop selectivity is the proportion of stopping falling inside the reward zone divided by the proportion of the track the zone occupies, so that 1.0 denotes stopping spread uniformly along the track. Sessions were then assigned to one of three strategies:

- cue-dependent - uncued selectivity below 2.0, or uncued hit rate below 0.25; - expert - uncued hit rate at or above 0.70; - partial - intermediate.

Across 79 sessions this gave 26 cue-dependent, 25 partial and 28 expert. The two extremes are behaviourally distinct in the way the task intends: cue-dependent sessions show selectivity of 5.9 when the zone is cued but 2.0 when it is not, with an uncued hit rate of 0.11; expert sessions show 5.2 and 4.8 with an uncued hit rate of 0.86. Both stop precisely at the zone when shown it; only one does so unaided.

Stopping rate is reported separately, not as a strategy. Six sessions exceed 15 stops per trial (median 19.5 against 3-8 elsewhere). These are not less selective once profiles are normalised within trial, and they span all three strategies (3 partial, 2 expert, 1 cue-dependent), so excessive stopping is an independent descriptor of a session rather than a stage of the ladder. It is nonetheless visible in the first stop of each trial, which for these sessions falls at 0.2-0.5 of chance - systematically away from the reward zone.

Independent confirmation. The strategies were validated by a model that was never told them. Stop counts were modelled per trial and position bin with three regressors - a constant, the reward zone on cued trials, and the reward zone on uncued trials - and the coefficient of partial determination computed for each. The uncued-zone regressor accounts for 6.7% of the variance in expert sessions and 0.09% in cue-dependent sessions, a difference of nearly two orders of magnitude, while the cued-zone regressor is substantial in both (11.0% and 8.9%).

### Local field potentials and theta phase

We compute the LFP per recording channel by applying a bandpass filter between 1 and 300 Hz to the raw 30 kHz recordings. To compute the theta phase of the LFP, we downsample the LFP to 60 Hz and then further bandpass filter between 5 and 10 Hz. We subsequently apply the Hilbert transform to obtain the phase. When linked to individual clusters, we use the phase of the channel where the template amplitude of the cluster was maximised.

### Theta and the population anchoring state

Signal and channel selection. Band power was computed from the band-limited (1-600 Hz) LFP sampled at 1 kHz, stored as 48 channel groups per probe, each the average of 8 adjacent contacts. Each session ships an anatomical assignment of every contact (CH1-CH384) to a brain region; a channel group was treated as entorhinal when the majority of its contacts were assigned to ENTm. Groups are contiguous blocks of the shank and therefore almost always fall within a single layer, making the majority vote unambiguous. A median of 26 of 48 groups per session were entorhinal; sessions with none (all of M22, consistent with their absence of entorhinal spatial cells) were excluded, leaving 58 sessions and 14,022 trials.

Depth was retained rather than averaged. Theta amplitude varies steeply through the entorhinal layers, so averaging across groups mixes signals of very different magnitude. Power was computed per trial and per channel group and, where a single value per trial was required, groups were combined only after within-group standardisation.

Band power. Power spectral density was estimated per trial with Welch's method and integrated over delta (1-4 Hz), theta (6-10 Hz), beta (10-20 Hz), gamma (30-80 Hz) and high (80-200 Hz) bands. Both relative and absolute power are reported. Relative power is compositional - the five bands sum to one, so a fall in theta mechanically raises the others and one effect can appear as four. Absolute power breaks that coupling and is the primary measure; it was z-scored within session and channel group before pooling, because absolute amplitude varies by an order of magnitude across probes and depths.

Separating a frequency shift from a weakening. Power in a fixed 6-10 Hz window falls if the rhythm weakens or if it moves out of the window, and these are different claims about the circuit. The two were separated using the stored theta phase: instantaneous frequency was taken as the derivative of unwrapped phase (computed as a wrapped per-sample difference, because a small fraction of NaN samples propagates through np.unwrap to the whole trace), and peak frequency as the spectral maximum within 4-12 Hz. Theta amplitude was taken as the Hilbert envelope of the 6-10 Hz signal.

Speed matching. Theta amplitude and frequency both track running speed, so every comparison between anchored and non-anchored trials was repeated on a speed-matched subset, formed by matching trials within deciles of trial mean speed. Comparisons were paired across sessions, so no session dominates.

Speed-theta coupling. Beyond a difference in level, we asked whether the coupling between speed and theta differs by state. Per session we fitted theta ~ speed x state by ordinary least squares and took the interaction term as the change in gain, and separately computed the within-state Pearson correlation. Because speed is genuinely less variable inside anchored epochs (SD 6.4 vs 8.4 cm/s, $p = 1 \times 10^{-8}$, 48/55 sessions), which attenuates a correlation on its own, the regression slope is the primary measure - it is unaffected by range restriction - with correlations reported alongside. Sessions contributed only when both states held at least 15 trials (55 of 58). Three controls were applied: (i) both states were trimmed to their common 5-95% speed support and the slopes refitted; (ii) the per-session change in correlation was tested against the per-session change in speed SD, to check for a dose-response with the confound; and (iii) a circular-shift null in which the state vector was rolled within session (200 shifts), preserving run structure and any within-session drift in theta.

Speed tuning of entorhinal cells. Firing rate and running speed were binned at 250 ms and restricted to running bins (>= 3 cm/s), each bin taking the anchoring state of its trial. All curated entorhinal cells were used (grid, non-grid spatial and narrow-spiking). Speed cells were defined pooled across states, so that the definition could not itself be state-biased, as cells whose rate-speed correlation exceeded a circular-shift null of the binned rate (200 shifts, $p < 0.05$, positive); this null preserves the autocorrelation of firing, which makes a parametric p meaningless at this bin size. Per cell and state we computed the regression slope of rate on speed (Hz per cm/s), the gain (slope divided by mean rate, which separates a change in speed sensitivity from a change in overall firing), and the correlation, each on all running bins and again after trimming both states to their common 5-95% speed support. Sessions, not cells, are the unit of analysis: cells within a session share one behavioural state, so a cell-level test would count the same state change hundreds of times.

Population speed decoding. Ridge regression from population binned counts to running speed, five-fold cross-validated, fitted separately within each state after matching bins across speed deciles and subsampling to equal count, so neither state received more training data or a wider speed range.

Regional and laminar specificity of the coupling change. To ask whether the decoupling is confined to entorhinal cortex, the speed->theta slope was computed for every channel group on the probe rather than only the entorhinal ones, with each group assigned to a region family by majority vote over its eight contacts, and with the anchoring state still defined by entorhinal cells. Theta amplitude per trial was taken as the RMS of the 6-10 Hz signal over running samples, regressed on trial mean speed within each state on the states' common 5-95% speed support. Slopes were normalised by mean amplitude within state, because absolute LFP amplitude varies by an order of magnitude across regions and depths and would otherwise dominate any cross-region comparison; raw slopes are reported alongside. Sessions contributed a region only when it held at least four channel groups, and only when both states retained at least 15 trials after trimming, which excluded seven otherwise usable sessions whose non-anchored blocks were too short (13-23 trials). Laminar comparisons use one value per session per layer: channel groups within a session share a behavioural state and a probe, and a per-group test counts the same session dozens of times.

Spike-theta phase locking. Each cluster was assigned the theta phase of the channel group containing its extremum channel, rather than a session average, because theta phase rotates through the entorhinal layers. Spikes were restricted to running samples in labelled trials, then matched across speed deciles between states and subsampled to equal count; locking strength is the mean resultant length averaged over 20 such draws. Both matching steps are necessary and run in the same direction: the mean resultant length is biased upward at small samples, and anchored firing rates are about 7% lower, so unmatched counts would make anchored cells appear more strongly locked. The speed dependence of locking was taken as the difference between the top and bottom speed terciles of the matched set, with counts equalised within each tercile.

Unexpected speed. Because anchored trials have both more reliable position coding and a more stereotyped speed profile, position can stand in for speed and inflate any apparent speed coding. Position was therefore partialled out of both sides: within each state, cyclic position (sin and cos of 2pi.position/L) was used to predict speed and, separately, each cell's spike count, both cross-validated (three folds; xgboost, 50 rounds at depth 3, count:poisson for rate and squared error for speed). The residuals - unexpected speed and unexpected rate - were then regressed on one another. The speed residual is orthogonal to position by construction, so a difference between states cannot arise from position substituting for speed. Sensitivity is reported both per cm/s (physical gain) and

per SD of residual speed, because the two answer different questions when the states differ in how much unexpected speed they contain.

Within-session time course. To ask whether the coupling changes while the state changes rather than only between pools of trials, the speed-theta correlation was recomputed in a sliding window of 25 consecutive trials (windows requiring >=60% valid trials) and compared with the anchored fraction of the same window. Neighbouring windows share trials, so the window-level correlation is autocorrelated and cannot be assessed parametrically; per-session significance was taken from a circular shift of the state vector alone (500-2000 shifts), which leaves the speed and theta series and the state run structure intact and breaks only their alignment. The analysis was repeated at window widths of 15 and 40 trials.

### The spectrum, theta-gamma coupling and gamma events by anchoring state

All of these read the gated labels in data/population_state/anchoring_trials.csv.*

The full spectrum rather than named bands. The band-power analysis above integrates over five bands fixed in advance, which cannot detect a change that reverses sign inside one of them - and 30-80 Hz "gamma" is exactly where that happens. Power spectral density was therefore estimated per trial and per entorhinal channel group with Welch's method at nperseg = 1024 on the 1 kHz signal, giving 0.98 Hz resolution over 1.9-150 Hz. The window length matters for the low end: nperseg = 512 gives 1.95 Hz bins whose first bin above 2 Hz is centred at 3.9 Hz, which leaves 2-4 Hz empty.

Mains. Bands at 48-52, 98-102 and 148-152 Hz were excluded from every band average. The 50 Hz line sits inside the slow-gamma band, so this is not cosmetic: removing it strengthens the slow-gamma effect (+0.109, p = 3.6 x $10^{-4}$ against +0.105), and the 48-52 Hz band tested on its own does not reach significance (p = 0.063), which establishes that the effect is not line noise tracking the state.

Whitening and normalisation. Spectra are reported as log□□(power) + log□□(frequency) - power multiplied by frequency - so that spectral peaks are visible against the 1/f background instead of being buried under it, and then mean-centred within session across frequencies so that no recording's absolute amplitude dominates the average. Per-band differences are computed on within-session z-scores and tested with a Wilcoxon signed-rank on per-session means, paired across states, with the session as the unit of analysis throughout.

Session inclusion. A session contributed when it held at least 8 trials in each state, giving 26 sessions. This threshold is lower than the 15 used for the speed-coupling analyses because a band-power mean is estimated from far fewer trials than a within-state regression slope; at 15 the same analysis gives 21 sessions and every reported effect holds (theta −0.099, p = 0.029; slow gamma +0.079, p = 0.018; mid gamma −0.125, p = 0.0025).

Theta-gamma phase-amplitude coupling. Coupling was quantified with Tort's modulation index: the gamma envelope was averaged into 18 bins of theta phase, normalised to a distribution, and the index taken as its Kullback-Leibler divergence from uniform, divided by log(18). A per-session null was built by shuffling the phase series in blocks and the reported quantity is the index minus its null mean, so that differences in the amount of data between states cannot produce a difference in coupling. Theta phase is stored in these files on 0-2pi, not −pi to pi; wrapping it into the symmetric range is required before binning, and omitting that step returns NaN for every session.

Gamma events. To ask whether discrete gamma events, rather than continuous power, carry the state or the task demand, the envelope of each gamma band was z-scored within session and thresholded at 2 SD; a supra-threshold excursion lasting at least one cycle of the band's centre frequency was counted as an event. Events were expressed per theta cycle, with the number of available theta cycles per trial as the denominator, so that a trial is not credited with more events simply for being longer or faster. Models are mixed models with a session random intercept, and the covariates added in sequence are trial mean speed, instantaneous theta frequency and band power - power last, because it is the quantity the event count is meant to be an alternative to, and an event effect that disappears on adding it is an effect on power.

Gamma events by track position. Event rate was additionally binned into 5 cm position bins, weighted by occupancy, separately by state and cue condition. Position binning here is a place where an identity comparison silently empties the result: comparing a numpy.bool_ state label to a Python bool with is is always false, so an earlier version returned empty position profiles for every session while reporting success. The binning is vectorised over per-trial arrays indexed by trial number, which also removes an apparent hang caused by a Python-level comprehension over ~1.8 million samples per session.

### Cell classification - spatial metrics

All spatial metrics are computed at the animal's true position; no temporal or spatial shift of the trajectory is applied.*

Rate maps. Open-field firing rate maps were constructed on a 40 x 40 bin grid spanning the 1 m x 1 m arena (2.5 cm bins), computed as the ratio of binned spike counts to binned occupancy, and smoothed with a Gaussian kernel of sigma = 1.5 bins. Bins with zero occupancy were treated as undefined and excluded from all subsequent calculations.

Spatial autocorrelogram. The spatial autocorrelogram (SAC) of each rate map was computed as the Pearson product-moment correlation between the map and a copy of itself displaced by every possible two-dimensional lag, evaluated over the region of overlap. For efficiency this was implemented with FFT-based convolution. Lags at which fewer than 3 bins overlapped, or at which either overlapping region had near-zero variance (< $10^{-12}$), were set to zero rather than being allowed to produce singular values, and the result was clipped to [−1, 1]. This implementation was verified against direct spatial-domain convolution (agreement to 1.8 x $10^{-13}$).

Grid score. Gridness was computed from the SAC using the annulus-sweep procedure of Banino et al. (2018), as implemented in the GridScorer class shared with the recurrent network model used elsewhere in this work, so that scores are directly comparable between recorded and simulated populations.

For a given annular mask M covering the SAC, let S be the SAC and S(theta) the SAC rotated by theta about its centre (cubic spline interpolation, output shape preserved). Writing A = ΣM for the ring area and

the rotational correlation at angle theta is

and gridness for that mask is

Two properties of this estimator should be stated rather than left implicit. First, the rotated map is centred by the mean and normalised by the variance of the unrotated masked SAC, not by its own; r(theta) is therefore a normalised cross-product rather than a symmetric Pearson correlation. It coincides with Pearson only insofar as rotation preserves the ring mean and variance, which holds approximately by the symmetry of the annulus but not exactly. Second, the $10^{-5}$ added to V regularises rings that are close to flat, and bounds r for cells

with negligible structure inside the annulus.

Because the appropriate annulus depends on grid spacing, which is not known in advance, gridness was evaluated over a sweep of 10 annuli and the maximum taken as the cell's grid score. All ten share an inner radius of 0.2 x 40 = 8 bins (20 cm), excluding the central peak; outer radii are linearly spaced from 0.4 to 1.0 in the same units, i.e. 16 to 40 bins (40 to 100 cm) in ten steps. The SAC extends to a maximum lag of 39 bins, so the widest annuli span the entire defined region of the autocorrelogram.

Unlike field-detection approaches, this procedure returns a grid score for every cell: no cell is excluded for failing to yield a minimum number of autocorrelogram peaks. Significance is established by the shuffle procedure below rather than by a structural precondition.

Spatial information. Spatial information was quantified as the Skaggs information rate in bits per spike

where p□is the occupancy probability of bin i, lambda□the firing rate in that bin, and lambdā the occupancy-weighted mean rate. This quantity is non-negative for all rate maps by Jensen's inequality; negative values cannot arise from data and indicate an implementation fault.

Border score. Firing fields were defined as connected components of the rate map exceeding 30% of the map maximum and covering more than 200 $cm^2$. The border score is (cM − dm)/(cM + dm), where cM is the largest proportion of any single wall covered by a single field, and dm is the mean distance of field firing from the nearest wall, weighted by firing rate and normalised by half the arena width.

Head direction score. Head-direction tuning curves were computed as occupancy-normalised firing rate as a function of head direction, and tuning strength was quantified as the mean resultant vector length.

Significance testing. Significance of each metric was assessed against a within-cell null distribution generated by circularly shifting the spike train relative to the behavioural trajectory by a random offset of at least 20 s, wrapping spikes past the end of the session to the beginning, and recomputing the metric. 100 such shuffles were generated per cell. A cell was scored as significant for a given metric if its observed value exceeded the 95th percentile of its own null distribution. Shuffle offsets were drawn from a generator seeded deterministically on (mouse, day, session, cluster, shuffle index), so that null distributions are exactly reproducible and can be extended with additional shuffles without recomputing existing ones.

Inclusion criteria. Only units labelled good or mua by bombcell were analysed; units labelled noise or non_soma were excluded. Units were required to have at least 500 spikes in the session to be classified.

Classification. Cells were classified from open-field activity as:

- Grid cell (GC) - significant grid score and significant spatial information; - Non-grid spatial cell (NGS) - significant spatial information but not classified as a grid cell; - Non-spatial (NS) - neither.

Classification was performed independently for the two open-field sessions (OF1 and OF2) recorded on each day, and both are reported.

To increase confidence that grid-like dynamics were genuinely captured and classified as grid cells, we analysed two open-field sessions acquired on the same recording day (OF1 and OF2). The motivation for using both sessions was to reduce the chance that cells with grid-like activity would be missed in one recording and subsequently misassigned to the non-grid spatial (NGS) population.

**Grid module identification**

Grid modules were identified within each open-field session, from the geometry of each cell's spatial autocorrelogram. Modules are defined over simultaneously recorded cells, so no analysis pools cells across sessions: a cluster formed across sessions could at best reflect a spacing band common to different animals, not a module.

Feature space. For every curated cell in a session - grid, non-grid spatial and non-spatial alike - the central disc of the spatial autocorrelogram was extracted, flattened and z-scored, giving a description of autocorrelogram shape independent of firing magnitude. All cells were embedded, not only grid cells: the question is where grid cells sit within the structure of the whole population, which cannot be asked if only grid cells are present.

Embedding and clustering. Cells were embedded in two dimensions with UMAP (Manhattan metric, spectral initialisation, min_dist = 0.05, n_neighbors scaled with session size) and clustered with DBSCAN (min_samples = 3). The DBSCAN neighbourhood radius was set per session by the k-distance knee - the point of the sorted k-nearest-neighbour distance curve lying furthest from the chord joining its endpoints - rather than by maximising a silhouette. Silhouette rewards many tight clusters and therefore collapses to the smallest radius available: it selected the sweep floor in 12 of 28 sessions, producing a median of 10 clusters (up to 25) and discarding 34% of cells as noise. On identical embeddings the knee criterion gave a median of 5 clusters and 2.5% noise.

A cluster is called a module only if it is both grid-enriched and geometrically coherent, requiring all of:

- at least 5 grid cells; - a grid-cell fraction at least 2x that of the session as a whole; - an interquartile range of grid spacing <= 10 cm across its grid cells; - a circular standard deviation of grid orientation <= 10°.

Enrichment alone is insufficient: a cluster can be grid-dominated while containing cells from several modules. The coherence criteria are what distinguish a module from a bin of grid cells, and in practice they reject clusters that an enrichment-only rule would have accepted.

Grid cells outside every qualifying cluster are left unlabelled. They remain classified as grid cells and simply carry no module assignment. Coverage - the proportion of grid cells receiving a label - is reported rather than maximised; a method that assigns modules confidently to some cells and abstains on the rest is preferable to one that forces every grid cell into a module.

Parameter choice. min_samples and the enrichment threshold were selected by sweeping min_samples □ {3,4,5,6} and enrichment □ {1.5, 1.75, 2.0, 2.5} on fixed embeddings, scoring each setting on module yield, coverage, cross-session reproducibility and module coherence. The 2x enrichment threshold is not a convention: cross-session reproducibility is 1.000 at 2.0 for every min_samples tested, falling to 0.897 at 1.75 and 0.768 at 1.5.

Validation. Because module identity is a property of a cell rather than of a recording, the two open-field sessions of each day provide an independent test. Module labels were computed separately in each session, and agreement assessed on the cells labelled in both, pairwise and label-invariantly (module indices are arbitrary per session): for every pair of such cells, whether the two sessions agree that they belong to the same module or to different modules. Agreement was 1.000 in 8 of 9 testable days (0.78 in the ninth). Module geometry also recurred across sessions without being constrained to - for example M25 D24 yielded modules at 51.5 cm and 37.2 cm spacing in the first session and 51.5 cm and 34.0 cm in the

second.

Yield. Across 28 open-field sessions this identified 31 modules in 24 sessions; 6 sessions yielded two or more. 480 of 962 grid cells (49.9%) received a module label, the remaining 482 being grid cells without a module assignment. Modules were geometrically tight - median interquartile spacing range 3.6 cm and median circular orientation SD under 4°, against admission thresholds of 10 cm and 10°.

### Classification of anchored and non-anchored trials

Firing on the linear track was classified, trial by trial and cell by cell, as anchored or non-anchored to the task reference frame.

This is not the spectral classifier used previously. Anchoring on the linear track has previously been assessed by testing whether firing is spatially periodic at a frequency commensurate with the track (Clark & Nolan). That method was built for, tuned on and validated against grid cells, and its published form computed each label over a rolling window of five trials either side - eleven trials of support. Neither suits this paper, which classifies grid and non-grid cells with the same instrument and evaluates labels on individual trials. The classifier used here asks only whether a trial's spatial firing profile resembles the same cell's other trials, so cells are judged by self-consistency rather than by periodicity and no firing-field shape is advantaged.

Procedure. For each cell:

1. Assemble trial-by-trial firing rate maps into an n<sub>trials</sub> x n<sub>bins</sub> matrix (2 cm bins over the 200 cm track). 2. Smooth each trial's profile (Gaussian, sigma = 1.5 bins) NaN-aware: unvisited position bins are missing data, not measured silence, so the smoothed map is normalised by a correspondingly smoothed validity mask. Zero-filling would assert a measurement never made, and because the unvisited pattern follows the animal's occupancy it is shared across simultaneously recorded cells - which would make trials correlate through the trajectory rather than through firing. 3. Mark trials whose summed smoothed activity falls at or below $10^{-6}$ as invalid; classify no cell with fewer than 6 valid trials. Invalid trials are assigned to the non-anchored class, not excluded. This operates on a substantial fraction of data - a median 12.3% of trials per cell - and because silence is correlated across simultaneously recorded cells it injects a shared component into the label matrix. Whether that component explains the population structure is tested directly below (Coordinated silence); it does not. 4. Correlate every valid trial's profile against every other (Pearson), giving a trial x trial correlation matrix, and summarise each trial by the mean of its correlations with all other valid trials, excluding the self-correlation on the diagonal. Note that this is a mean of pairwise correlations, not a correlation against the cell's average firing rate profile: the two are related but not identical, and the pairwise form avoids each trial being scored partly against itself through its own contribution to the mean profile. 5. Partition those one-dimensional values by 2-means clustering, and label the cluster with the higher centre - trials that are mutually self-consistent - as anchored. There is therefore no fixed correlation threshold; the boundary is set per cell by the data, which is what allows cells with very different field structure and signal-to-noise to be classified on the same footing. Empirically the two centres sit near −0.01 and +0.08, and the implied boundary falls at a median mean-correlation of 0.033 (IQR 0.015-0.063, 5th-95th percentile 0.003-0.126; 221 cells from three sessions), with a median silhouette of 0.60 for the two-cluster partition. 6. Median-filter the label sequence (size 5, not the firing data) to remove phase-aliasing false positives, where a non-anchored trial of a periodic cell aligns in phase with other non-anchored trials by chance. This raises specificity from 0.808 to 0.884 at negligible cost to sensitivity (0.990 to 0.993).

Because trials are compared by Pearson correlation, the classifier is invariant to the scale and offset of the input: scaling activity by $10^{-3}$ or $10^{3}$ returns bit-identical labels, so the same code applies unchanged to spike counts, calcium fluorescence and simulated network activations. The only scale-dependent parameter is the validity floor in step 3. The median filter in step 6 erases any run shorter than three trials, which sets the temporal resolution at three trials - against eleven for the spectral classifier. Recall of a non-anchored epoch is 0.02 at one trial and 0.07 at two, rising to 0.64 at three and above 0.80 from five onward. Analyses concerned with briefer events can disable the filter and accept the lower specificity.

Validation. Accuracy was measured against a simulated recurrent-network session with known anchoring ground truth that the classifier never saw: the position cue was present for 35 trials, withdrawn for 30 so that the network's position estimate drifted off the task reference frame, then restored for 35. Across 1,024 simulated units, median accuracy was 0.990, sensitivity 0.993 and specificity 0.884. Accuracy depended only weakly on how grid-like a unit was (Spearman $r = +0.219$, $p = 1.5 \times 10^{-12}$; least grid-like quintile 0.970 against 0.990 for the most), supporting its use on grid and non-grid populations alike. The same classifier, unmodified, recovers experimenter-controlled cue visibility in two-photon calcium imaging of hippocampus (Pettit et al.), so agreement across three signal types is evidence about the phenomenon rather than about a shared measurement pipeline. On recorded data, where no ground truth exists, cluster separation was quantified by the silhouette score of the 2-means partition and reported per cell.

Which cells define the population state. Population-level anchoring - the per-trial fraction of cells labelled anchored, from which every population measure in this paper is derived - is computed over medial entorhinal spatial cells only, defined as cells classified as grid or non-grid spatial whose reconstructed position falls within ENTm.

This restriction is necessary rather than cosmetic. The recordings are not confined to MEC: across 11,611 spatial cells, 66.1% are in ENTm, with the remainder in visual cortex (14.9%), parasubiculum, presubiculum and subiculum (11.8%), and other structures (5.8%). The per-session proportion varies far more than that average suggests, from 0 to 100%, and 23 of 81 sessions are less than half ENTm. In 19 sessions there are no ENTm spatial cells at all: those probes sat in subiculum, presubiculum, parasubiculum, visual cortex or, in a few cases, cerebellum. Anatomical reconstruction is available for every one of these cells, so this reflects genuine electrode placement rather than failed registration. A population state computed over all recorded cells would therefore not be a measure of the medial entorhinal cortex in a substantial minority of sessions.

Restricting to ENTm removes 22 of 81 sessions at the 25-cell threshold, including all sessions from one animal (M22) whose probe did not reach MEC, and leaves a median of 112 ENTm spatial cells per retained session. The region filter is applied in one place (spatial_cell_ids, with REGION_PREFIX = 'ENTm') and inherited by every downstream analysis.

It was previously duplicated inline across notebooks, which had allowed the copies to diverge in their handling of unvisited bins.

Population anchoring state and its structure. The per-trial fraction of cells labelled anchored summarises the population state, and its major axis of variation is the first principal component of the cell x trial label matrix, taken as a per-trial score. PC1's sign is arbitrary and is fixed by orienting it to correlate positively with the anchored fraction.

Whether this reflects a genuinely shared state, rather than the

arithmetic of averaging, requires a null: cells anchored on most trials agree with one another most of the time by construction. Each cell's label sequence was therefore shifted circularly and independently, preserving every cell's anchored fraction and its run structure while destroying only the alignment between cells. Two statistics were compared against this null - the mean pairwise correlation between cells, and PC1's share of the label variance. Across 59 qualifying sessions PC1 exceeds its null in the large majority, typically by roughly a factor of two.

Major population transitions. Several figures mark the trials at which the population changes state. The raw thresholded state (anchored fraction > 0.5) changes too often to mark directly: these sessions carry ten or more runs, half of them only a few trials long, and drawing every change buries the raster under lines. A transition is therefore defined as a sign change of the thresholded population state after a 9-trial median filter (scipy.ndimage.median_filter, mode='nearest'), which removes runs shorter than about five trials while leaving the block structure intact. The same definition and the same filter width are used in every figure that marks transitions, so the lines are comparable across them.

Two alternatives were rejected. Requiring both runs adjacent to a boundary to exceed a length silently drops a boundary whenever a long block is interrupted by a brief one, so clear block edges went unmarked; filtering the state first does not have this failure. Marking the raw changes is faithful but unreadable.

Note that this filter is separate, and applied after, the size-3 median filter inside the per-cell classifier (step 6 above). The classifier's filter smooths each cell's label sequence and sets the method's temporal resolution; this one smooths the population summary and exists only to decide which transitions are worth drawing. No statistic in the paper is computed on the filtered population state - it is used for annotation, and the switch-timing analysis locates transitions on the unfiltered state recomputed without the cell in question.

Coordinated silence: tested, and not the explanation. The classifier assigns trials on which a cell emits no spikes to the non-anchored class, and silence is correlated across simultaneously recorded cells because it largely reflects trials on which the animal did not run. This injects a shared component into the label matrix that has nothing to do with anchoring, and the circular-shift null cannot remove it, since shifting preserves each cell's marginal and therefore the same proportion of forced labels. Every population measure was therefore recomputed with silent trials held as missing and dropped pairwise rather than forced to zero.

The structure is unaffected. Across 9,464 cells the mean correlation with the population axis is 0.164 with silent trials forced and 0.165 with them excluded; by region the largest change in either direction is 0.003. Cerebellar cells, which have the lowest silent-trial fraction of any group (2.4% against 6.7% for entorhinal cells), change by −0.003. The population measures in this paper are therefore not measuring coordinated silence, and the control previously listed as outstanding is resolved.

Per-cell agreement with the population axis. Each cell's label sequence was correlated against the MEC PC1 score. For cells that are themselves part of the MEC population the axis was recomputed by SVD with that cell removed, since a cell otherwise helps define the axis it is scored against; cells outside MEC never enter the decomposition and need no such correction, and applying one to them as well would bias the two groups differently. Each cell was additionally compared against its own circular-shift null, and results are reported as the correlation minus that null, because a cell anchored on most trials correlates with a population that is also anchored on most trials irrespective of any coordination. Roughly a quarter of MEC spatial cells are individually locked to the axis (median 27% per session, against a 5% false-positive rate), so the state is carried by a minority subpopulation rather than entered by the whole of MEC together.

Mediolateral position. Within MEC, cells were split at 3400 μm. Scoring both halves against the joint MEC axis is not a fair comparison, because each half helps define the axis it is measured against and the larger half dominates it. The axis was therefore built from each half separately, with cell counts matched by subsampling before the decomposition, and used to score both halves - giving a medial->medial, medial->lateral, lateral->lateral, lateral->medial matrix. Because grid cells constitute 10.9% of medial but only 3.3% of lateral cells and follow the axis more strongly, every comparison was repeated on non-grid cells alone.

### Major transitions in the population state

The population anchoring state is summarised per trial by the first principal component of the cell x trial label matrix, and the figures mark the trials on which that state turns over. A transition is a sign change of PC1: it is the variable the panels plot, so its zero crossing is the boundary, and no threshold on the anchored fraction and no smoothing width has to be chosen.

One constraint is applied. Runs of fewer than three trials on one side of zero are absorbed into the preceding epoch, on the grounds that a trial or two across the axis is a crossing rather than a state the population entered. The floor matters only for sessions that hover near zero: in M27 D23, the most-switching session in the dataset, it takes the count from 22 transitions over 393 trials to 12, while leaving the other example sessions unchanged or nearly so.

An earlier definition thresholded the anchored fraction at 0.5, put it through a 9-trial median filter, and in one figure absorbed blocks shorter than 5% of the session. That is three free parameters where the present rule has one, and the three were set by eye.

### Theta index

Rhythmicity at theta frequency is scored per cell from the open field, not from the virtual-reality session, for the same reason grid score and speed score are: a measure taken on the session being explained cannot be used to explain it. OF1 is used for the reported values and OF2 is computed as well, as a reliability check.

The index follows Kornienko et al. Spike times are converted to an instantaneous firing rate - counts in 2 ms bins, smoothed with a Gaussian of 5 ms standard deviation - and a power spectrum of that rate is estimated by Welch's method, using 2048-sample segments at 500 Hz with half overlap, which resolves 0.24 Hz and so separates the theta band cleanly from the flanks it is compared against. The index is then

T = (q - baseline) / (q + baseline)

where q is the mean power in the theta band, 6-10 Hz, and baseline is the mean power across the two adjacent bands, 3-5 and 11-13 Hz. Being a contrast between a band and its own flanks, T is bounded between −1 and +1, is dimensionless, and does not scale with firing rate - which matters here because the comparison runs across cell classes whose rates differ several-fold, putative interneurons firing roughly five times as fast as principal cells. The frequency of the theta-band peak is stored alongside the index; its median across entorhinal cells is 8.5 Hz, which is a check that the measure finds theta rather than some other rhythm.

Reliability. Scored independently in the two open-field sessions of the same day, the index agrees at Spearman rho = 0.865 across 11,810 cells.

No shuffle null, and no threshold. The question the index is used for - whether theta-rhythmic cells follow the population axis more than others - is better asked of a continuous variable than of a binary split at an imported cut-off, so the index enters a mixed model with random intercepts for mouse and for session within mouse. A median split is much weaker on these data (p = 0.085 pooled) because it discards the within-session variation that carries the effect, and is not used.

A note on which index. An earlier version of this analysis used the other standard form, the power spectrum of the spike-train autocorrelogram scored as peak power in 5-11 Hz over mean power in 0-50 Hz. The two indices are not interchangeable: the autocorrelogram form is a peak-to-broadband ratio rather than a contrast against the immediately adjacent bands, is unbounded, and gives a cell with a wide low-frequency shoulder a high score without theta rhythmicity. It is also the less reliable of the two here, agreeing between open-field sessions at rho = 0.746 against 0.865, and it reverses the conclusion: under it the within-class relationship disappears entirely.

### Clustering position profiles by shape

Populations of cells are grouped in Figure 6 by the SHAPE of their mean firing profile against track position, not by where that profile peaks. Sorting a set of profiles by peak position imposes a diagonal on the resulting image whether or not the population has any structure, so a sorted heat map cannot be used as evidence that structure exists; clustering lets the groups declare themselves, and the dendrogram shows how separable they are.

Each cell contributes one profile: its trial-averaged rate against position, smoothed and then min-max normalised to [0, 1] so that cells with the same field shape at different firing rates group together and the clustering is not dominated by a handful of high-rate cells. Profiles are compared with correlation distance, 1 − r between two profiles, which again is a distance between shapes rather than between magnitudes. Agglomerative clustering uses average linkage, chosen over single linkage, which chains, and over Ward, which assumes Euclidean geometry that correlation distance does not provide. The tree is cut at a fixed four clusters in every panel, so that the three populations compared in one row are cut identically and any difference between them is a difference in the data rather than in the cut.

Cluster identity is assigned by the order in which clusters first appear reading the dendrogram leaves from top to bottom, so the top block is cluster 1, and the same colour is used for a cluster's branches, its block of the heat map, its mean profile and its box in the correlation panels beside it. Branch colours are set explicitly from the cluster labels rather than through a colour threshold on the linkage, because a threshold and a cluster cut are different operations and can disagree about which branch belongs to which group.

Cluster means are drawn with the standard error over member cells, each on its own axes rather than sharing one, so that a flat cluster reads as flat instead of being stretched to fill a band. A fixed cut at four produces one or two near-singleton clusters in some panels; their n is printed beside every mean, and no claim rests on a cluster of fewer than about twenty cells.

### Single-unit identity and its relation to the population state

Identity comes from the open field, by design. Every cell-type label used in this figure - grid, non-grid spatial, non-spatial - is taken from the open-field recordings (cell_class_of1), and the speed score likewise from open-field running, never from the virtual-reality session under analysis. The linear-track task is the setting whose dynamics are in question: firing there varies with engagement, motivation and task demand across a session, and those are precisely the variables the anchoring state tracks. Classifying cells on the same data would let the effect define the categories and then recover itself. Assigning identity in a different environment makes the categories independent of the state they are used to explain, at the cost of any cell whose tuning differs between environments being mislabelled for the track - a cost we accept as the conservative direction, since it can only blur differences between classes, not manufacture them. The one exception is the interneuron call below, which is derived from the spike waveform and is therefore environment-independent.

Putative interneurons. No interneuron label exists for this dataset, so one was derived from the spike waveform. A two-component Gaussian mixture was fitted to log peak-to-trough duration and the threshold taken as the antimode, rather than set by convention. The distribution is convincingly bimodal - component means 0.246 and 0.690 ms, separated by 4.03 SD - giving a threshold of 0.423 ms. Firing rate was fitted the same way and is not comparably bimodal (components 2.3 and 8.8 Hz, 1.81 SD apart), so rate cannot define the group and is used only as a second requirement: putative fast-spiking cells are narrow-waveform and in the top quartile of firing rate (>= 10.2 Hz). This yields 842 of 1,495 narrow cells; the 653 narrow-but-slow cells excluded are an acknowledged ambiguity. As an independent check on a property they were not selected for, putative interneurons are the most speed-modulated group in the open field ($p = 7 \times 10^{-31}$).

Open-field speed score. Each session ships a time-shifted speed GLM (time_shifted_S_glms.parquet), and it is not used: its median score is negative for nearly every cell and only ~1% of entorhinal cells reach FDR significance, so a null result against it could not distinguish absent speed modulation from an insensitive statistic. Speed modulation was instead computed directly, in the standard form: the Pearson correlation between a cell's instantaneous firing rate and running speed in 200 ms bins (boxcar smoothed over 2 bins) restricted to moving periods (>= 2 cm/s), tested against a circular-shift null of the full rate trace (200 shifts). This reaches significance in ~51% of cells and replicates across OF1 and OF2 at r = 0.75, which is what licenses the negative results that rest on it. OF1 is used where both sessions exist, with OF2 retained as the reliability estimate rather than averaged.

Dissent and locking. A cell can disagree with the population in two ways that the correlation cannot separate - it can be anchored when the population is not, or non-anchored when the population is. Both were computed separately: stay = P(cell anchored | population non-anchored) and drop = P(cell non-anchored | population anchored), each requiring at least 5 trials in the conditioning state. Cells whose labels do not vary at all (zero label SD, i.e. the absolute gate assigned every trial to one class) have an undefined correlation and are analysed separately as locked. Analyses of locking are restricted to sessions where the population itself switches (>= 10 trials in each state, 35 of 62 sessions), because in a single-state session every cell is locked by construction; the restriction strengthens the reported asymmetries rather than producing them.

What a non-anchored trial contains. Each trial's smoothed map was cross-correlated against the cell's anchored template - the mean map over its anchored trials only, so the template is not contaminated by the trials scored against it - at every circular offset, recording the correlation at zero offset, the best correlation over offsets, and its offset. Cells with fewer than 8 anchored trials were not given a template. The best-over-offsets statistic is a maximum over 100 offsets and is therefore large by construction, so two nulls were computed on a 10-session subset: a bin-shuffled trial map, which destroys spatial structure while preserving the rate distribution (+0.189),

and a different cell's template from the same session, which preserves real spatial structure on both sides (+0.500). The second is the operative floor, and reporting the best-offset correlation without it inverts the conclusion.

Firing rate by state. Rate was computed two ways, and the distinction matters. A rate averaged over the position bins of a rate map divides by occupancy, which running speed sets; a rate computed as spike count divided by running duration does not. The two correlate at r = 0.96 across cells but differ in mean (6.84 against 9.03 Hz), which is enough for a ~3% state difference to change sign between them. Rates are reported time-averaged, over running samples only, with trial duration also taken as running duration so that a trial on which the animal stopped often is not scored as low-rate for that reason. Speed matching within session, in deciles of trial mean speed with equal counts per decile, is reported alongside. Every split uses the population state, never the cell's own label: a trial with more spikes yields a better-estimated map, which is more likely to match the others, which earns the anchored label - scored that way the effect reverses.

Agreement between identity groups. Whether two identities are coordinated with each other, rather than each independently tracking the population, was tested by correlating every pair of cells' label sequences within session and averaging by identity pair. The diagonal is the mean within-group pairwise correlation excluding self-pairs. Because two cells that are each anchored on most trials correlate for that reason alone, every entry is reported as excess over a null in which each cell's sequence is circularly shifted independently (20 draws per session), which preserves each cell's anchored fraction and run structure and destroys only the alignment between cells. Restricted to sessions in which the population switches and to cells whose labels vary; locked cells have no correlation to compute. The matrix is symmetric by construction and is cached to identity_agreement.csv.

Switch timing. Implemented. Each cell's label sequence was cut in a +/-6-trial window around every population state transition and oriented so that positive denotes entering the anchored state; the cell's switch point is the steepest rise of its averaged profile and the lag is that point minus the population's. Transitions were found on the population state recomputed without the cell in question. Note that "do cells lead the population" is not a testable question as posed: the population state is the fraction of cells anchored thresholded at one half, so it crosses when half the cells have switched and ~50% of cells must lead whatever the biology. Only the difference between identities is informative, and it is tested by likelihood ratio between mixed models with and without an identity term; it is not significant (p = 0.15). Figure 2F shows the aligned profiles over +/-5 trials.

Reliability of the agreement score. Trials were split into odd and even - which keeps the halves matched for slow within-session drift in a way a first/second split does not - and the agreement score recomputed within each half using the leave-one-out population state of that half. The split-half correlation across cells is 0.939, or 0.969 after the Spearman-Brown correction for full-session length, and is uniform across identities (0.952-0.985). Attenuation of the identity comparisons is therefore negligible, and no identity is measured appreciably less reliably than another.

Anatomical axes. coord_SCs_x is the mediolateral coordinate (stored negative; magnitude 2610-3774 μm, and the axis on which the 3400 μm medial/lateral boundary is defined), coord_SCs_y is dorsoventral and coord_SCs_z is anterior-posterior. Positions were z-scored within mouse before modelling, since animals differ in probe placement and raw coordinates would let between-animal placement appear as a within-animal gradient. Each axis was tested alone and with cell identity added, since identity is itself anatomically non-uniform.

Hierarchical statistics. Cells are nested in sessions in mice, so comparisons over cells treat correlated observations as independent. All identity and anatomy comparisons in this figure are linear mixed models with random intercepts for mouse and for session within mouse, with pairwise contrasts Holm-corrected. The cell-level rank tests were four to eight orders of magnitude smaller, and one conclusion changes: grid cells and putative interneurons differ over cells but not once the hierarchy is modelled (p = 0.54).

**Putative monosynaptic connections**

Reported in Figure 4.

Implemented. Spike trains were cross-correlated in 1 ms bins over +/-50 ms and compared against a hollow-Gaussian predictor (sigma = 5 ms, 60% hollow) that preserves slow co-modulation while removing the millisecond structure a synapse produces. Detection is a Poisson test with continuity correction on the causal window 0.7-4.7 ms, Šidák-corrected across the bins examined, with the zero-lag bin excluded and minimum spike counts enforced on both cells.

This is the Stark & Abeles / English et al. convolution-predictor family, and differs from the common variant that convolves with a plain Gaussian and thresholds each bin at a fixed percentile in three respects. The predictor is hollow - 60% of the kernel mass at the centre is removed - so a genuine peak does not contribute to the baseline it is tested against; with a plain kernel a strong peak inflates its own predictor and the test becomes conservative in exactly the cases of interest. Multiplicity is handled by Šidák correction across the causal bins rather than by requiring consecutive significant bins; at 1 ms binning the two are close to equivalent, since the usual "two consecutive 0.5 ms bins" rule asks the deviation to span 1 ms, which one of our bins already does. And the per-bin alpha is $3 \times 10^{-4}$ rather than the conventional $10^{-3}$, set from a jitter null rather than by convention (below).

Rejection rules and their correction. The published criteria reject a pair if any non-peak bin exceeds 2.5 SD, or if any anticausal bin reaches $p < 0.01$. Applied per-bin and uncorrected across the 101 bins of the window, these fire by chance on ~45% and ~39% of pairs respectively and together reject roughly two thirds of all pairs regardless of content - on one test session they returned zero connections in 17,366 pairs, discarding pairs with z = 10.4 causal peaks. Both rules are given the same Šidák correction the detection receives, across the number of bins each actually examines. Nothing else about the criteria was changed.

Setting the threshold from a null, not from convention. Relaxing a rejection can only increase the count, so a higher count is not evidence the correction is right. The operating point was set from a jitter null: the same spikes displaced by +/-10 ms, which destroys millisecond timing while preserving firing rate, slow co-modulation and the coarse correlogram shape. Sweeping the peak threshold on one session (20,052 pairs), alpha = $3 \times 10^{-4}$ is where the null is essentially empty while the real count is still near its maximum. Across the full dataset that threshold gives a false-discovery rate of 0.09 for excitatory and 0.10 for inhibitory connections, estimated from the jitter null run in every fourth session (15 of 60).

Inhibitory connections. The criteria as published detect a causal peak and therefore excitatory connections only; an inhibitory connection appears as a trough and cannot be found by them at any threshold. detect(kind='inh') mirrors every criterion onto the lower tail of the same Poisson test, taking the causal extremum as a minimum, with the rejection rules, corrections, window and minimum counts unchanged, so the two signs are held to identical standards and each carries its own jitter null. Two properties external to the detector validate the result:

inhibitory latencies are slower than excitatory ones (median 3.0 against 1.0 ms), and inhibitory connections originate from waveform-classified interneurons at 3.1x their population share while excitatory connections do not - the waveform classification played no part in detection.

A third check addresses the specific reason inhibitory detections are often discarded, namely that they are the far side of an excitatory connection in the opposite direction, produced by a short-timescale oscillatory process rather than by a synapse. That pattern is essentially absent here: of 7,775 inhibitory connections, 19 (0.24%) have a detected excitatory connection in the reverse direction and 7 (0.09%) in the same direction. The +/-10 ms jitter null speaks to the same concern from the other side, since it preserves the coarse correlogram shape and any slow co-modulation while destroying millisecond timing.

Connection probability by identity. Reported as a percentage of the ordered pairs actually testable, counted per session from the number of cells of each identity present, rather than as a share of detected connections, so that a group's abundance does not inflate its apparent connectivity. Enrichment was tested by permuting identity labels among the cells of each session (2,000 shuffles), which holds cell counts, connection counts and probe geometry fixed and moves only the labelling.

Do connected pairs share the anchoring state? Every within-session pair of cells whose labels vary was scored by the Pearson correlation of their label sequences (>= 20 shared trials). Connected pairs were compared against unconnected pairs drawn from the same session, of the same identity pair, at a similar probe distance (+/-40 μm, up to 20 controls per connected pair). All three constraints are necessary: a monosynaptic peak is only detectable between cells the probe records well, which are usually neighbours; neighbouring cells share anchoring through common input, shared local field and spike-sorting bleed, which manufactures the connection and the correlation together; and identity alone predicts agreement, so an unmatched comparison would recover the identity effect of §2 and report it as a wiring effect. Cells with constant labels are excluded throughout - under the absolute gate about a third of cells are all-anchored or all-non-anchored, and a constant vector has no correlation with anything. Comparisons are Mann-Whitney on connected against pooled matched controls. Cached to
data/monosyn/pairs_by_identity.csv.

Power, and which nulls are reportable. These comparisons are the paper's only substantive null, so each is accompanied by the minimum effect it could detect (two-sided alpha = 0.05, 80% power, pooled SD). The benchmark is the effect the null is denying - the identity effect of §2, ~0.11. Grid-interneuron (n = 193, MDE 0.045) and interneuron-interneuron (n = 116, MDE 0.043) resolve effects roughly a third that size, so their nulls carry information; the grid-interneuron result holds even under the conservative assumption that the matched controls contribute no more information than the connected pairs themselves (MDE 0.060). Grid-grid does not: 27 connected pairs give MDE 0.157, larger than the effect being tested for, so its p = 0.31 is reported as underpowered and is excluded from the claim. The remaining heterogeneous pool reaches significance (p = 0.032) on an effect of +0.006, an order of magnitude below the identity differences in the same panel and resolvable only because it contains 3,238 pairs; it is not interpreted as a wiring effect.

### Eye tracking: pupil size

Reported in Figure 4 ( -> fig5_pupil_arousal_M21D19.pdf).

The eye contralateral to the probe was filmed at 30 Hz during virtual-reality sessions and the pupil tracked with DeepLabCut, giving eight perimeter landmarks (n, ne, e, se, s, sw, w, nw), each with a likelihood, plus a derived centroid. Eye tracking is available for 80 of 81 recorded virtual-reality sessions.

DeepLabCut training. A single network was trained across all animals rather than one per mouse: 200 frames were labelled from one session per animal, and the pooled set used to train one model. Training on every animal together means the same network, with the same learned notion of where the pupil boundary lies, is applied to every session, so a between-animal difference in pupil size cannot arise from a between-animal difference in the labeller or the model. It also avoids a per-animal model overfitting the handful of sessions it was trained on. Each frame carries the eight pupil-perimeter landmarks described below, and the per-frame likelihoods the network returns are used as the tracking-quality criterion.

Pupil size. Pupil size is the mean Euclidean distance from the eight perimeter landmarks to their centroid. The centroid was verified to be exactly the centroid channel supplied in the file (r = 1.00000, maximum absolute difference 0.0000 px). We use this landmark-derived radius rather than the eye_dilation channel supplied alongside it, because that channel is not reproducible from the landmark geometry - it correlates with the landmark-derived mean diameter at only r = 0.56-0.86 across sessions and is approximately twice its magnitude, so its definition cannot be recovered from the data and it cannot be audited.

Blink and tracking-loss rejection. Frames whose mean landmark likelihood falls below 0.5 are discarded (typically 1-3% of frames; sessions exceeding 20% are excluded entirely). The centroid itself carries no likelihood, being derived, so the mean over the landmarks it is computed from is used instead.

Head-movement artifacts. Although the animal is head-fixed, the head can shift within the headplate, and tracking can fail transiently. Frames in which the landmark centroid departs from its local baseline (median over a +/-2 s window) by more than 0.8 pupil radii and returns within a few frames are discarded: these are tracking failures and partial occlusions by the lid or a whisker, and they corrupt the landmark geometry from which the radius is computed. Thresholds are expressed in units of each session's own median pupil radius rather than in pixels, because camera distance and zoom differ between sessions (median radius 7.8-12.7 px), so a fixed pixel threshold would correspond to a different physical displacement in each.

The pipeline additionally detects sustained steps, in which the head slips to a new position within the headplate, and subtracts the cumulative offset. That correction does not affect any result reported here, because pupil radius is the mean distance from the perimeter landmarks to their own centroid and is therefore invariant to translation of the eye within the camera frame. It is retained in the code, and described here only so that the processing applied to the recordings is on record.

Per-trial measures. Pupil radius is binned onto the same trial x position-bin grid as the neural rate maps (2 cm bins, moving epochs only), so that each column corresponds to the same stretch of track in both. Two per-trial quantities are then formed:

- Pupil radius: the mean binned radius over the trial. Values are z-scored within session before pooling across sessions, because the raw pixel scale depends on camera placement and is not comparable between sessions.

Session inclusion. A session contributes if fewer than 20% of its frames fail the likelihood threshold, it retains at least 40 trials with position-bin coverage of 0.5 or better, and - the operative criterion - it has at least 30 trials in each of the anchored and non-anchored classes. Fifty-seven of eighty sessions qualify.

The per-class minimum was calibrated rather than assumed. We took the 24 sessions with at least 90 trials in both classes, subsampled them to n trials per class, and measured the

spread of the resulting estimate around that session's own full-data value. The estimator is unbiased at every n (|bias| <= 0.01 z), so the criterion concerns precision alone:

Against an effect of approximately 0.55 z, a session with 20 trials per class has a 95% error band of +/-0.55 - as wide as the effect itself, so such a session cannot establish even the sign of its own estimate. At 30 per class the SD is about half the effect, which is where a per-session value becomes individually interpretable; 40 per class buys little further precision at the cost of five more sessions.

The pooled result does not depend on this choice: the median session effect runs from −0.561 with no per-class minimum to −0.507 at a minimum of 50, with $p <= 3 \times 10^{-9}$ throughout. The threshold determines which individual session values may be read, not whether the effect exists. It does, however, matter for the per-session picture, because the excluded sessions carry the largest magnitudes (down to −2.2 z) - the expected signature of noise inflating |effect| in small samples, and the reason the unfiltered per-session distribution has a heavier tail than the effect warrants.

Track profiles. To locate where along the track the states differ, the binned quantities were averaged across trials within each state and within session, and then across sessions, retaining the position axis. Profiles are reported both in absolute terms - alongside the all-trial mean, computed across every trial rather than as a weighted average of the two states, since the two classes do not share the same valid bins - and as each state's departure from that all-trial mean. The second form is necessary because the position profile is much larger than the state effect: pupil radius varies by approximately 2.5 z across the track, against a between-state difference of 0.24, so on an absolute axis the two states are visually indistinguishable.

Video frames and their alignment. Example figures showing the eye require individual video frames, and these align to the behavioural data exactly, with no synchronisation step: for each session the number of video frames, the number of rows in the camera capture log, and the number of eye samples in the NWB file are identical (58,925 for M21 D19, 62,634 for M28 D18), all at 30.000 fps. Video frame i is therefore eye sample i, and the NWB timestamps that sample on the ephys clock.

Two details cause silent errors if overlooked. First, trial numbering: the anchoring analysis uses trials clipped to the ephys window and renumbered from 1, whereas the NWB trials table retains the raw numbering (M21 D19's clipped trials are raw 10-116), so an unmapped lookup is displaced by nine trials. Second, coordinate frames: the DeepLabCut landmark coordinates in the NWB are relative to a cropped region, whose origin is given per session in all_eye_crops.csv. Files named *_eye_zone.avi are already cropped, but more tightly than that table records, so their landmarks require a further fixed offset. That offset was solved from the data - matching the centroid of the dark pupil to the NWB centroid across sampled frames - and verified visually before use: (−30.4, −31.4) px for M28 D18 and (−31.0, −31.7) px for M29 D17, with a between-frame SD of under 2 px in each case.

Error bars on position profiles. Pupil radius binned by track position is summarised as the mean +/- SEM across trials, not across camera frames. Frames within a trial are highly correlated - the pupil changes little in 33 ms - so treating the several hundred frames contributing to a position bin as independent understates the standard error by a factor of two to three (0.03 px against 0.11 px in a representative session). The trial is the independent unit.

Bin-occupancy correction. The tuning-curve routine used for binning computes bin occupancy from the dense (~60 Hz) position trace but averages the sparse (~30 Hz) eye-camera samples, and initialises empty bins to zero rather than to NaN. A position bin the animal visited but which contains no eye-camera sample therefore retains a placeholder value of exactly 0. Since a true pixel coordinate of 0 is impossible here (raw values never fall below ~90), such bins are identified by exact equality with zero and set to NaN. Approximately 3-4% of bins are affected.

Peri-transition analysis. To test whether pupil radius tracks the anchoring state itself rather than some session-long covariate of it, trials were aligned to the first trial of a new population state, separately for entries into the anchored and into the non-anchored state.

Windows were required to be state-pure: the population state must be constant across the whole of the four trials preceding the transition and constant across the four trials following it. This is stricter than requiring a minimum run length either side, and the distinction is consequential. With a three-trial run criterion the state outside that run flickers substantially - eight trials before a transition, 46% of trials are already in the state being entered, and only 15% of windows are free of flicker across +/-8 trials. Because that contamination pulls the pre-transition plateau toward the post-transition value, it both understates the total change and overstates the fraction of it occurring in a single trial: entry into the non-anchored state appears as a 95% one-trial step under the run criterion, but as 31% of a change twice as large on state-pure windows. A window of +/-4 trials retains 196 and 198 transitions (entries into the anchored and non-anchored state respectively) across 56 sessions; +/-6 trials would retain fewer than half as many.

Segments were averaged within session before averaging across sessions, because switch rate varies several-fold between sessions and pooling raw transitions would weight sessions by how often they switched. Pre- and post-transition means were compared by Wilcoxon signed-rank test across sessions.

On state-pure windows pupil radius changes in the expected direction at the transition - −0.414 entering the anchored state and +0.493 entering the non-anchored state ($p = 1.7 \times 10^{-7}$ and $4.4 \times 10^{-8}$) - and does not change instantaneously: the single trial spanning the transition accounts for only 16-31% of the total change. The classifier's size-5 median filter smooths the label sequence over +/-2 trials, so the data cannot distinguish a genuinely graded change from an abrupt one whose detected onset is imprecisely timed, and we make no claim about which it is.

Pupil size against performance, and the test for an inverted U. Whether pupil size indexes drowsiness at one end and distraction at the other - the classical inverted-U - or varies monotonically with engagement over the range an animal actually occupies is an empirical question, and it decides how the anchored state should be described. Trials from eye_anchoring_trials.csv were merged with the behavioural trial table (14,011 trials, 58 sessions) and pupil radius z-scored within session, since absolute radius differs with camera placement. Hit rate was then computed by decile of within-session pupil z, and modelled as a mixed model with a session random intercept, with and without a quadratic term in pupil z. A genuine inverted U requires the quadratic coefficient to be negative with a turning point inside the data; here it is positive (+0.018, $p = 3 \times 10^{-15}$) with a minimum at z = +3.4, so within the observed range the relationship is monotonically decreasing (−0.123 per z). Because pupil size and running speed covary in general, the model was refitted with within-session z-scored trial mean speed as a covariate; mean speed is flat across pupil deciles (37-40 cm/s) and the pupil coefficient is unchanged (−0.128, quadratic +0.011), so the relationship is not a behavioural artifact. The same analysis restricted to uncued trials gives the same signs.

### Anchoring Dependence Index

The Anchoring Dependence Index (ADI) quantifies how much more the population anchoring state matters to behavioural success when the reward zone is not cued - that is, when the animal must locate it by path integration rather than by sight.

Definition. For each session, the per-trial population anchored fraction (the proportion of simultaneously recorded spatial cells labelled anchored on that trial, from the classifier above) was used to discriminate rewarded from unrewarded trials, separately within each trial type, and the discriminability quantified as the area under the ROC curve. ADI is the difference

Positive values mean the anchoring state carries more information about success when the reward zone is not cued. Sessions entered the analysis if they had at least 25 classified spatial cells and at least 25 trials of each type with both outcomes represented; 78 sessions qualified.

Why AUC rather than correlation. The obvious metric - the correlation between anchored fraction and trial outcome - is the wrong one here, because the two trial types have very different base rates (median hit rate 0.85 cued, 0.55 uncued). With little outcome variance to explain, r on cued trials is attenuated for reasons that have nothing to do with anchoring. AUC is formally invariant to class prevalence and does not have this defect. Consistent with that, the correlation-based index is roughly 50% larger than the AUC-based one (+0.148 vs +0.100) and correlates twice as strongly with the cued hit rate (rho = +0.449 vs +0.221), which is the signature of the artifact rather than of a larger effect.

Result. Across all 78 sessions the anchored fraction discriminated outcome better on uncued than on cued trials: median AUC 0.733 uncued against 0.593 cued, ADI = +0.100, Wilcoxon signed-rank $p = 1.0 \times 10^{-4}$, positive in 52 of 78 sessions.

Caveat - the measure is unreliable near ceiling and floor. Prevalence invariance is a property of AUC as an estimand, not a guarantee about its estimate. When performance on a trial type approaches ceiling or floor, the AUC for that type rests on a handful of trials of the minority outcome, and those trials are a selected sample: the few failures in a session with an 0.85 hit rate are unusual events - a disengaged trial, a missed stop on an otherwise correct approach - not representative ones. The quantity being estimated is then no longer comparable between the two trial types, and since cued trials sit closer to ceiling than uncued trials in almost every session, this asymmetry runs in the same direction as the effect. The all-session ADI is positively correlated with the cued hit rate (rho = +0.221, p = 0.052), which is what this concern predicts.

Control - restricting to sessions with power on both trial types. We therefore repeated the analysis on sessions where both trial types had hit rates within [0.2, 0.8], so that both AUCs are estimated from a substantial minority class. This is a power criterion, not a data-quality one: no session is excluded for being badly recorded. Fifteen sessions met it, and the effect was unchanged in size and remained significant - cued AUC 0.626, uncued AUC 0.701, ADI = +0.103, p = 0.0012, positive in 13 of 15. The result is not an artifact of asymmetric ceiling effects. It is also not sensitive to where the bounds are placed: [0.15, 0.85] gives ADI +0.110 (n = 24, $p = 3.2 \times 10^{-4}$) and [0.1, 0.9] gives +0.110 (n = 39, $p = 3.5 \times 10^{-5}$).

The criterion must be applied to both trial types. Restricting only on the cued hit rate (< 0.8), which is the intuitive move since cued trials are the ones near ceiling, admits sessions where the uncued hit rate is near floor and gives ADI = +0.035 (n = 31, p = 0.23) - the effect appears to vanish. This is the same failure at the other end of the scale: a session in which the animal almost never succeeds on uncued trials supplies an uncued AUC estimated from a few atypical successes.

Prevalence matching was considered and rejected. The obvious alternative to excluding sessions is to subsample the majority outcome within each session so that the two trial types have equal hit rates. This does not work, and we state it because it is the first thing a reader will propose. Matching leaves ADI essentially unchanged (+0.103 matched vs +0.100 raw, $p = 8.5 \times 10^{-5}$, positive in 41 of 60 sessions) and leaves its correlation with the cued hit rate slightly worse (rho = +0.298 matched vs +0.221 raw, p = 0.021). Equating the rate does not equate what a miss is: a miss drawn from a session at 0.85 remains an atypical event after subsampling, because subsampling discards hits and cannot manufacture representative misses. The inclusion criterion addresses the problem at its source; matching only rescales it.

Regional specificity. The index is not specific to the medial entorhinal cortex. Recomputing the population anchored fraction from MEC cells alone, in the 59 sessions where both an all-cell and an MEC-only population state could be formed, gives ADI = +0.101 (p = 0.001, 37/59) against +0.099 for all cells in the same sessions; the paired difference is null (p = 0.81, MEC larger in 23 of 59). This is reported as a negative result, not a control that passed.

### xgBoost encoding models

Neuronal firing was modelled with gradient-boosted regression trees (XGBoost) configured for Poisson output, using the MLencoding framework of Benjamin et al. (2018).

Binning and covariates. Spike trains and behavioural variables were binned at 10 ms. The base covariate was track position (the animal's cumulative travel modulo the 200 cm track length). Neural covariates were the binned spike counts of simultaneously recorded cells. All covariates were supplied with temporal history: each covariate was projected onto a basis of raised-cosine temporal filters spanning the preceding 1000 ms (100 filters at 10 ms resolution). Spike history of the target cell itself was not included, so that predictions cannot be trivially satisfied by autocorrelation.

Model parameters. Objective count:poisson; learning rate 0.05; maximum depth 5; 580 estimators; subsample 0.6; minimum child weight 2; gamma 0.4.

Cross-validation. Models were fit with 10-fold cross-validation using contiguous folds: the recording was divided into 10 consecutive blocks, each serving once as the test set with the remainder as training data. Contiguous rather than randomly interleaved folds are required here because the temporal history filters span 1000 ms, and random bin assignment would leak training data into the test set.

Scoring. Model performance was quantified as the Poisson pseudo-$R^2$

where $L_?$ is the log-likelihood of the model, $L_?$ the log-likelihood of a homogeneous-rate null model (mean firing rate of the training data) and L_S the log-likelihood of the saturated model. A value of 0 indicates performance equal to the mean-rate null and 1 indicates a perfect fit. Where performance is reported for a subset of trials or for a behavioural condition, $pR^2$ was recomputed on the time bins belonging to that subset using the same fitted predictions and the session-wide mean rate as the null, so that values remain comparable across conditions.

Covariate populations. For the analyses of coordination, covariate cells were drawn from a defined population (co-modular grid cells, non-co-modular grid cells, or non-grid spatial cells) and added incrementally, ordered by physical distance from the target cell (nearest first, farthest first, or in random order as stated for each analysis). Distance was computed in the probe-plane coordinates (coord_SCs_x,

coord_SCs_y) of the reconstructed recording sites. Reported comparisons are made at matched numbers of covariate cells.

Medial and lateral covariate sets. For the mediolateral analyses, non-grid spatial cells were divided into medial (coord_SCs_x < 3400 μm) and lateral (coord_SCs_x >= 3400 μm) populations, and each was used as the covariate set in otherwise identical models.

**Attractor state on the torus, and whether it survives the non-anchored state**

The account in Figure 8 and the simulation in Figure S13 both assume that the non-anchored state is a continuous attractor still in motion and merely uncorrected, rather than one that has come apart. The two possibilities differ in how concentrated the module's activity is on the torus, which is what this measures.

Module and phase offsets. Modules here were identified spectrally, within the VR session, by spectral_analysis in spatial_manifolds.toroidal - the routine used for the toroidal figures - and not by the open-field UMAP/DBSCAN procedure described under Grid module identification above. The two are distinct and are not interchangeable: the quantity needed is each cell's phase offset on the module's lattice in the task, which only the spectral decomposition provides. Each cell's unrolled trial-by-position rate map, Gaussian-smoothed (sigma = 2.5 bins) and z-scored, was transformed to a spectrogram; cells were clustered on those spectrograms; and the module's three lattice axes were taken from the two most prominent spectral peaks and their beat. Each cell's offset on each axis is the circular mean over the windows whose phase pattern agrees with some other window. An axis for which no window passes that test yields no offset and is dropped, and a session in which no axis survives is excluded entirely - this applies to the whole session and so cannot favour either state. Position maps were restricted to the trials lying wholly inside the ephys window, since a block of empty trials places spectral content where there is no recording.

Two axes, not three. A bump on a two-dimensional sheet has two degrees of freedom, and the three offsets returned are not independent: the third is the beat of the other two. Following Gardner et al., the two axes whose phases are most consistent across windows were kept and the third discarded, consistency being the proportion of windows whose phase pattern agrees with some other window. This selection also runs on the whole session and cannot favour either state. Using all three axes instead changes nothing material: anchored +0.0611 against +0.0610, non-anchored +0.0421 against +0.0412, on the same 24 sessions.

The attractor state, moment to moment. Rates were binned at 100 ms, Gaussian-smoothed (sigma = 2.5 bins, i.e. 250 ms) and z-scored per cell, as the toroidal figures prepare them. Following the same projection the toroidal figures use, the population was projected onto the cells' phase offsets, and the attractor's state on axis j at time t is the argument of Σ_c r_c(t).exp(i φ_{j,c}). How concentrated the population is - the bump's population vector - is the modulus of that sum divided by the population's own norm, which is 1 when every cell fires in register with a single bump and 0 when the pattern bears no relation to the lattice. Values were averaged over the axes that resolved. Only samples with running speed above 3 cm/s, the task's own stop criterion, were used.

The null is a phase shuffle, and the comparison is the excess over it. A population vector grows with the number of cells and with how uneven the firing rates happen to be, and firing rates differ between the two states. Shuffling which phase offset belongs to which cell (100 permutations) leaves every rate in the session untouched and destroys only the lattice organisation, so the difference between the real projection and the shuffled one isolates the part of the bump attributable to the attractor. All reported quantities are this excess. A bare resultant is not usable here: it is high whenever each cell's phase is merely stable, and a state in which maps are more stable is what anchoring means, so it would restate the premise rather than test it.

Speed matching. Samples were sorted into ten speed bins with edges at the deciles of the pooled distribution, and the same number drawn from each state within every bin, averaged over 40 draws. The population vector is already normalised by the population's own norm, so a uniform change in rate cancels; this removes any residual.

Smoothing series (reported here, not plotted). The whole analysis was repeated with the temporal smoothing set to 100 ms and to none, since a bump that drifts while unpinned is smeared by smoothing and would appear weaker for that reason alone. Retention was compared across smoothing levels by a paired Wilcoxon test over sessions, the three levels being the same sessions.

Comparison with the simulation (reported here, not plotted). The identical measure - 100 ms bins, the same smoothing, z-scored per unit, projected onto the ring phase and referred to the same phase shuffle - was applied to the simulated population of the model in Figure S13, in both gate states. Only the proportion of the bump retained is compared between model and recordings; the absolute values are not comparable, since the model has 128 noiseless units evenly tiling the ring against a median of 18 recorded cells.

Notation. Three symbols are involved and they are not interchangeable. φ are the per-cell phase offsets: cell i has one offset φ_a,i on each lattice axis a, fixed for the session, and they are what the population is projected onto. ψ are the population's firing-rate-weighted mean phases, ψ_a(x) = atan2(Σ_i sin φ_a,i . r_i(x), Σ_i cos φ_a,i . r_i(x)), one per axis, which move from moment to moment. theta□ and theta□ are the bump's centre of mass on the unit cell of the sheet - a population quantity, obtained from ψ by the transformation below, which carries the three ψ to the two degrees of freedom a bump on a sheet actually has. theta and ψ therefore describe the same thing, the population, in two coordinate systems; theta is used here because it is the coordinate system of the sheet itself, which panels D and E are drawn.

A fourth symbol belongs to the per-cell side and is easy to confuse with theta: applying the same transformation to each cell's offsets, s□,i = φ□,i − φ□,i/2 and s□,i = (sqrt3/2).φ□,i, gives that cell's fixed location on the sheet. The sheet displays bin activity into those locations. So φ and s describe where the cells sit, ψ and theta where the population's bump sits, and only the latter pair varies with position and trial in the way Figure S14B and S14C show.

The sheet coordinates, and which axes they use (Figure S14B-E). The population's firing-rate-weighted mean phase on each lattice axis, ψ_a, was carried to the two coordinates of the unit cell by the transformation of Gardner et al., theta□ = ψ□ − ψ□/2 and theta□ = (sqrt3/2).ψ□; Panels B and C show ψ, not theta. The shear is a planar coordinate system for the sheet, which is what the displays in D and E need; a phase along one cycle of the torus is not improved by it. It would also be applied unevenly: for theta□ it is a pure scale that drawing one full period undoes exactly, so that panel would be unchanged, while theta□ would mix the two axes and no longer correspond to the cycle its icon shows. All four display panels use the two lattice axes that advance along the track, which need not be the two the measure uses: one axis of a hexagonal lattice can lie nearly perpendicular to a linear track, and then it carries no positional signal and only injects noise into theta□, which mixes the two. In M25 D24 the three axes advance by −1.40, −0.00 and −1.34 revolutions per trial, so the stationary one is dropped. The quantitative measure keeps the two

best-resolved axes, as above; the two rules give nearly the same answer, since all three axes here are well resolved (window consistency 0.97, 0.93 and 0.91) and dropping to any two moves the result by less than 0.001.

The path, unwrapped (Figure S14D). The phases of each trial were unwrapped along the track with np.unwrap before the transformation, following Gardner et al., who note that unwrapping is for visualisation only and enters no analysis; the same holds here. Twelve trials were sampled evenly from the anchored epoch preceding the chosen switch and twelve from the non-anchored epoch following it; the dashed lines are the unit cell, and the marked points are the mean start and end across the sampled trials.

The bump on the sheet (Figure S14E). The two phases were carried onto the rhombic unit cell of a hexagonal lattice by the transformation of Gardner et al., s□ = φ□ − φ□/2 and s□ = (sqrt3/2).φ□, applied to each cell's offsets. The bump traverses the sheet as the animal runs, so an average over a whole trial smears it over the whole unit cell; each frame is therefore one trial at one place on the track, 10-30 cm. With a few dozen cells the quantile binning of Gardner et al. leaves most bins empty, so activity was instead kernel-smoothed onto the cell, each pixel taking a Gaussian-weighted average (sigma = 0.62 rad in toroidal distance) of the cells near it, and each frame was z-scored for display. Frames are drawn one lattice vector apart, so they tile. The transition shown is the one with the longest pair of adjacent opposite epochs, chosen automatically.

The accompanying statistic is the across-trial concentration of the bump's phase at that same track position: the resultant length over trials of each phase, averaged over the two, which is 0.806 over anchored trials against 0.206 over non-anchored ones in this session ($p < 0.0005$ by permutation of the state labels, 2,000 draws). This is a property of one session shown alongside the population measure, not a separate population claim.

A bump-centred check, not shown. The same sheet was also averaged in the bump's own frame - subtracting the bump's centre of mass from every cell's phase at each 100 ms sample, so that a bump which is intact but moving still produces a peak. Because centring on an estimate the population itself provides concentrates activity even when there is no bump, the phase shuffle was put through the identical steps and subtracted; the shuffle alone peaked at 1.67 anchored and 2.98 non-anchored, so the uncorrected maps are not interpretable. The excess was 1.74 anchored against 1.07 non-anchored, agreeing with the population-vector result.

Does the bump deteriorate on the way into the non-anchored state (Figure S14H)? The bump was also taken trial by trial, as the mean of the population vector minus its shuffle over the moving samples of that trial (trials with fewer than ten qualifying samples were dropped). This per-trial measure reproduces the session-level result (+0.0614 anchored against +0.0427, $p = 4.3 \times 10^{-4}$). Trials were then aligned to anchored-to-non-anchored switches, defined exactly as elsewhere in the paper, with runs of fewer than three trials absorbed into the preceding epoch; 150 switches had eight scorable trials either side. There was no step at the switch (−0.0053 comparing three trials either side, $p = 0.186$) and no decay once in the new state (slope −0.00008 per trial within non-anchored epochs of at least eight trials, $p = 0.905$; the anchored epochs likewise, $p = 0.19$). Scoring each trial instead by its distance from the nearest switch (computed but not plotted), the difference between states is +0.0107 within two trials ($p = 0.037$), +0.0104 at three to five ($p = 0.241$), +0.0225 at six to ten ($p = 0.012$) and +0.0235 beyond ten ($p = 2.1 \times 10^{-4}$). The weaker bump is therefore a property the state holds throughout rather than a progressive deterioration, and it is expressed within a few trials of a switch - the interval over which the anchoring classifier's own three-trial resolution makes the labels least certain.

Yield and outcome. 24 sessions and 440 grid cells met the criteria (>= 8 grid cells in the module, >= 60 trials, between 15% and 85% of trials anchored, >= 100 qualifying samples per state, and at least one resolved torus axis). The bump exceeded its shuffle by +0.0610 when anchored and +0.0412 when not, both $p = 1.2 \times 10^{-7}$ (Wilcoxon signed-rank over sessions), with the states differing by +0.0198 ($p = 1.3 \times 10^{-4}$); with speeds equated, +0.0190 ($p = 6.4 \times 10^{-5}$). Median retention in the non-anchored state was 82% with no smoothing, 84% at 100 ms and 67% at 250 ms. The paired contrast between none and 250 ms was +3.9 points, lower in 18 of 24 sessions, $p = 0.029$; none and 100 ms did not differ ($p = 0.49$). The smearing interpretation is therefore supported at the longest smoothing only and is not graded, and is offered as a reading rather than an established result. The simulation keeps 101% under the same measure.

**Statistics**

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## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work the authors used Claude (Anthropic) to write and debug analysis code, to compute the statistics reported here from the authors' own recordings, to generate the figures, and to draft and revise manuscript text. The scientific questions, experimental design, data collection and analysis decisions were the authors'. Every reported result was produced by code that the authors reviewed, and the authors reviewed and edited all generated text. The authors take full responsibility for the content of this publication.

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*Citations for works named in the text. The source manuscript carries unresolved superscript placeholders and no bibliography; these were resolved against the published record rather than reconstructed from memory, and should still be checked against the originals before submission:*


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22. Target selectivity of septal cholinergic neurons in the medial and lateral entorhinal cortex (2018), Proc. Natl. Acad. Sci. USA - the source for septal cholinergic projections being denser to MEC than LEC. Title, journal and year verified; the author list could not be retrieved and is left outstanding rather than reconstructed.

## SUPPLEMENTAL FIGURES

*Supplemental figures are referenced from the Results and are presented here in the order their main figure appears.*

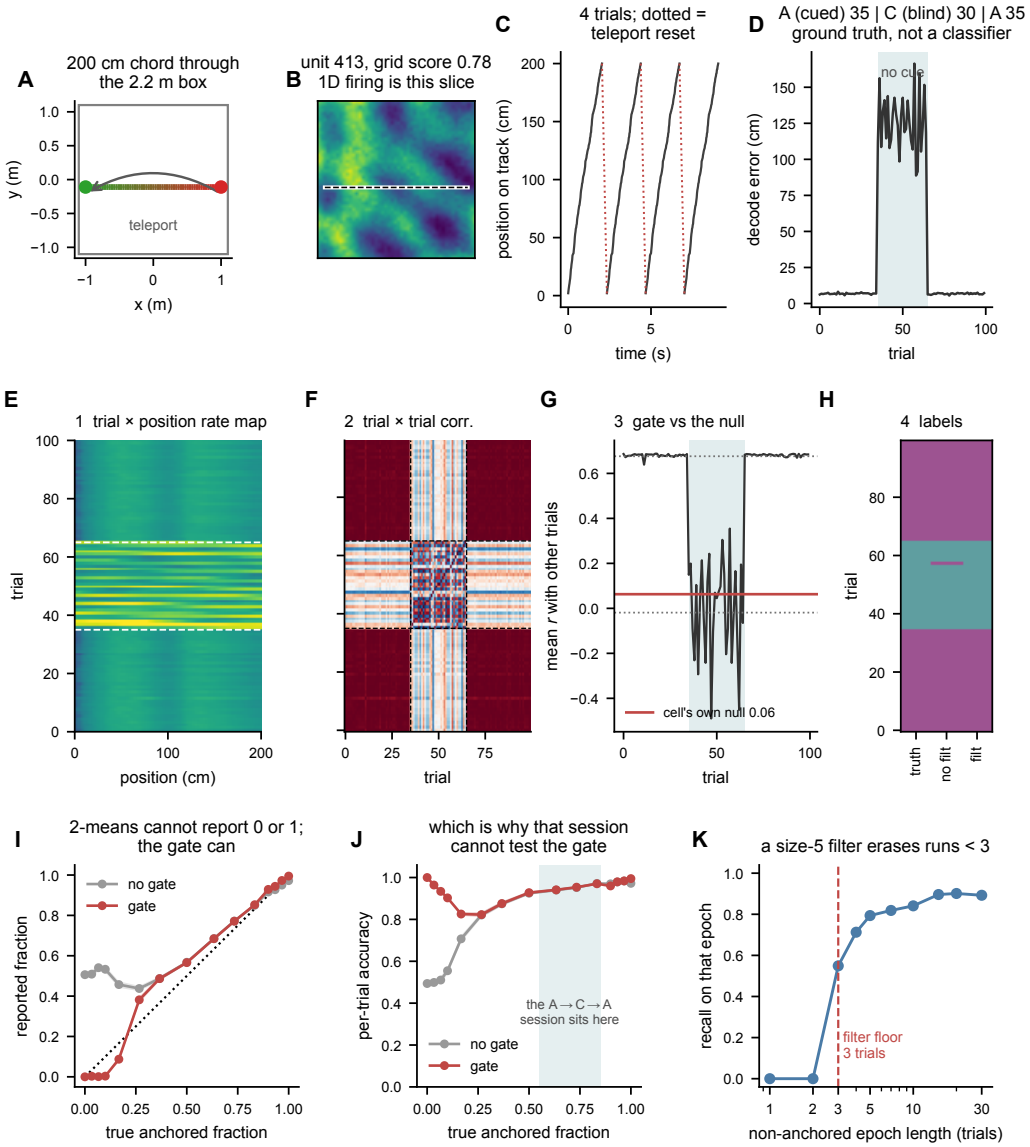



**Figure S1. The anchoring classifier, step by step, against RNN ground truth, related to Figure 1**

The classifier validated against ground truth, using an RNN trained on a 1D slice of a 2D environment with a teleport, where the trial labels are known rather than inferred. (A) The 200 cm chord through the 2.2 m box that the network runs along. (B) One network unit's 2D rate map with that chord drawn on it: the 1D firing the classifier sees is this slice. (C) Position on the track against time; the dotted lines are teleport resets. (D) Decode error per trial, which shows the manipulation worked before any classifier is applied: the blind block is re-anchored at neither end. (E-H) The four steps of the classification on one cell, in order: the NaN-free trial x position rate map, the trial x trial correlation matrix, the cell's mean correlation against its own circular-shift null, and the resulting label sequence against truth, with and without the median filter. (I) Reported against true anchored fraction, with and without the gate. Ungated, a session that is 0% anchored is reported as 50.6% anchored; gated, the curve tracks the identity line. (J) Per-trial accuracy over the same range; the band marks where the validation session itself sits, which is why that session cannot test the gate. (K) Recall against the length of a non-anchored epoch: the size-5 median filter erases runs shorter than three trials.

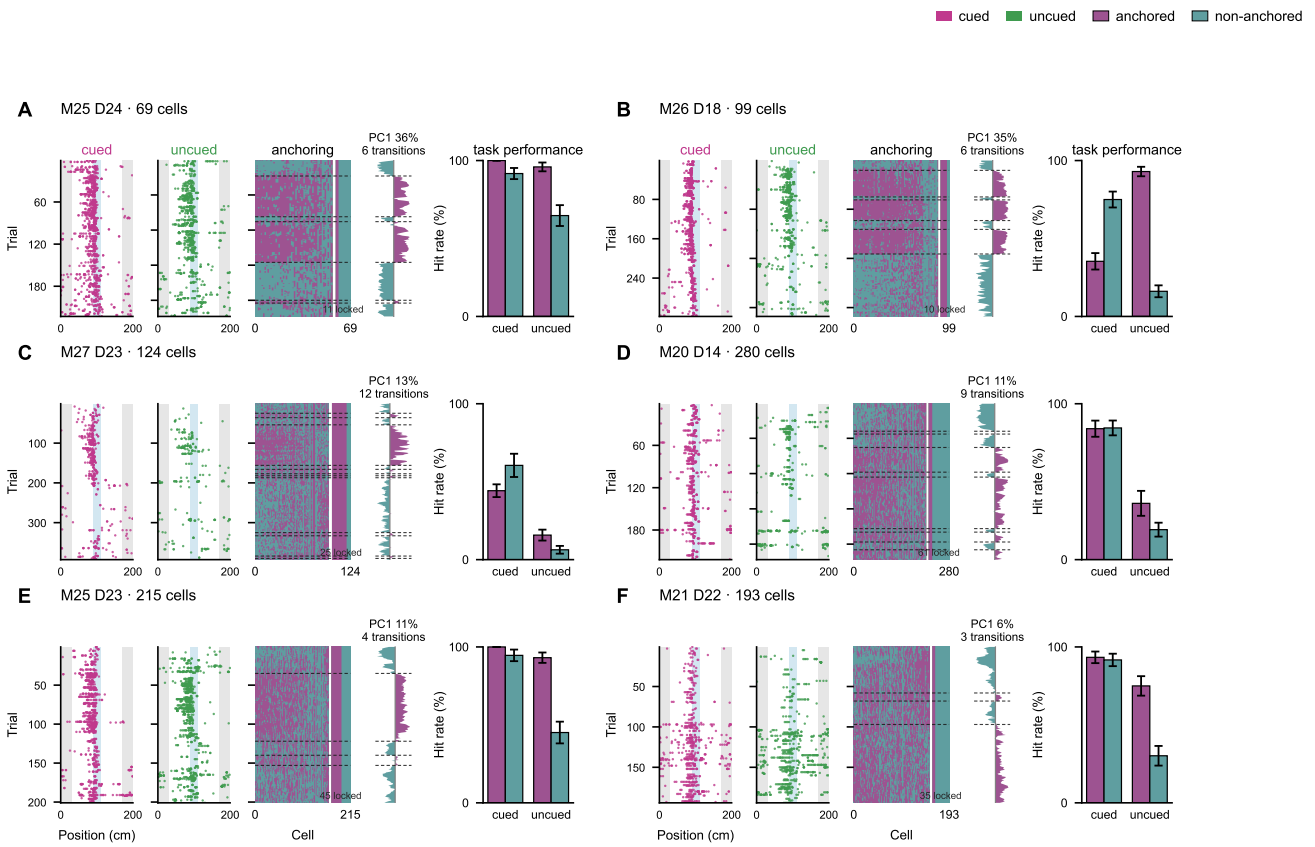



**Figure S2. Population anchoring states across sessions, related to Figure 1**

Six further sessions, ordered by the share of label variance the first principal component explains, from sessions where nearly all cells move together to sessions where they do not. (A-F) One session each, as five panels on a shared trial axis: stops on cued trials, stops on uncued trials, the anchoring raster with cells on the x axis, PC1, and hit rate by state and cue condition. Error bars are binomial standard errors, the unit here being the trial within one session.

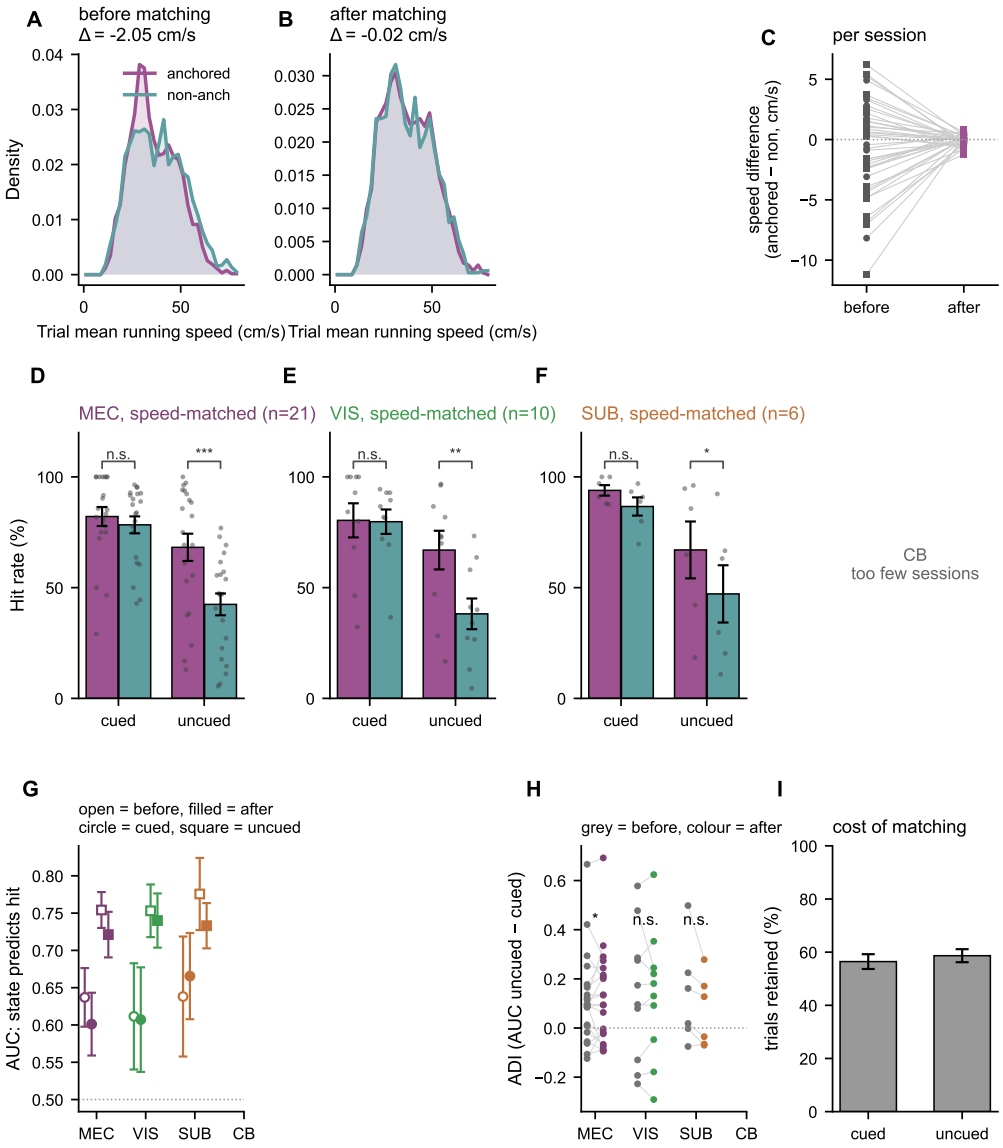



**Figure S3. Behavioural controls for the anchoring state, related to Figure 1**

Behavioural controls for the anchoring state, and the speed-matched subsets used wherever a comparison could otherwise be carried by a difference in locomotion. (A, B) Trial mean running speed by state, before and after matching within session in deciles of speed. (C) The same per session, before and after. (D-F) Hit rate by state and cue condition after matching, for MEC, visual cortex and the subicular complex; the cerebellar sessions are too few to match. (G) How well the state predicts a hit, as AUC, before and after matching, for each structure and cue condition. (H) The Anchoring Dependence Index per session, before and after matching. (I) The cost of matching, as the percentage of trials retained.

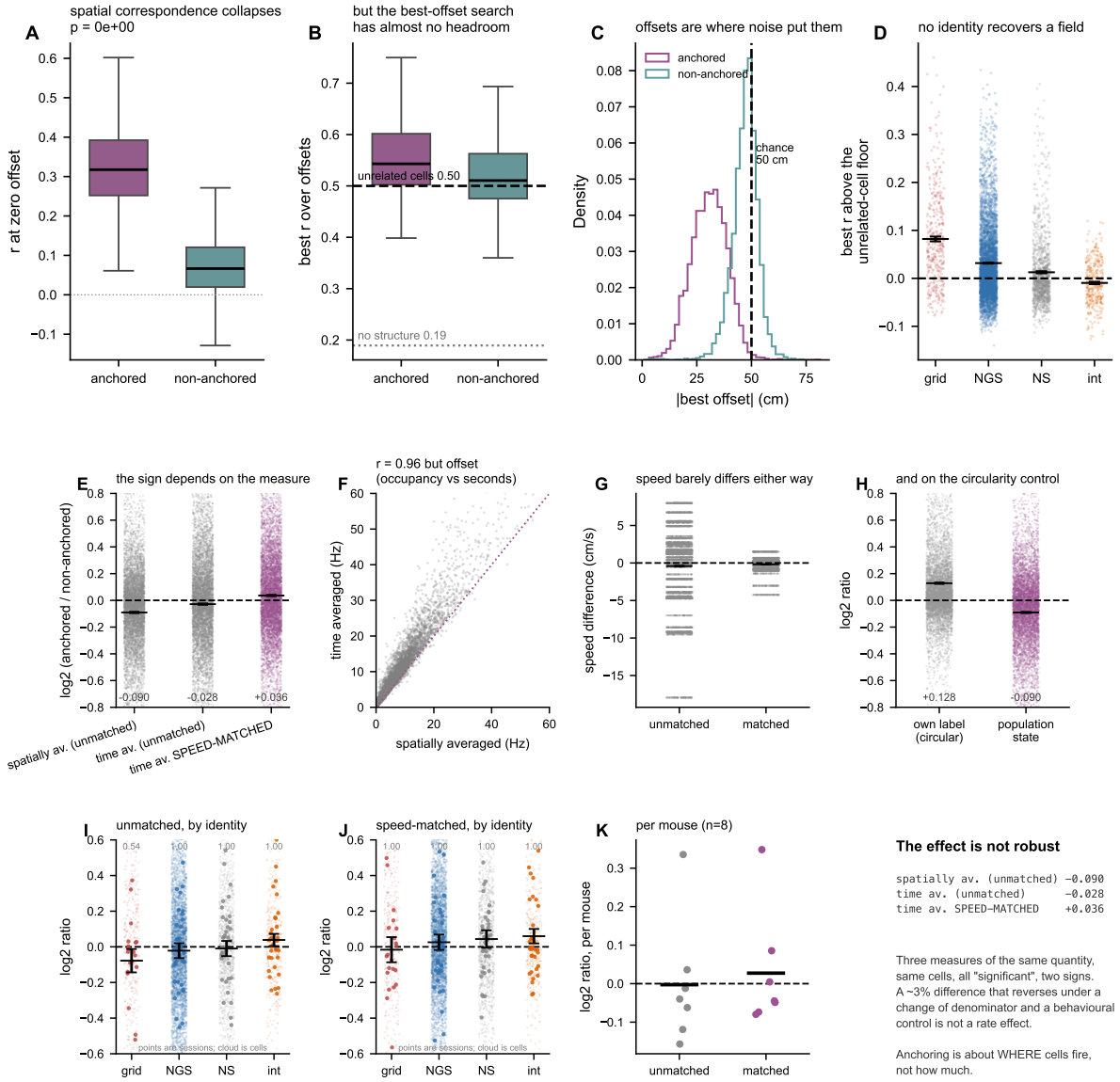



**Figure S4. What a non-anchored trial is, and whether firing rate changes with the state, relatedto Figure 2**

Two halves of one question. (A-D) Has the field moved, or has it gone? (A) Each trial's map correlated against the cell's anchored template at every circular offset. (B) Zero-offset correlation by state: +0.33 to +0.07. (C) Best-over-offsets correlation by state, which falls only from +0.56 to +0.53 and looks like displacement until the null is applied. (D) That null: a maximum over 100 offsets is large by construction, two unrelated cells reach +0.500 under the same search, and non-anchored trials sit 0.029 above that floor with a mean best offset of 47 cm against the 50 cm a uniform choice would give. Non-anchored trials lose spatial correspondence with no evidence of a coherent field elsewhere. (E-K) Does rate change with the state, or only where cells fire? (E) The same cells under three measures: log2 anchored over non-anchored is -0.090 spatially averaged, -0.028 time averaged, and +0.036 time averaged and speed-matched. All three are significant as cell-level tests and they disagree about direction. (F) Why: a rate read off a rate map divides by occupancy, which running speed sets, while a rate in spikes per running second does not; the two correlate at r = 0.96 and differ in mean. (G) Trial mean speed by state, before and after matching. (H) The circularity control: splitting a cell by its own label gives +0.128 against -0.090 for the population state, so every split here uses the population state. (I, J) The same question by identity, unmatched and speed-matched, with one point per session over the cell cloud and a Holm-corrected p per class. Sessions are the unit because cells within a session share one state. On that test no class differs from zero on either measure; putative interneurons are +2.7% unmatched and +4.2% speed-matched over 34 sessions, the largest value and the only class that does not go down, but not significant. (K) Per mouse, unmatched and matched. Nothing that reverses sign under a change of denominator should be reported as a rate effect: anchoring concerns where cells fire rather than how much.

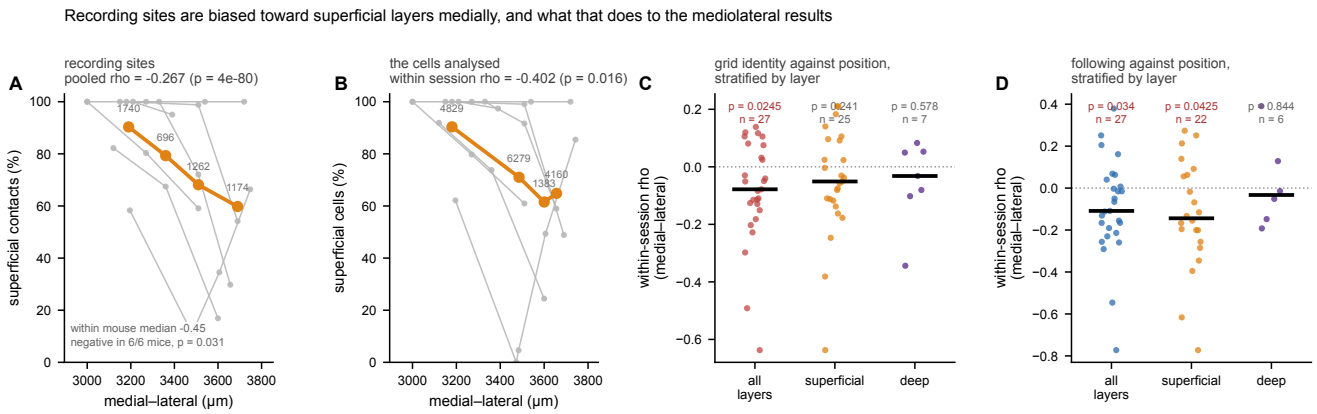



**Figure S5. Recording sites are biased toward superficial layers medially, and what that does to the mediolateral results, related to Figures 3 and 4**

MEC is curved and a four-shank probe is flat, so shanks at different mediolateral positions do not enter the tissue at the same depth relative to the layers. (A) They do not: superficial layers are 90.3% of entorhinal recording sites in the most medial quartile and 59.8% in the most lateral (pooled rho = -0.267, p = 4e-80), a gradient present in every animal (within-mouse median rho = -0.447, p = 0.031). (B) The same bias in the neurons actually analysed (within-session median rho = -0.402, p = 0.016). (C, D) The mediolateral tests stratified by layer. The identity gradient does not survive within superficial cells (median rho = -0.051, p = 0.24, 25 sessions) against -0.078 (p = 0.025) across all layers; the following gradient does and is if anything stronger (-0.144, p = 0.043, 22 sessions against -0.109, p = 0.034). Deep layers are too sparsely sampled to test on their own.

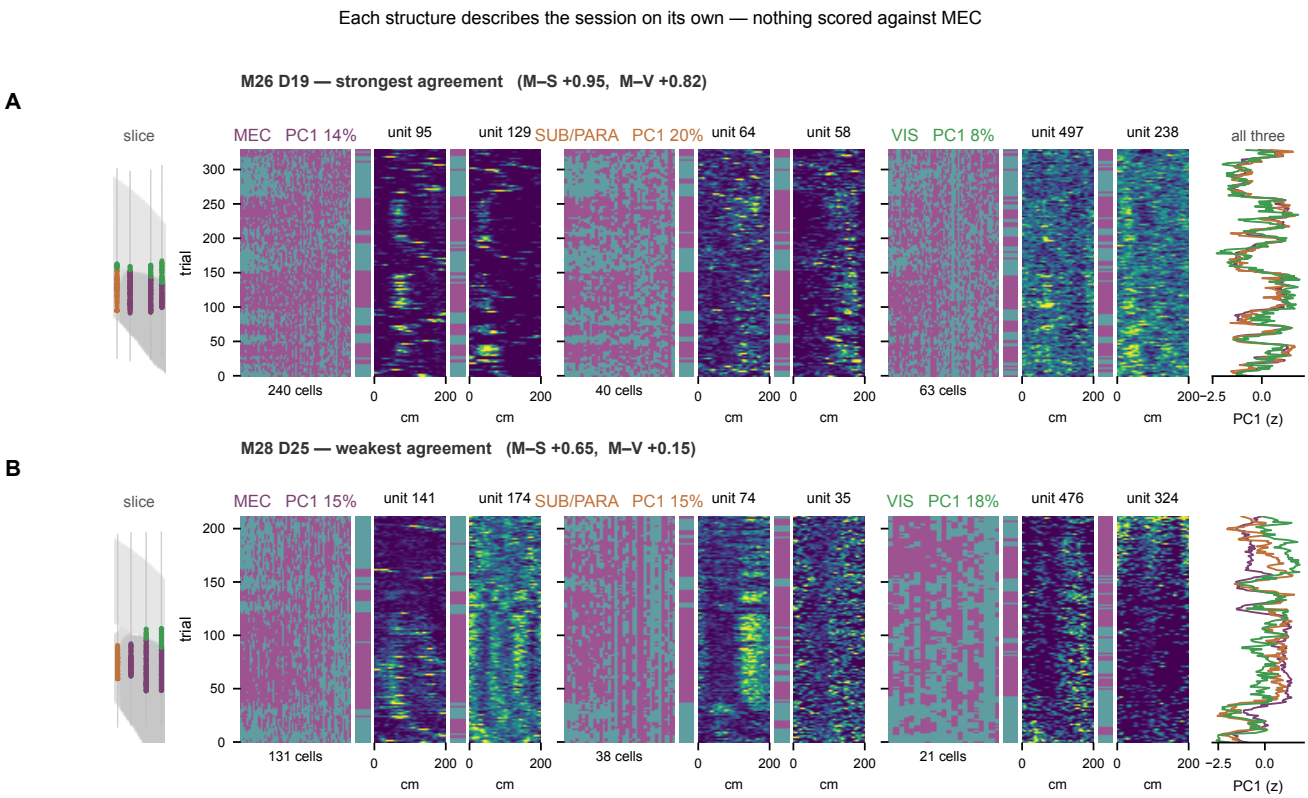



**Figure S6. The same state, described separately by each structure, related to Figure 7**

Two sessions bracketing the range among those recording all three structures. (A, B) One session each, read left to right as entorhinal, subicular and visual cortex. Everything within a structure is built from that structure's own cells, the raster ordered by its own PC1 loading and the component signed to its own anchored fraction, so agreement between the groups is agreement between independent descriptions rather than a consequence of a shared frame.

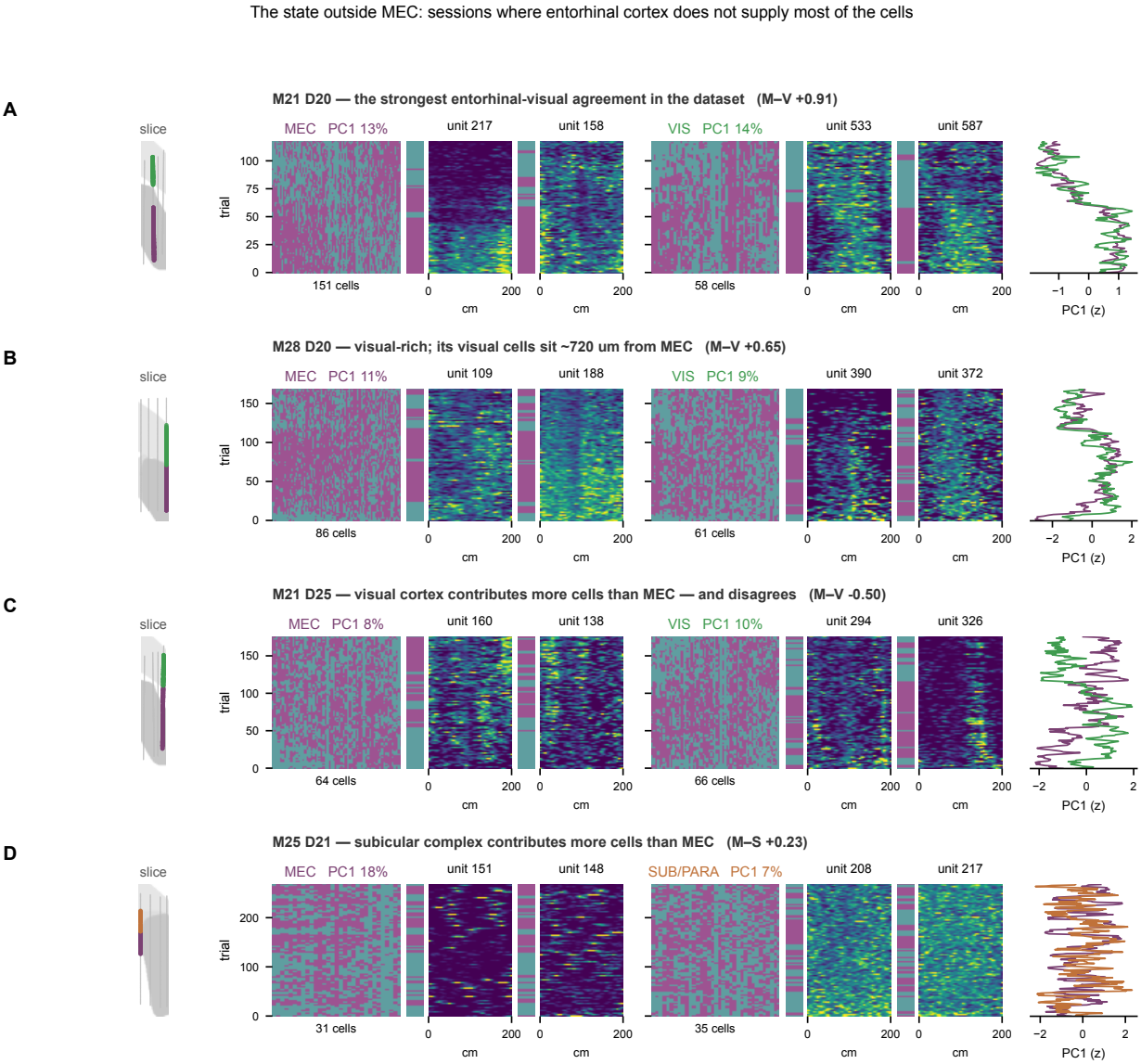



**Figure S7. The state outside MEC, where MEC does not dominate the sample, related to Figure 7**

Four sessions chosen for their sampling rather than their result. (A, B) Two visual-rich sessions. (C) A session in which visual cortex contributes more cells than entorhinal and disagrees with it. (D) A session in which the subicular complex does. Each row is built as in Figure S6, every structure described by its own cells. A cerebellar session was removed: it rested on 15 to 18 cells, too few for an axis to be estimated with confidence.

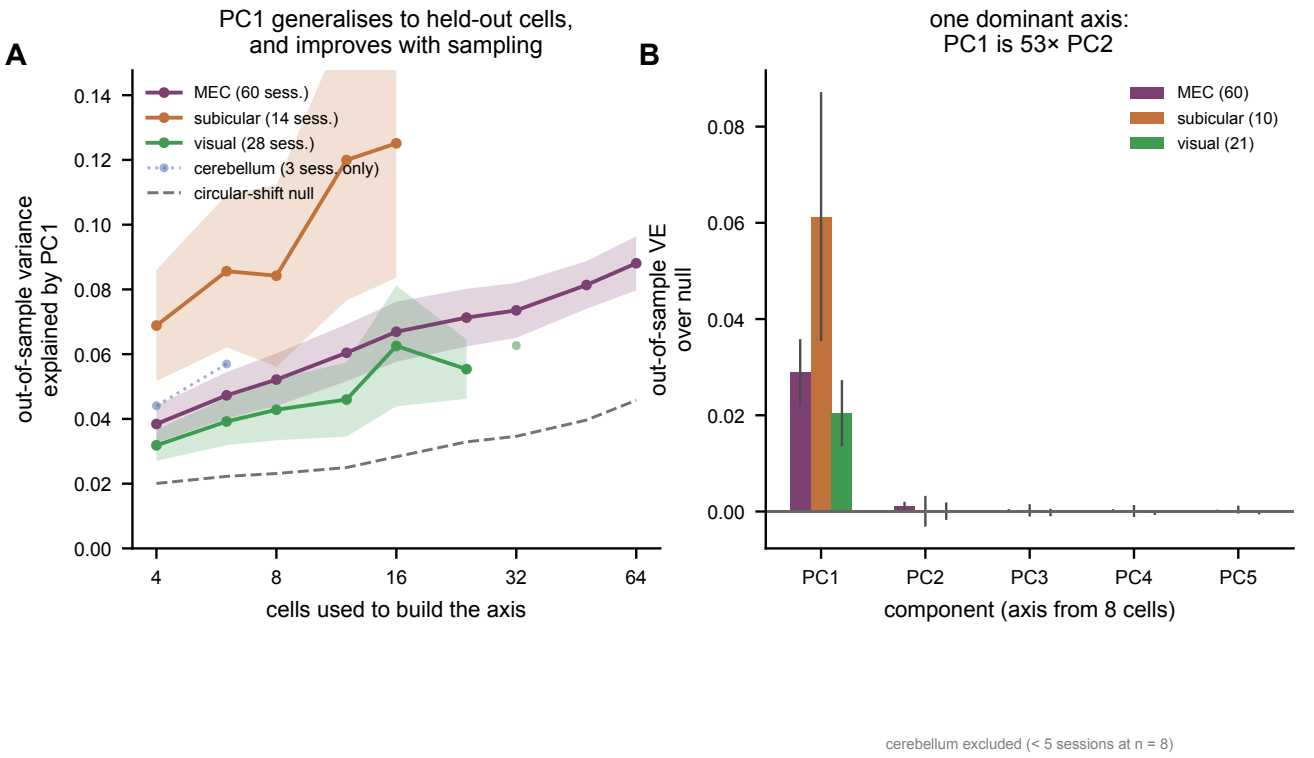



**Figure S8. One axis is an adequate description of the state, related to Figure 7**

Out-of-sample variance explained, with the axis built from one half of a region's cells and scored on the other. (A) Neuron-dropping curve per region: how much of the held-out half one axis explains as cells are added. (B) The same at the full cell count, per region. One axis is an adequate description of the state in every structure recorded.

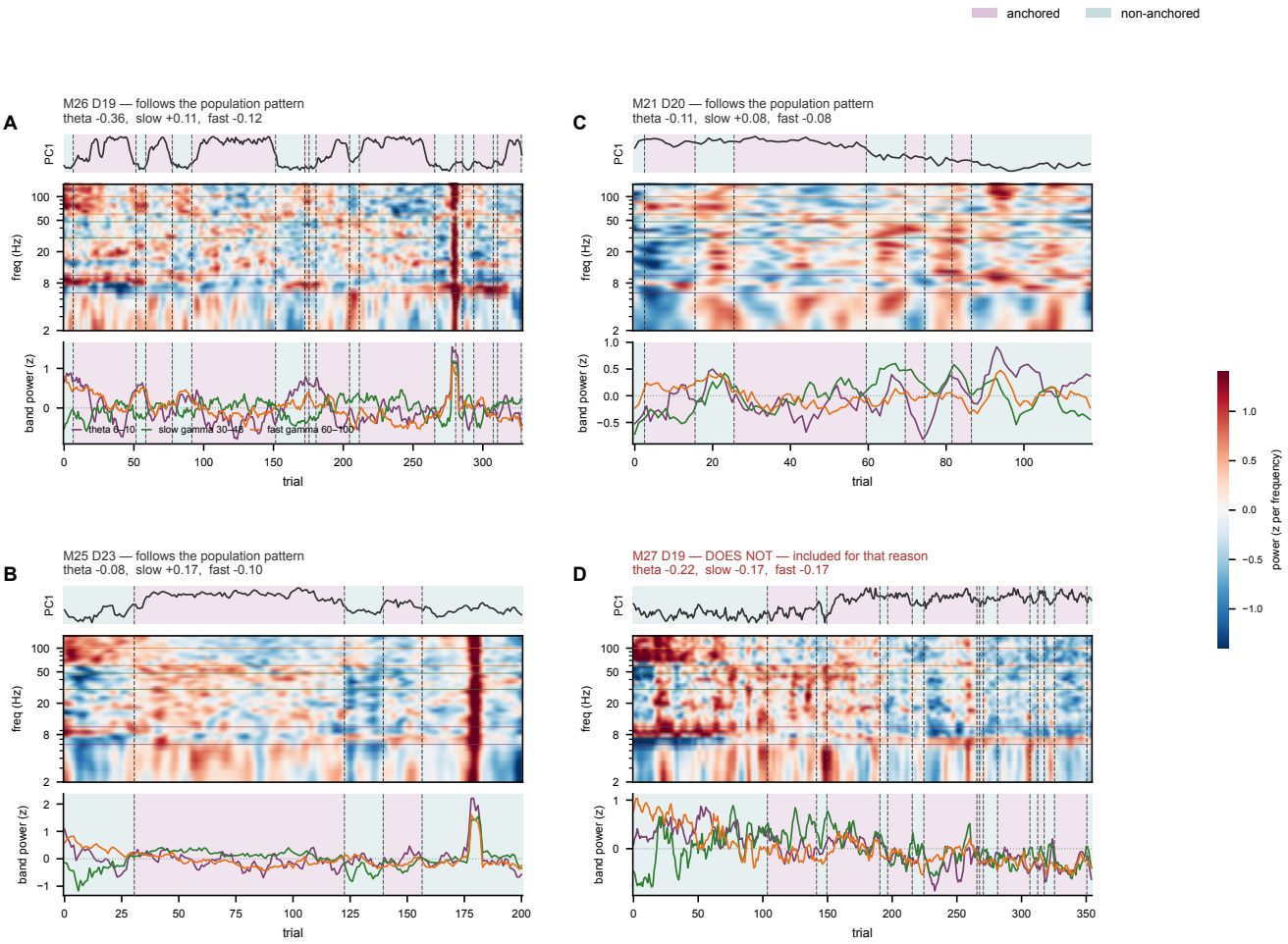



**Figure S9. Band power tracking the population state in individual sessions, related to Figure 8**

Four sessions as the population axis, the log-spaced spectrogram and the three band traces on a shared trial axis. (A-C) Sessions showing the population pattern, slow gamma rising and fast gamma falling across anchored blocks. (D) A counterexample, drawn in red: a session with clear state transitions whose band traces do not follow them. It is included because the population effect is a mean over sessions that individually scatter widely -- slow gamma rises in 21 of 26 and fast gamma falls in 19 of 26, not in all of them.

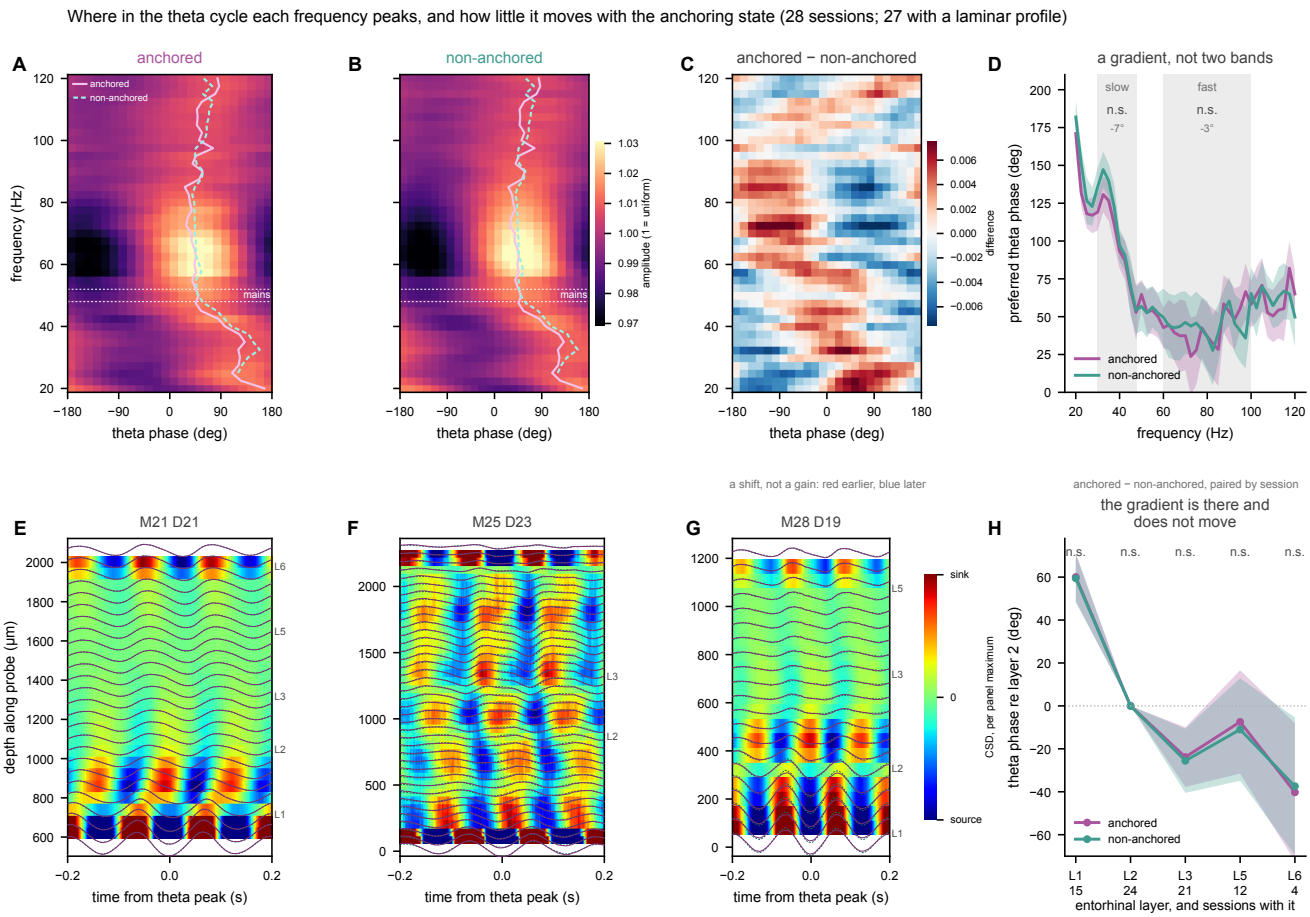



**Figure S10. Where in the theta cycle each frequency peaks, and how little it moves with the state,related to Figure 8**

Figure 8 fixes two gamma bands in advance and asks how strongly each is nested in theta; the modulation index discards phase, so it cannot ask where in the cycle they peak. This does, as a continuous function of frequency. (A, B) The phase-frequency map for each state over 28 sessions, each frequency row normalised to sum to one so colour is where a frequency peaks and not how much power it carries. Both states' preferred-phase ridges are drawn on both maps. (C) Their difference, which is where a state effect would live. The two maps share 94% of their pixel variance. (D) The same reduced to preferred phase against frequency. The relationship is a gradient, not a step between two bands: phase runs from about 130 degrees at 20-30 Hz to about 41 degrees at 60-100 Hz, and that separation holds in each state on its own (+34 degrees anchored, p = 5e-4; +52 degrees non-anchored, p = 1e-5). Paired within session the state moves it very little and not reliably: 6.6 degrees earlier in the slow band (p = 0.33) and 2.8 degrees in the fast (p = 0.16). (E-G) The same question against depth: theta-triggered LFP and CSD down one shank for three example sessions, both states overlaid. (H) The laminar phase gradient over 27 sessions by layer. Theta advances about 100 degrees across the entorhinal layers and the gradient is identical in the two states: referenced to each session's own layer 2, every layer differs by less than 3 degrees with p >= 0.06.

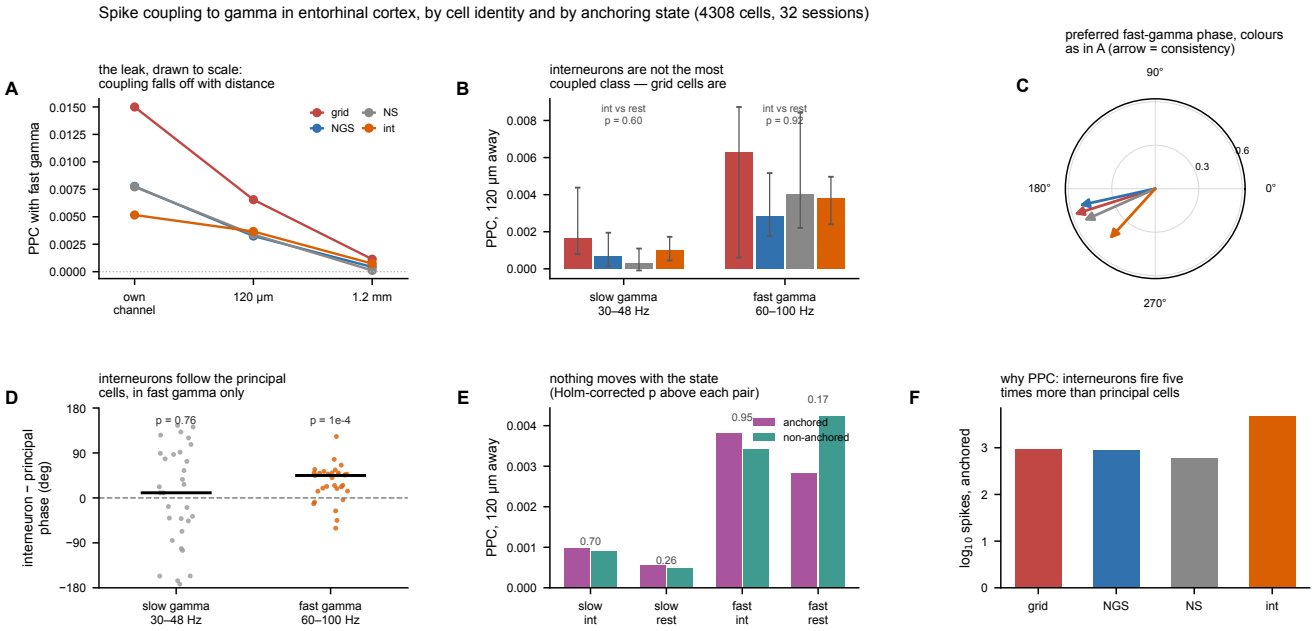



**Figure S11. Spike coupling to gamma, by cell identity and by anchoring state, related to Figure 8**

Whether fast-spiking interneurons are the cells coupled to gamma, and whether that coupling moves with the anchoring state. Coupling is pairwise phase consistency, which is unbiased by spike count. (A) The distance control. Coupling measured on a cell's own channel group is 2 to 4 times the value measured 120 um away, which is spike waveform leaking into the LFP, so every other panel uses the nearest group at least 120 um from the cell. (B) Coupling by cell identity and band. Grid cells are the most coupled class, not fast-spiking cells. (C) Preferred gamma phase by identity, on polar axes. In the slow band the interneurons have no consistent preferred phase at all (R = 0.05, against 0.22 to 0.28 for the three principal classes), so that band is one its putative generators are blind to. (D) The interneuron-minus-principal phase lag per session: a median +45 degrees in the fast band (p = 9.6e-5, later in 25 of 31 sessions) and nothing in the slow band (+10 degrees, p = 0.77). The interneurons therefore FOLLOW the principal cells, the direction PING predicts, and do so in one band only. (E) Coupling by anchoring state. No band and no class moves with the state (Holm p >= 0.17). (F) Spike counts per cell and state, to show the null in E is not a power difference.

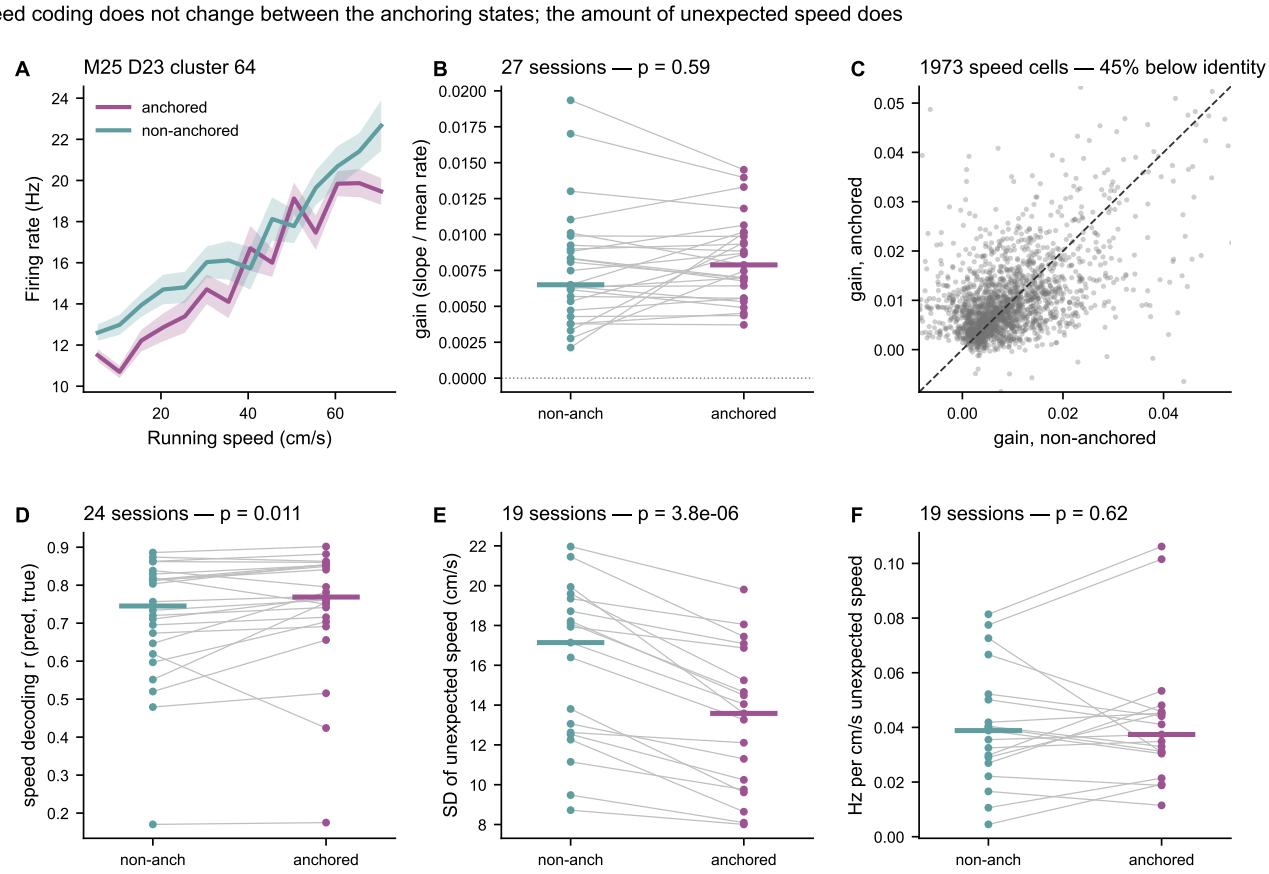



**Figure S12. Speed coding does not change between the anchoring states, related to Figure 8**

Theta loses much of its entrainment to running speed in the anchored state, and the account in Figure 9 turns on whether the cells that carry the speed signal lose it too. (A) One session's speed and firing rate on the trial axis, with both states shaded. (B) Rate against speed for open-field speed cells, by state, on the two states' common speed support. (C) The regression slope by state across sessions: 0.1523 anchored against 0.1361, +11.9%, p = 0.121, 29 sessions. (D) The same as rate-normalised gain, which separates a change in speed sensitivity from a change in overall firing: +6.8%, p = 0.565. (E) The other cell groups, none of which changes either. (F) The power of the comparison: it resolves a 17% change in slope and an 18% change in gain at 80% power, so a halving of speed drive of the size seen in the field potential is excluded in the cells that carry it.

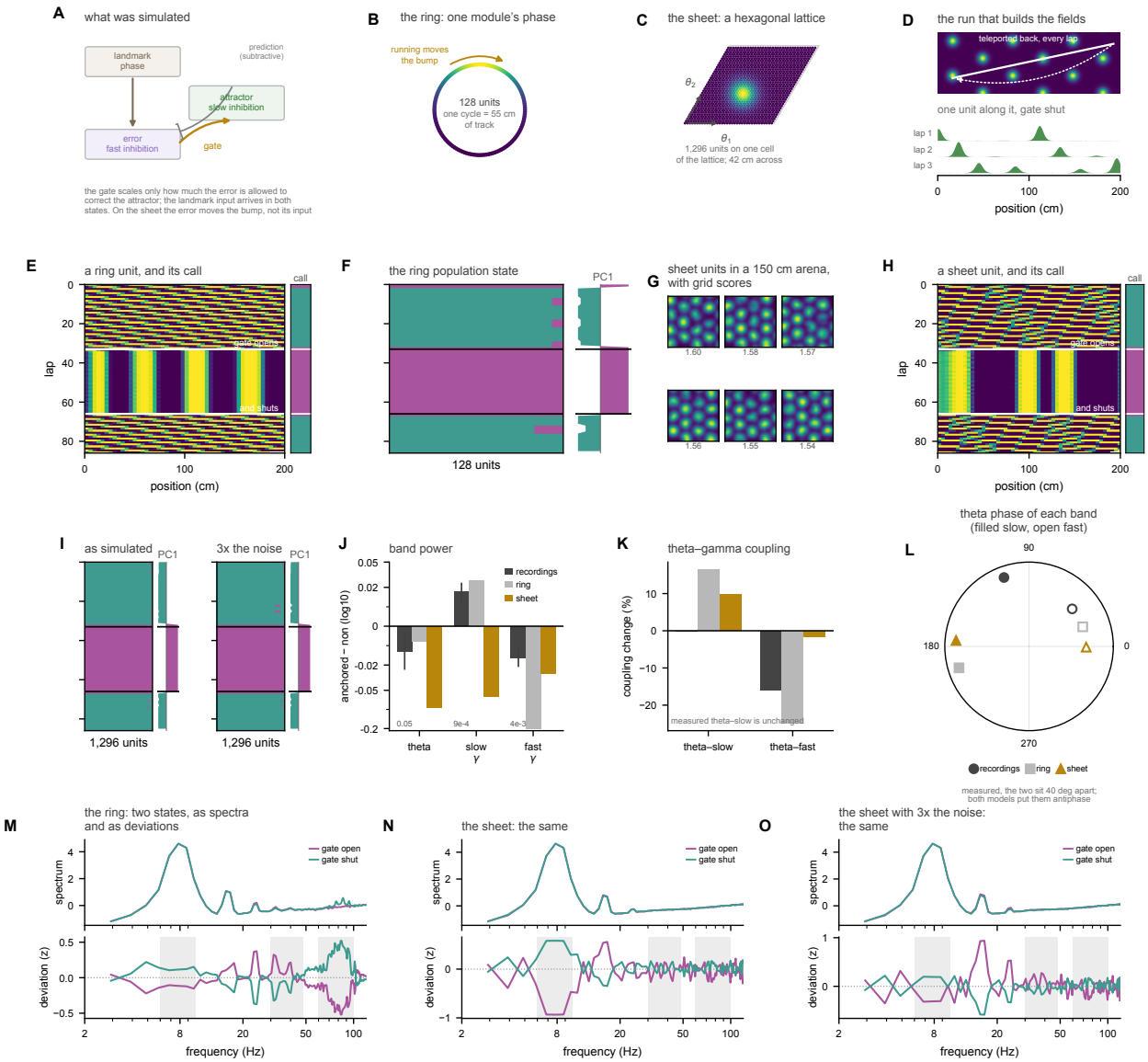



**Figure S13. The gated attractor, simulated as a ring and as a two-dimensional grid network, related to Figure 9**

The mechanism Figure 9 proposes, built as a circuit to see what it produces rather than to illustrate what it claims, and built twice: as a ring carrying one module's phase along the track, which is enough for the field potential but assumes what a grid model ought to produce, and as a two-dimensional sheet whose units fire in hexagonal grids in an open arena. In both the gate is shut, then open, then shut again over about a hundred laps, because the recorded population state comes in blocks of tens of trials rather than as one switch. (A) The architecture. A landmark phase drives an error population with fast inhibition, which compares it against a subtractive copy of the attractor; the gate scales only how much that error is allowed to correct the attractor, which has slow inhibition. The landmark input arrives in both states, so the gate controls whether the error is acted on and not whether it is transmitted. No geometry is named here because the same circuit is built both ways below, and the one difference the geometry forces -- how the error reaches the attractor -- is described under H. (B, C) What each network is made of. The ring is 128 units carrying one module's phase, so one turn around it is 55 cm of track; the sheet is 1,296 units covering one cell of a hexagonal lattice 42 cm across, drawn in the coordinates in which the bump is round. Both hold the bump in a slow synaptic variable and both are moved by an exact spectral translation, so the geometry its units tile is the only thing that differs between them. (D) The run that builds the fields, and what the teleport does. The sheet is driven along a straight line through the arena at 12 degrees to a lattice axis and the animal is then put back at the start, while the integrator is not. Behind the line is one unit's firing in the plane; the three traces are that same unit along that same path on three successive laps with the gate shut. The path is identical every lap and the fields it crosses are not, because the integrator has carried on by the per-lap phase step G gives. That drift is what the gate has to cancel, and the diagonals in H are the same thing drawn over 86 laps. (E) One ring unit over the session, with the anchoring classifier's call on each lap beside it. Its fields recur in the same places while the gate is open and walk when it shuts, and nothing is reset at the transitions, so the attractor runs straight through them. The call is right on every open lap and on 90% of shut ones, and the strip shows what the remainder are: laps on which the drifting phase happens to pass close to where the fields sat when anchored. Those coincidences are periodic rather than scattered -- the phase slips a median 131 degrees per lap, so it comes back into register every 2.7 laps, and with the classifier's five-trial median filter switched off the shut block shows exactly that rhythm and a false-positive rate of 0.37 rather than 0.10. Their regularity is itself evidence that the bump drifts at a constant rate rather than wandering, which is what Figure S14E also finds. (F) The same session through the recordings' own anchoring classifier and its first principal component, the code that produces Figure 1E, applied unchanged: every unit is called anchored on every lap of the open block, and PC1 is positive on 33 of 33 open laps and 4 of 53 shut. (J) The band changes, measured against both models, in the same units: a difference of log10 power between the states over the same bands, with the mains lines excluded. (G) Six sheet units in the arena with their grid scores, from the same scorer used for the recorded cells: median 1.53 over all 1,296 units, every one above 0.4. Two geometric choices matter and neither is the obvious one. How many fields a unit has per lap is set by the track's heading through the lattice rather than by the period: a track that grazes a lattice row gives one long field however short the period is. And what carries the bump between laps is the per-lap step in PHASE, which is larger for a shorter period, not a longer one. Because path integration here is exact, that step is identical every lap, so if either lattice axis steps only a little the unit fires in closed blobs spanning several laps instead of fields that shift. At 42 cm and a heading of 12 degrees the two axes step 0.38 and 0.30 of a cycle per lap, 89% of units have two or more fields per 200 cm, and the unit shown has six. (H, I) The same two panels as E and F for the sheet, which is driven along a line through the arena and teleported back to the start each lap, as a virtual

linear track does. The teleport is what gives the gate work: the animal's position jumps back and its path integrator does not. The sheet also forces the one change of mechanism in this figure. On the ring the error reaches the attractor as a drive, and that drive moves the bump because it is concentrated on 128 units; spread over 1,296 it cannot. Scanned from a gain of 5 to 300 it never bought a lock without destroying what it was locking: the mean phase mismatch fell only from 54 to 38 degrees, while the drive -- which is shaped like a bump at the LANDMARK phase -- painted a second bump on the sheet beside the attractor's own, taking the population vector from 0.27 to 0.047 and, at a gain of 150, the grid score to 0.29. An earlier version of this panel ran at the low end of that scan, and its open block repeated only every few laps rather than every lap. On the sheet the error therefore corrects the bump's POSITION instead, driving the phase error towards zero with a time constant of 0.25 s, which is what a correction to a path integrator does and costs the bump nothing: the population vector is 0.430 in both states. The mismatch is then 6 degrees with the gate open against 139 with it shut, successive laps' maps correlate at +0.98, and every unit is called anchored on every lap of the open block against 4% of laps outside it, and PC1 is positive on 33 of 33 open laps and 2 of 53 shut. I shows the sheet twice: as simulated, and run again with three times the noise on both populations. The noisier sheet is no worse, every unit anchored in the open block and 5% outside it. The comparison is a clean one now: what holds the phase is the correction above, whose rate does not scale with the noise, so tripling the noise does not quietly strengthen the gate along with it. It shows that the state classification is not an artefact of a quiet simulation. (K) Theta-gamma coupling in both models against the recordings. (M, N, O) Each model's two states as spectra and as deviations, drawn as Figure 8E draws the recorded ones, with one spectrum per lap standing in for one per trial: the ring, the sheet, and the sheet with three times the noise. (L) Where in the theta cycle each band peaks, the quantity Figure S10 measures. Taken together the two models settle different things and neither settles all. The circuit works when its units are real grid cells and the anchoring switch survives in two dimensions, which the ring could not show. The ring gets the fall in theta-fast coupling, and generates the prediction that was confirmed in the recordings (within the gate-shut state, fast-gamma amplitude against mismatch, r = +0.51); the sheet gets neither (r = 0.00), because its error population's inhibitory pool averages over 1,296 units rather than 128 and never reaches the strongly oscillating regime that carried them. The same division runs through J. The ring reproduces all three measured band changes in sign -- theta -0.008, slow +0.025, fast -0.204, against -0.013, +0.018 and -0.016 in the recordings -- while the sheet puts all three downwards, theta -0.093, slow -0.062 and fast -0.027, and so gets the theta and fast-gamma falls but not the slow-gamma rise. What does rise on the sheet is theta-slow COUPLING, by 9.8% as simulated and 22.1% with three times the noise (K), which the recordings leave unchanged. The attractor's own pool barely moves between the two states -- its mean square differs by 1% -- so what band differences the sheet does produce are carried by the error population falling quiet once the correction succeeds, which is the fast-gamma fall itself, and not by any change in the attractor's rhythm. And both put the two gamma bands nearly antiphase in theta where the recordings place them about 40 degrees apart; sweeping a phase offset on the error population's theta gate over a full cycle does not move it, because the two pools' envelopes are driven in antiphase by the construction itself, one tracking how concentrated the bump is and the other how large the error is.

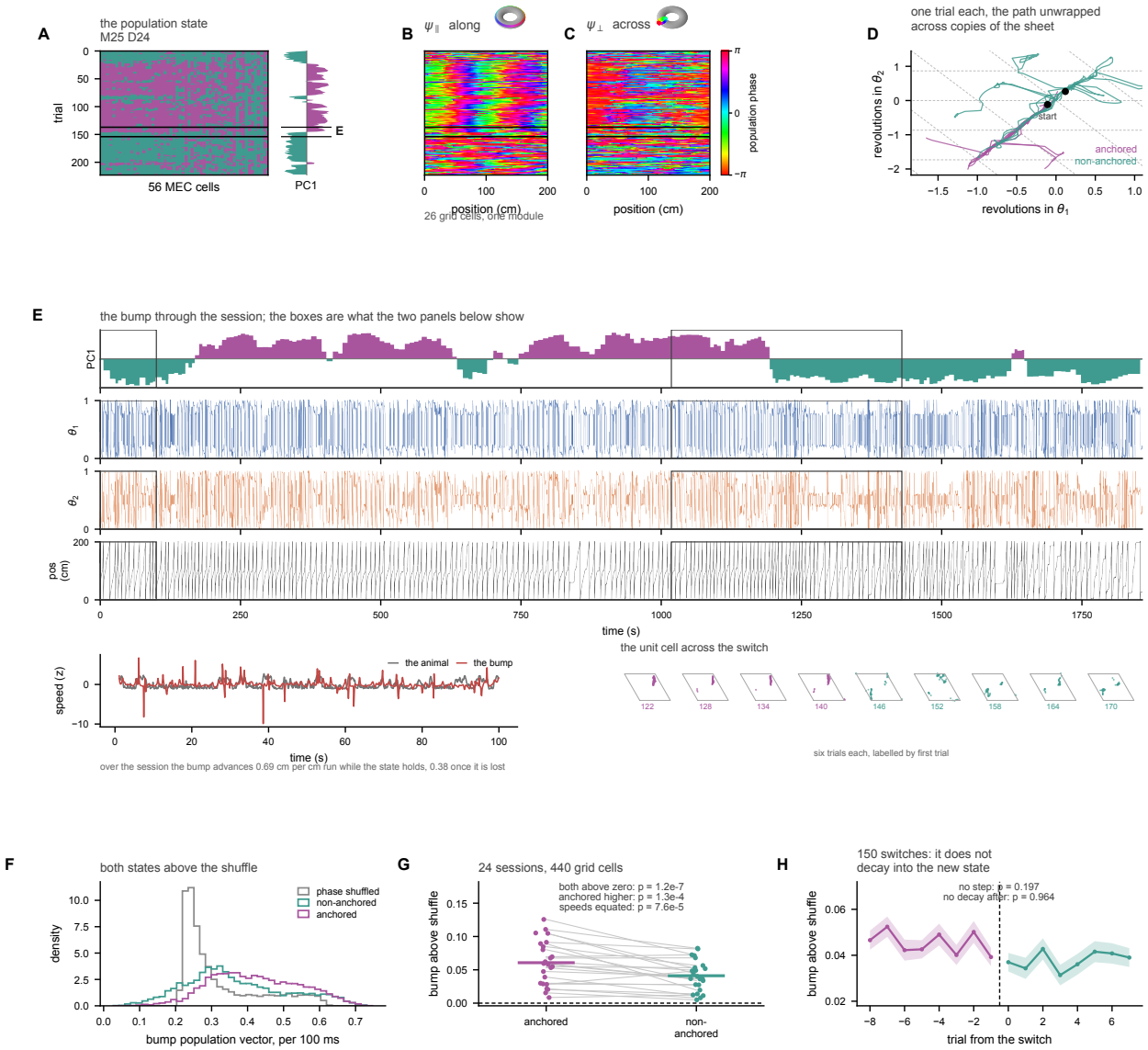



**Figure S14. The attractor bump survives while non-anchored to the task reference frame, related to Figure 9**

Figure 9 and the model in Figure S13 both assume the non-anchored state is an attractor that is still running and merely uncorrected, rather than one that has come apart. That assumption is tested here. Each cell is given its phase offsets on the module's torus by the spectral method, the population is projected onto them, and every result is referred to a shuffle of which offset belongs to which cell, which leaves all firing rates untouched and destroys only the lattice organisation. (A) The population anchoring state in M25 D24, cell by cell and as PC1; the two lines mark the trials shown in E. (B, C) The population's phase along two of the module's lattice axes at each position of each trial. Which two matters: on a linear track the bump has one degree of freedom, so the track direction projects onto two of the three axes almost equally and onto the third hardly at all, and those two are then nearly the same phase, correlated at +0.96 and 12 degrees apart on average. B therefore shows the axis running along the track and C the one running across it, which is the independent pair (circular r = +0.05); the torus above each shows which cycle it is. (D) The bump's path over single trials, unwrapped across copies of the sheet, twelve trials either side of the switch, with the unit cell dashed and the mean starting point marked. This panel uses the two axes that advance, since an unwrapped path needs coordinates that move. (E) The bump through the whole session and at the switch. Above, its two sheet coordinates and the animal's position as continuous traces. Below left, a hundred-second window in which the bump's own speed is plotted against the animal's: the two move together. Over the session the bump advances 0.69 cm per centimetre run while the state holds and 0.38 once it is lost, and the same shortfall appears in the position domain, where the non-anchored phase advance is 0.55 to 0.73 of the anchored one. That is the mechanism rather than a random walk: over a 200 cm lap it accumulates three to five radians, half to three quarters of a cycle, so the phase at a given place on the track is effectively re-drawn by the next lap, which is also why lap-to-lap phase displacement looks uniform and cannot be unwrapped. Below right, the unit cell itself in successive windows of six trials across the switch, each a scatter of where the bump sat at 10-30 cm on the track: tight while the state holds, filling the cell once it is lost. (F) The same session's bump population vector per 100 ms, both states above the shuffled distribution. (G) Every session. The bump exceeds its shuffle by +0.061 when anchored and +0.041 when not, both p = 1.2e-07, so the attractor does not collapse; the states differ by +0.020, p = 1.3e-04, so it is modestly weaker, and equating the two states' running speeds changes nothing (+0.019, p = 7.6e-05). (H) The bump aligned to the 150 anchored-to-non-anchored switches, mean and SEM over switches. It does not fall abruptly at the switch (-0.005 over three trials either side, p = 0.20) and does not decay once there (slope -0.00004 per trial within non-anchored epochs, p = 0.96), so the weaker bump is a property the state holds throughout rather than a progressive deterioration.

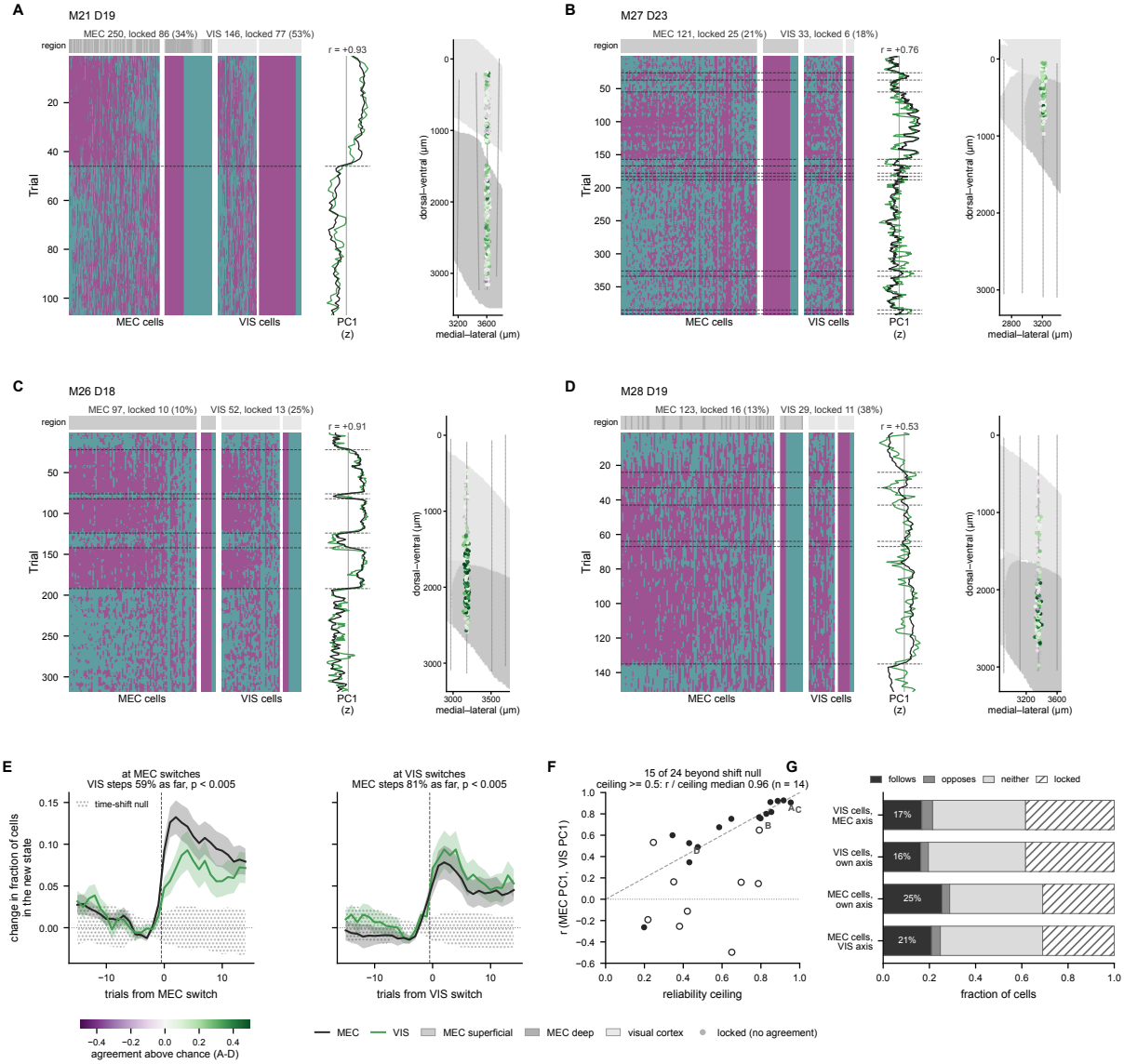



**Figure S15. Entorhinal and visual anchoring in four sessions, related to Figure 7**

Four sessions, drawn as Figure 4 draws its examples: the two of Figure 7 (A, B) and the two others with at least 25 cells in each structure and the largest difference between them in the share of locked cells, from mice not already shown (C, D). Left, one anchoring raster split into an entorhinal (MEC) and a visual block, each under a strip giving every cell's region in the grey key of Figure 4; within a block, cells that vary are ordered by their correlation with the session PC1 and locked cells sit past a gap, with each structure's cell count and locked share above. Cells with no usable label on any trial are left out. Middle, PC1 computed separately from each structure's own cells, z-scored, one line per structure, with their correlation above. Dashed lines are the trials on which the session PC1 changes sign for at least three trials. Right, the session's best-fit atlas slice at true aspect, with every device contact drawn small and grey as in Figure S7 and depth measured from the most dorsal contact, and with every cell coloured by its agreement above chance with the session PC1, on the diverging scale of Figure 4 (white at chance), with 15 um of display jitter along the mediolateral axis. Locked cells have no agreement, since a constant label correlates with nothing, and are drawn small and grey. The visual cells that vary follow the entorhinal axis closely: the two structures' PC1s correlate at +0.93, +0.76, +0.91 and +0.53 in A-D. What separates visual cortex is its locked block, larger than the entorhinal one in A, C and D (53% against 34%, 25% against 10%, 38% against 13%) but not in B (18% against 21%). (E-G) Whether the two structures share one state, in every session with at least ten varying cells in each (24 sessions). Each structure's state is its own PC1, and its switches are that PC1's sign changes with epochs shorter than three trials absorbed, as for the dashed lines. (E) Switch-triggered averages: the change in the fraction of each structure's varying cells in the anchored mode, signed so that the new state is up and referred to the five trials before the switch, averaged per session and then across sessions (mean and SEM), around MEC switches (left, 167) and visual switches (right, 234). Stippled, the middle 95% when the other structure is shifted in time 200 times. Visual cortex steps at MEC switches, 59% as far as MEC over the first five trials, and MEC steps at visual switches, 81% as far as visual cortex, both beyond every shift ($p < 0.005$), at the same trial and with no lead in either direction. (F) The correlation of the two PC1s against the ceiling set by their own reliability, the geometric mean of each axis's split-half reliability (Spearman-Brown corrected): a session on the diagonal shares as much as its noise allows. Filled, beyond a null shifting one axis in time; letters, the sessions of A-D. Where both axes are reliable (ceiling at least 0.5, 14 sessions) the correlation is a median 0.96 of its ceiling, with three exceptions (M21 D25, M28 D25, M29 D25); most sessions without a correlation are ones whose visual axis is too noisy to show one. (G) Every cell against each axis: following it (r above the 95th percentile of the cell's own circular-shift null), opposing it (below the 5th), neither, or locked, with a cell's own structure's axis rebuilt without it. Visual cells follow the MEC axis as often as their own (17% against 16% of 1,355) and MEC cells follow the visual axis nearly as often as their own (21% against 25% of 3,657). The state is shared between the structures; what distinguishes visual cortex is the share of its cells that never switch.

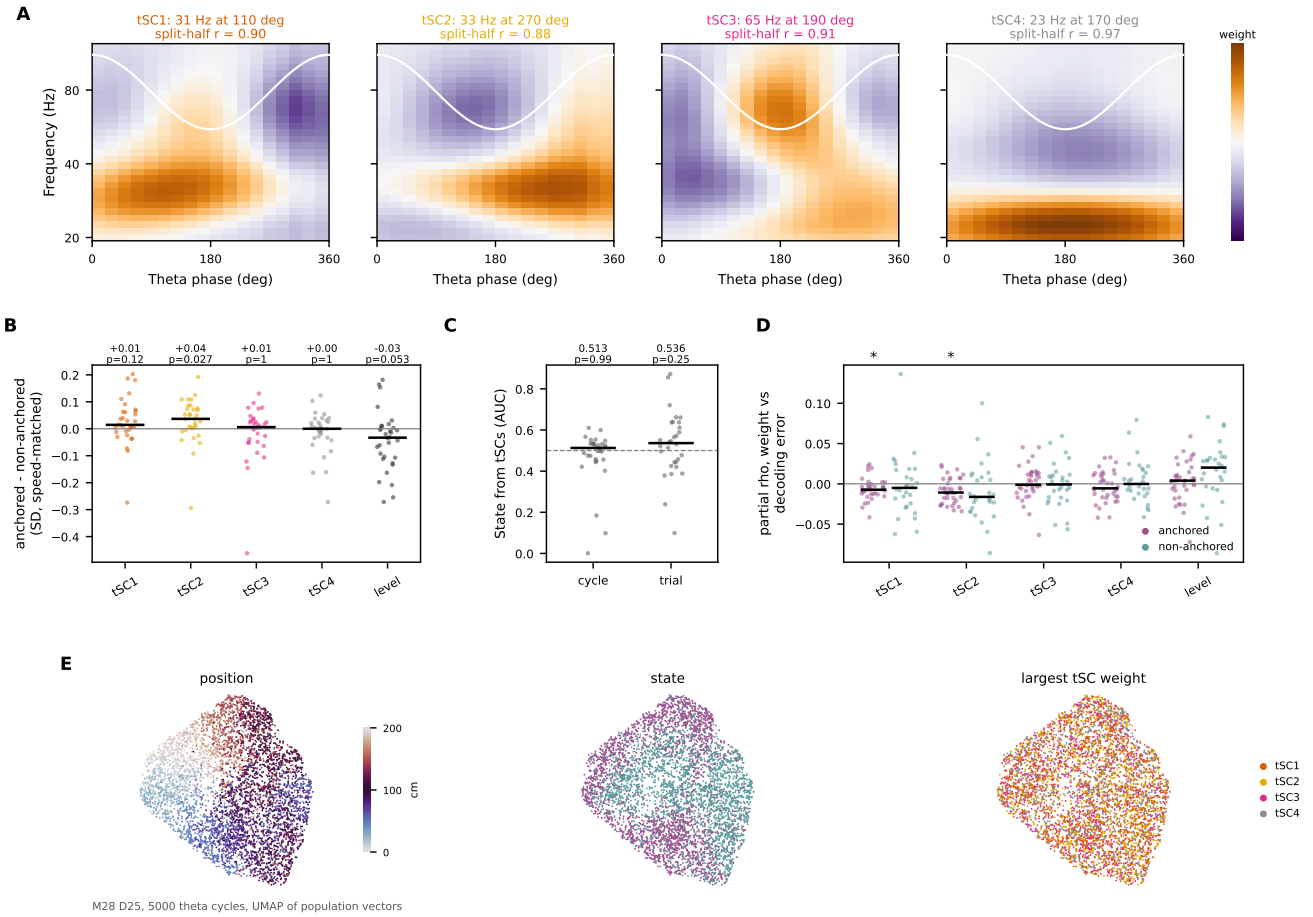



**Figure S16. Anchored and non-anchored trials are built from nearly the same kinds of theta cycle, related to Figure 8**

Following Lopes-dos-Santos et al. (2018) and Esparza et al. (2023), each theta cycle of the entorhinal LFP is described by its 20-120 Hz amplitude across 18 theta-phase bins, after subtracting the theta-locked waveform so that harmonics of non-sinusoidal theta cannot pose as nested gamma. Maps are normalised per session and frequency, the cycle's overall level is kept separately, and equal numbers of cycles from each of 44 sessions are decomposed by PCA and independent component analysis into theta-nested spectral components (tSCs). (A) The four components, the number at which every one reproduces across two disjoint halves of the sessions (split-half r above each), with a theta cycle drawn over them: slow gamma on the rising and on the falling phase, fast gamma at the trough, and a 23 Hz component across the cycle that may be residual theta harmonic. (B) Anchored minus non-anchored weight on each component and the cycle's gamma level, per session, after matching the two states on running speed, in units of the session's SD; Holm-corrected p above. Only the falling-phase slow-gamma component differs (+0.04 SD, p = 0.027) and the level is if anything lower when anchored (-0.03 SD, p = 0.053). (C) How well state can be read from a cycle's four weights and level: cross-validated AUC with folds by trial, per session, for single cycles (median 0.51) and for trial means (0.54), neither above chance. (D) Within each state, the partial correlation of each weight with how far the MEC population decoded on that cycle lies from the true position, controlling for speed, theta amplitude, spike count and position; stars mark Holm p < 0.05. Both slow-gamma components go with slightly better decoding in the anchored state (median rho -0.007 and -0.011), too small to carry the state difference. (E) One session's per-cycle MEC population vectors embedded with UMAP and coloured by position, state and the cycle's largest tSC: the manifold is organised by position and by state, and cycle type is scattered across it.